ON THE INTEGRABLE THEORY OF CHIRAL YANGIANS AND RELATED CHIRAL QUANTUM GROUPS

RAEEZ LORGAT

Abstract. For a chiral algebra $\mathcal{A}$ on a smooth algebraic curve X, we define the *ordered chiral homology* $\int_X^{\mathrm{ord}} \mathcal{A}$ as the derived pushforward of the ordered factorisation $\mathcal{D}$-module $\mathcal{F}^{\mathrm{ord}}(\mathcal{A})$ on the ordered Ran space $\mathrm{Ran}^{\mathrm{ord}}(X) = \coprod_n U_n(X)$, the moduli of ordered tuples of distinct points. The Beilinson–Drinfeld Ran space $\mathrm{Ran}(X)$ is the $\Sigma_\bullet$-coinvariant quotient, and the averaging map $\mathrm{av}\colon R\Gamma(\mathrm{Ran}^{\mathrm{ord}}(X), \mathcal{F}^{\mathrm{ord}}) \to R\Gamma(\mathrm{Ran}(X), \mathcal{F}^{\Sigma})$ recovers the Beilinson–Drinfeld chiral homology. For E_∞-chiral algebras (all standard vertex algebras), the Kontsevich–Tamarkin formality of the little-disks operad gives a quasi-isomorphism between the ordered and symmetric theories; for genuinely E_1-chiral algebras (Yangians, Etingof–Kazhdan quantum vertex algebras), this formality fails and the descent is lossy, depending on the choice of a Drinfeld associator.

The main structural result is the chiral quantum group equivalence: on the Koszul locus, three structures on an E_1-chiral algebra (the vertex R-matrix, the chiral A_∞-structure, and the chiral coproduct) determine each other, forming an equivalence triangle mediated by the ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ and its deconcatenation coproduct.

Five computations span the $G/L/C/M$ shadow classification: Heisenberg (class G, regular KZ, ordered = symmetric), $V_k(\mathfrak{sl}_2)$ (class L, KZ conformal blocks), $\beta\gamma$ (class C, nilpotent R-matrix), Virasoro (class M, irregular KZ connection with Stokes data), and the Yangian $Y(\mathfrak{sl}_2)$, which is the first example where ordered chiral homology is strictly richer than symmetric: the kernel $\ker(\mathrm{av})$ is non-trivial at every degree $n \geq 2$, with $\dim \ker(\mathrm{av}_n) = d^n - \binom{n+d-1}{d-1}$ on the d-dimensional representation (Proposition 2.6).

For E_∞-chiral algebras, the symmetric bar $\overline{B}^{\Sigma}(\mathcal{A})$ carries an E_2-coalgebra structure from the curve geometry. The derived chiral centre $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ is therefore an algebraic E_2 closed-sector object, and together with the boundary algebra it forms the universal $\mathrm{SC}^{\mathrm{ch,top}}$ pair. This package does not extend to genuinely E_1-chiral algebras, where the symmetric bar does not exist and the ordered bar yields only E_2 on the ordered derived centre; passage to a physical bulk then requires the categorical Drinfeld-centre step and an open–closed comparison. For $V_k(\mathfrak{g})$ at non-critical level, the Sugawara element topologises the affine derived centre on BRST cohomology, giving an $\mathsf{E}_3^{\mathrm{top}}$-algebra. The zero-differential cohomology complex gives an unconditional chain-level $\mathsf{E}_3^{\mathrm{top}}$ model quasi-isomorphic to the original complex, while the lift to the original cochain complex is conditional on the coherence equation $[m, G] = \partial_z$. The same formal parameter space $(k + h^\vee)H^3(\mathfrak{g})[[k + h^\vee]]$ classifies the Costello–Francis–Gwilliam perturbative Chern–Simons $\mathsf{E}_3^{\mathrm{top}}$-algebra. Formal disk restriction matches the associated graded and the P_3 bracket; identifying the full deformation families remains conjectural.

The construction extends to genus 1 and genus 2: the ordered chiral homology $\int_{E_\tau}^{\mathrm{ord}} Y(\mathfrak{sl}_2)$ on an elliptic curve E_τ is computed degree by degree, with the KZ connection replaced by the Knizhnik–Zamolodchikov–Bernard connection (Bernard [5], Felder [32]) and the B-cycle monodromy producing the quantum group parameter $q = e^{2\pi i}$. At genus 2, the period matrix is a 2×2 symmetric matrix in the Siegel upper half-space, the bar propagator is $d \log E(z, w)$ where E is the prime form, and the degree-2 Euler characteristic jumps to -12 (from -4 at genus 1). The kernel $\ker(\mathrm{av})$ grows strictly with the genus.

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1. Introduction

The ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with its deconcatenation coproduct is the level-2 object before symmetrisation. Every construction here begins with this E_1 chart datum; the symmetric theory is recovered by applying the lossy coinvariant projection $\mathrm{av}\colon \mathfrak{g}^{\mathsf{E}_1}_{\mathcal{A}} \to \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$. This is the E_1 primacy principle: construct the ordered story first, then derive the symmetric story by averaging.

The ordered bar complex leads naturally to the *ordered chiral homology* $\int_X^{\mathrm{ord}} \mathcal{A}$ of a chiral algebra $\mathcal{A}$ on an algebraic curve X: the derived pushforward of an ordered factorisation $\mathcal{D}$-module on the moduli of *labelled* tuples of distinct points, with no symmetric-group quotient imposed. The Beilinson–Drinfeld chiral homology [4] requires E_∞-commutativity of the factorisation product; the ordered theory does not.

Every standard vertex algebra (Heisenberg, affine Kac–Moody, Virasoro, $\mathcal{W}$-algebras, $\beta\gamma$, lattice) is E_∞-chiral: the locality axiom gives a symmetric factorisation product descending to $X^{(n)} = X^n/\Sigma_n$. The genuinely E_1-chiral objects (Yangians $Y(\mathfrak{g})$, Etingof–Kazhdan quantum vertex algebras [18], any algebra with a non-trivial vertex R-matrix $S(z)$) have a *braided* factorisation product: swapping inputs introduces a twist by $S(z)$.

The ordered Ran space $\mathrm{Ran}^{\mathrm{ord}}(X) = \coprod_n U_n(X)$ accommodates both: the Beilinson–Drinfeld Ran space is its $\Sigma_\bullet$-coinvariant quotient, and the commutative factorisation structure on $\mathrm{Ran}(X)$ is the equivariant shadow of the ordered structure that exists for any chiral algebra. This paper constructs the ordered factorisation $\mathcal{D}$-module $\mathcal{F}^{\mathrm{ord}}(\mathcal{A})$ on $\mathrm{Ran}^{\mathrm{ord}}(X)$ and defines ordered chiral homology as its derived pushforward.

1.1. Main results.

Three theorems and a conjecture emerge from the ordered theory. The first constructs the $\mathcal{D}$-module and defines ordered chiral homology; the second separates it from topological Hochschild homology; the third identifies the derived-centre and topologization package that the ordered picture feeds but does not itself carry. At the representation-theoretic level, Proposition 2.6 proves that $\dim\ker(\mathrm{av}_n) = d^n - \binom{n+d-1}{d-1}$ on $V^{\otimes n}$ for any d-dimensional representation V of any simple $\mathfrak{g}$, with generating function $\Delta P_d(t) = (1-dt)^{-1} - (1-t)^{-d}$; Conjecture 2.4 assembles the expected global structure of the ordered derived chiral centre of the chiral Yangian, including the E_2-algebra structure and the recovery of $\mathfrak{g}$ from the Gerstenhaber bracket.

Theorem 1.1 (Ordered chiral homology)**.** *Let $\mathcal{A}$ be a chiral algebra on a smooth proper curve X of genus g, and let $r(z)$ denote the classical r-matrix (the degree-2 projection of the Maurer–Cartan element $\Theta_{\mathcal{A}}$, the universal element of the E_1 convolution L_∞-algebra $\mathfrak{g}^{\mathsf{E}_1}_{\mathcal{A}}$ that packages the full bar differential of $\mathcal{A}$; see §6).*

(i) (Ordered factorisation $\mathcal{D}$-module on $\mathrm{Ran}^{\mathrm{ord}}(X)$.) *The OPE of $\mathcal{A}$, extended meromorphically across the diagonals, defines a factorisation $\mathcal{D}$-module $\mathcal{F}^{\mathrm{ord}}(\mathcal{A})$ on the ordered Ran space $\mathrm{Ran}^{\mathrm{ord}}(X) = \coprod_n U_n(X)$. Its restriction to the n-th stratum is the holonomic $\mathcal{D}$-module*
$$\mathcal{F}^{\mathrm{ord}}(\mathcal{A})\big|_{U_n} = \mathcal{A}^{\boxtimes n}\big|_{U_n}, \qquad \mathcal{F}^{\mathrm{ord}}_n(\mathcal{A}) := j_* j^*(\mathcal{A}^{\boxtimes n}) \in \mathcal{D}\text{-mod}(X^n),$$
where $j\colon U_n = X^n \setminus \bigcup_{1\le a<b\le n} \Delta_{ab} \hookrightarrow X^n$ is the inclusion of the ordered configuration space. No Σ_n-equivariance is imposed. The factorisation product, the KZ connection, and the chiral differential are encoded in the single object $\mathcal{F}^{\mathrm{ord}}(\mathcal{A})$ on $\mathrm{Ran}^{\mathrm{ord}}(X)$.

(ii) (Ordered chiral homology.) *The* ordered chiral homology *of $\mathcal{A}$ over X is the derived pushforward*
$$\int_X^{\mathrm{ord}} \mathcal{A} \;:=\; R\Gamma_{\mathrm{dR}}\big(\mathrm{Ran}^{\mathrm{ord}}(X),\ \mathcal{F}^{\mathrm{ord}}(\mathcal{A})\big).$$
The degree-by-degree decomposition $\bigoplus_{n\ge 0} R\Gamma_{\mathrm{dR}}(X^n,\ \mathcal{F}^{\mathrm{ord}}_n(\mathcal{A}))$ computes this pushforward stratum by stratum.

(iii) (Symmetric descent.) *The quotient* $q\colon \mathrm{Ran}^{\mathrm{ord}}(X) \to \mathrm{Ran}(X)$ *induces the averaging map*

$$\int_X^{\mathrm{ord}} \mathcal{A} = R\Gamma\big(\mathrm{Ran}^{\mathrm{ord}}(X),\ \mathcal{F}^{\mathrm{ord}}\big) \xrightarrow{\ \mathrm{av}\ } R\Gamma\big(\mathrm{Ran}(X),\ \mathcal{F}^{\Sigma}\big) = \int_X \mathcal{A},$$

which is a quasi-isomorphism for E_∞*-chiral* $\mathcal{A}$*. For* E_1*-chiral* $\mathcal{A}$*, the descent is lossy: the kernel of* $\mathrm{av}\colon R\Gamma_{\mathrm{dR}}(X^n, \mathcal{F}^{\mathrm{ord}}_n) \twoheadrightarrow R\Gamma_{\mathrm{dR}}(X^n, \mathcal{F}^{\mathrm{ord}}_n)_{\Sigma_n}$ *is non-trivial at* $n \geq 2$*: at* $n = 2$ *the R-matrix twist produces* $\ker(\mathrm{av}_2) \neq 0$ *(Example 2.3), and at general degree* $\dim\ker(\mathrm{av}_n) = d^n - \binom{n+d-1}{d-1}$ *on the d-dimensional representation (Proposition 2.6). At* $n \geq 3$ *the Drinfeld associator contributes* GRT_1*-dependent data.*

Theorem 1.2 (Topological Hochschild vs. ordered chiral homology of $D^\times$)**.** *Let* $\mathcal{A}$ *be a chiral algebra on a curve* X*, and let* $D^\times = \mathrm{Spec}\ \mathbb{C}((z))$ *be the punctured formal disk.*

(i) (Topological Hochschild.) *Viewing* $\mathcal{A}$ *as an* E_1*-topological algebra by forgetting the holomorphic structure, the factorisation homology* $\int_{S^1} \mathcal{A}$ *is the topological Hochschild homology* $\mathrm{HH}_*(\mathcal{A})$*, computed by the cyclic bar complex with the topological Hochschild differential using the associative product* $m_2\colon \mathcal{A} \otimes \mathcal{A} \to \mathcal{A}$ *and the higher* A_∞ *operations* m_k *(the chiral* A_∞*-structure maps; defined formally in Definition 6.2).*

(ii) (Ordered chiral homology of $D^\times$.) *The ordered chiral chain complex at degree n on* $D^\times$ *is*

$$C_n^{\mathrm{ord}}(D^\times, \mathcal{A}) = \Big(\Omega^*_{\log}\big(\overline{\mathrm{FM}}{}_n^{\mathrm{ord}}(D^\times)\big) \otimes (s^{-1}\bar{\mathcal{A}})^{\otimes n},\ \ d_{\mathrm{total}}\Big)$$

where $\Omega^*_{\log}$ *denotes logarithmic differential forms on the Fulton–MacPherson compactification,* s^{-1} *is the desuspension* ($|s^{-1}v| = |v| - 1$)*,* $\bar{\mathcal{A}} = \ker(\varepsilon)$ *is the augmentation ideal, and*

$$d_{\mathrm{total}} = d_{\mathrm{dR}} + d_{\mathrm{bar}} - \nabla_{\mathrm{KZ}}$$

with d_{dR} *the de Rham differential,* d_{bar} *the bar differential encoding the OPE, and*

$$\nabla_{\mathrm{KZ}} = \sum_{i<j} r_{ij}(z_i - z_j)\, dz_{ij}, \qquad dz_{ij} = d(z_i - z_j),$$

the Knizhnik–Zamolodchikov connection. The Arnold forms $\omega_{ij} = d\log(z_i - z_j)$ *are generators of the cohomology algebra* $H^*(\mathrm{Conf}_n(\mathbb{C}))$ *and appear as coefficients in the bar differential* d_{bar}*; they are* not *the connection 1-form of* ∇_{KZ}*.*

(iii) (Formality bridge.) *For* E_∞*-chiral* $\mathcal{A}$*, the formality bridge* (*Theorem* 7.4) *gives a quasi-isomorphism* $C_n^{\mathrm{ord}}(D^\times, \mathcal{A}) \xrightarrow{\sim} \mathrm{HH}_*(\mathcal{A})$*: ordered and symmetric chiral homologies agree as chain complexes, though the transferred* A_∞*-structure may carry non-trivial higher operations for class* $L/C/M$*. For genuinely* E_1*-chiral* $\mathcal{A}$ (*where* $S(z) \neq \mathrm{id}$)*, this quasi-isomorphism does not exist.*

Theorem 1.3 (Derived centre, topologization, and Chern–Simons)**.** *Let* $\mathfrak{g}$ *be a simple finite-dimensional Lie algebra and* $V_k(\mathfrak{g})$ *the affine Kac–Moody vertex algebra at level k.*

(i) (The algebraic bulk package.) *The curve geometry provides the* E_2 *structure on* $\overline{B}^{\Sigma}(V_k(\mathfrak{g}))$ (*Warning* 3.5)*. Its chiral Hochschild cochains*

$$Z^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g})) = \mathrm{HH}^*\big(\overline{B}^{\Sigma}(V_k(\mathfrak{g})),\ \overline{B}^{\Sigma}(V_k(\mathfrak{g}))\big)$$

carries the algebraic E_2 *structure of the derived centre. Together with the boundary algebra* $V_k(\mathfrak{g})$ *it forms the universal* $\mathrm{SC}^{\mathrm{ch,top}}$ *pair; the bar complex remains the* E_1 *coalgebraic engine, not the bulk object itself.*

(ii) (Affine topologization at non-critical level.) *If* $k \neq -h^\vee$*, the Sugawara element upgrades the derived centre on BRST cohomology to an* $\mathsf{E}_3^{\mathrm{top}}$*-algebra. More precisely,* $H_Q^\bullet\big(Z^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g}))\big)$ *carries* $\mathsf{E}_2^{\mathrm{top}} \otimes$

$\mathsf{E}_1^{\mathrm{top}} = \mathsf{E}_3^{\mathrm{top}}$, *the zero-differential cohomology complex gives an unconditional chain-level* $\mathsf{E}_3^{\mathrm{top}}$ *model quasi-isomorphic as a chain complex to the original derived centre, and the lift to the original cochain complex is conditional on* $[m, G] = \partial_z$.

(iii) (Matching deformation parameter space.) *The affine package and the perturbative Costello–Francis–Gwilliam package are both governed by the same formal parameter space* $(k + h^\vee)H^3(\mathfrak{g})[[k + h^\vee]]$. *For simple* $\mathfrak{g}$, $H^3(\mathfrak{g})$ *is one-dimensional, so there is a single deformation direction at each order in* $(k + h^\vee)$.

(iv) (Comparison with perturbative Chern–Simons.) *Costello–Francis–Gwilliam* [12] *identify* $(k + h^\vee)H^3(\mathfrak{g})[[k + h^\vee]]$ *with the deformation space of perturbative Chern–Simons quantisations, filtered* $\mathsf{E}_3^{\mathrm{top}}$*-algebras deforming* $C^*(\mathfrak{g})$, *braided monoidal deformations of* $\mathrm{Rep}_{\mathrm{fin}}(\mathfrak{g})$, *and quasi-triangular quasi-Hopf deformations of* $U(\mathfrak{g})$. *Formal disk restriction matches the affine and CFG constructions at the associated-graded and* P_3*-bracket level. Identifying the full deformation families as chain-level* $\mathsf{E}_3^{\mathrm{top}}$*-algebras remains conjectural.*

(v) (Topological enhancement.) *By Khan–Zeng* [28], *the Sugawara Virasoro element at non-critical level upgrades the holomorphic-topological symmetry of the 3d gauge theory to fully topological symmetry. Khan–Zeng* [28] *prove this at the classical* (*PVA*) *level; the quantum extension holds because the Sugawara element exists as an exact operator in* $V_k(\mathfrak{g})$ *at every non-critical level, satisfying the full Virasoro commutation relations without quantum corrections to the BRST-exactness argument* (*see Proposition 8.11*). *At the critical level* $k = -h^\vee$, *the Sugawara element is undefined and the topological enhancement does not apply; the topological nature of the perturbative* E_3 *in the formal neighbourhood of critical level holds order-by-order in* $(k + h^\vee)$ (*since the Sugawara element exists at any nonzero departure from critical level*).

1.2. **Architecture of the paper.** §2: the Yangian motivating example (three structures, ker(av), kernel formula). §3: the $\mathsf{E}_1/\mathsf{E}_\infty$ hierarchy. §4–5: $\mathcal{D}$-modules and ordered chiral homology. §6: the chiral quantum group equivalence (the main structural result). §7: formality bridge. §8: the E_3 structure. §9: explicit computations $(G/L/C/M)$. §10–11: genus 1 and genus 2. §12–14: comparison with BBJ/Latyntsev and open problems.

2. The motivating example: $Y(\mathfrak{sl}_2)$

The Yangian $Y(\mathfrak{sl}_2)$ is the simplest genuinely E_1-chiral algebra: it is the first example where ordered chiral homology is strictly richer than the Beilinson–Drinfeld (symmetric) theory. Before developing the general machinery, we present this example in full to exhibit the three structures (vertex R-matrix, chiral A_∞-structure, chiral coproduct) and the non-trivial kernel ker(av) that motivate the entire construction.

Proposition 2.1 (Yangian presentations)**.** *The following are isomorphic as filtered associative algebras:*

(i) *The Yangian* $Y(\mathfrak{g})$ *in the RTT presentation: generated by* $T_{ij}(u)$ *with* $R_{12}(u - v)\, T_1(u)\, T_2(v) = T_2(v)\, T_1(u)\, R_{12}(u - v)$, *where* $R(u) = 1 + \quad P/u$ *and* $\quad = 1/(k + h^\vee)$.

(ii) *The Yangian* $Y(\mathfrak{g})$ *in the Drinfeld first presentation: generated by* $\{x_i, \xi_i : i \in I\}$ *with Serre-type relations* (*Drinfeld* [10]).

(iii) *The* E_1*-chiral Koszul dual* $\mathcal{A}^!_{\mathrm{ord}}$ *of* $V_k(\mathfrak{g})$: *the algebra obtained from the ordered bar coalgebra* $B^{\mathrm{ord}}(V_k(\mathfrak{g}))$ *by spectral strictification, with RTT relations recovered from collision residues on* $\mathrm{Conf}_n^{\mathrm{ord}}(\mathbb{C})$.

The isomorphism (i) ≅ (ii) *is classical* [10]. *The isomorphism* (ii) ≅ (iii) *is the spectral Drinfeld strictification* ([34, §III]*; proved for all simple* $\mathfrak{g}$).

Remark 2.2 (Spectral parameter locus). The spectral parameter z lives on the structures (the coproduct Δ_z, the R-matrix $R(z)$, the transfer matrix $T(z)$, the evaluation functor ev_z), not on the algebra symbol. The algebra $Y\ (\mathfrak{g})$ is a single filtered algebra; z parametrises representations and coalgebra maps.

Example 2.3 (Yangian $Y\ (\mathfrak{sl}_2)$ as a chiral quantum group: three structures and the ordered chiral center)**.** The Yangian $Y\ (\mathfrak{sl}_2)$ is the first fully worked-out E_1-chiral quantum group: an E_1-chiral algebra equipped with chiral coproduct Δ_z, spectral R-matrix $R(z)$, and quasi-triangularity, making $\mathrm{Mod}_{Y\ (\mathfrak{sl}_2)}$ a braided monoidal category (equivalently, E_2 in Cat). It is the first example in which ordered chiral homology differs from Beilinson–Drinfeld (symmetric) chiral homology. The R-matrix is independent input, not derived from any E_∞-chiral OPE. The formality bridge (Theorem 7.4) does not apply, because the factorisation $\mathcal{D}$-module is not Σ_n-equivariant. We exhibit all three structures of the chiral quantum group equivalence (Theorem 6.5) concretely, then compute the ordered chiral center at low degrees and identify $\ker(\mathrm{av}) \neq 0$.

Data. Let $\mathfrak{g} = \mathfrak{sl}_2$ with standard basis $\{e, f, h\}$, brackets $[e, f] = h$, $[h, e] = 2e$, $[h, f] = -2f$, and normalised Killing form $(e, f) = 1$, $(h, h) = 2$. The dual Coxeter number is $h^\vee = 2$. The deformation parameter is $\ = 1/(k + h^\vee) = 1/(k + 2)$ (Remark 8.7). The inverse Casimir is

$$\Omega = e \otimes f + f \otimes e + \tfrac{1}{2}\, h \otimes h \ \in\ \mathfrak{g} \otimes \mathfrak{g}. \tag{2.1}$$

In the fundamental representation $V = \mathbb{C}^2$: $(\rho \otimes \rho)(\Omega) = P - \frac{1}{2} I$, where P is the permutation operator on $V \otimes V$.

Structure (I): the R-matrix. The Yang R-matrix on $V \otimes V$ is

$$R(u) = u\, I + \quad P \ \in\ \mathrm{End}(V \otimes V)[u], \tag{2.2}$$

where u is the spectral parameter (the difference of evaluation points on $\mathbb{C}$). This satisfies the quantum Yang–Baxter equation

$$R_{12}(u - v)\, R_{13}(u)\, R_{23}(v) = R_{23}(v)\, R_{13}(u)\, R_{12}(u - v)$$

in $\mathrm{End}(V^{\otimes 3})[u, v]$, and the unitarity condition $R_{12}(u)\, R_{21}(-u) = (\ ^2 - u^2)\, I$ (scalar: invertible for $u \neq \pm\ $).

At the classical level, $R(u) = u\, I + \quad P = u(I + \quad \Omega/u + O(\ ^2))$ after restoring the Casimir from $P = \Omega + \frac{1}{2} I$ on $V \otimes V$. The classical r-matrix $r(z) = \quad \Omega/z$ is in KZ normalisation; at $\ = 0$ (i.e. $k \to \infty$), $r = 0$; at the critical level $k = -2$, diverges.

The eigenvalues of $R(u)$ on $V \otimes V$ are $u +\ $ on $\mathrm{Sym}^2 V$ (dimension 3) and $u -\ $ on $\bigwedge^2 V$ (dimension 1); the R-matrix has a zero on $\bigwedge^2 V$ at $u =\ $, selecting the antisymmetric channel.

Structure (II): the A_∞ structure. The chiral A_∞-structure maps $m_k^{\mathrm{ch}} : Y^{\otimes k} \to Y$ are the operations extracted from the ordered bar complex via the bar-cobar correspondence. The transfer matrix

$$T(u) = \begin{pmatrix} A(u) & B(u) \\ C(u) & D(u) \end{pmatrix} \ \in\ \mathrm{End}(V) \otimes Y\ (\mathfrak{sl}_2)[[u^{-1}]]$$

encodes the generating functions of the algebra.

The binary operation m_2^{ch} is the RTT relation: for any two entries $T_{ij}(u)$ and $T_{kl}(v)$ of the transfer matrix,

$$R_{12}(u - v)\, T_1(u)\, T_2(v) = T_2(v)\, T_1(u)\, R_{12}(u - v), \tag{2.3}$$

where subscripts indicate the auxiliary space factor. Expanding this 4×4 matrix equation produces the Yangian relations in RTT form. The ten independent component relations include:

- diagonal commutativity: $[A(u), A(v)] = 0$, $[D(u), D(v)] = 0$;
- off-diagonal commutativity: $[B(u), B(v)] = 0$, $[C(u), C(v)] = 0$;

- the cross-relations, e.g. $(u-v)[B(u), C(v)] = \big(D(u)A(v) - D(v)A(u)\big)$.

The higher operations m_k^{ch} for $k \geq 3$ involve the Drinfeld associator Φ_{KZ}. The binary product m_2^{ch} is strictly associative only up to the holonomy of the KZ connection on $\overline{\mathrm{FM}}_3^{\mathrm{ord}}(\mathbb{C})$: the correction is $m_3^{\mathrm{ch}}(a, b, c) = \Phi_{\mathrm{KZ}} \cdot m_2(m_2(a, b), c) - m_2(a, m_2(b, c))$, where Φ_{KZ} is the regularised holonomy between the two maximal boundary strata of $\overline{\mathrm{FM}}_3^{\mathrm{ord}}(\mathbb{C})$.

The Gauss decomposition $T(u) = F(u)\,K(u)\,E(u)$ (lower-triangular, diagonal, upper-triangular) translates the RTT relations into the Drinfeld presentation: generators $\{E_r, F_r, H_r\}_{r\geq 0}$ with relations

$$\begin{aligned} [H_r, H_s] &= 0, \\ [H_0, E_s] &= 2E_s, \quad [H_0, F_s] = -2F_s, \\ [E_r, F_s] &= H_{r+s}, \\ [H_{r+1}, E_s] - [H_r, E_{s+1}] &= \ (H_r E_s + E_s H_r), \end{aligned}$$

and the Serre relation $[E_{r+1}, [E_r, E_s]] - [E_r, [E_{r+1}, E_s]] = 0$ (and similarly for F). The leading modes $E_0 = e$, $F_0 = f$, $H_0 = h$ generate a copy of $U(\mathfrak{sl}_2)$ inside $Y(\mathfrak{sl}_2)$; the PBW filtration has $\operatorname{gr} Y(\mathfrak{sl}_2) \cong U(\mathfrak{sl}_2[t])$.

Structure (III): the chiral coproduct. The Drinfeld coproduct on $Y(\mathfrak{sl}_2)$ is

$$\Delta_z(T(u)) = T(u) \dot{\otimes} T(u - z), \tag{2.4}$$

where $\dot{\otimes}$ denotes matrix multiplication in the auxiliary space $\mathrm{End}(V)$ and ordinary (completed) tensor product in the Y factor. The spectral parameter z is the difference coordinate $z_1 - z_2$ on $\mathrm{Conf}_2(\mathbb{C})$. On Drinfeld generators:

$$\begin{aligned} \Delta_z(E_0) &= E_0 \otimes 1 + 1 \otimes E_0 + \ \ H_0 \otimes E_0 \cdot z^{-1} + O(z^{-2}), \\ \Delta_z(H_0) &= H_0 \otimes 1 + 1 \otimes H_0 + 2 \ \ (E_0 \otimes F_0 - F_0 \otimes E_0) \cdot z^{-1} + O(z^{-2}). \end{aligned}$$

The coproduct is coassociative up to the Drinfeld associator Φ:

$$(\Delta_{z_1} \otimes \mathrm{id}) \circ \Delta_{z_2} = \Phi_{123}(z_1, z_2) \cdot (\mathrm{id} \otimes \Delta_{z_2}) \circ \Delta_{z_1} \cdot \Phi_{123}(z_1, z_2)^{-1},$$

where $\Phi_{123}(z_1, z_2) \in \exp(\widehat{\mathrm{Lie}}(\Omega_{12}, \Omega_{23}))$ is the KZ associator with spectral parameters.

The R-matrix is recovered from the coproduct: $R(u) = \sigma \circ \Delta_z \circ (\Delta_z^{\mathrm{op}})^{-1}$ evaluated on $V \otimes V$, where σ is the flip. This is the content of the equivalence (I)$\Leftrightarrow$(III) in Theorem 6.5.

The three structures are equivalent. The ordered bar complex $\overline{B}^{\mathrm{ord}}(Y(\mathfrak{sl}_2)) = T^c(s^{-1}\overline{Y}\ )$ (desuspended augmentation ideal, deconcatenation coproduct) is the universal datum from which all three structures are recovered:

(a) The bar differential encodes the A_∞ maps (coderivations of T^c biject with multilinear operations): the RTT relation (2.3) is the degree-2 bar differential, and the Serre relations are the degree-3 component.
(b) The deconcatenation coproduct on T^c induces the Drinfeld coproduct (2.4) via cobar dualisation.
(c) The R-matrix (2.2) is the degree-2 KZ monodromy: the parallel transport of the connection $\nabla = d - \ \ \Omega/(z_1 - z_2) \cdot d(z_1 - z_2)$ around the diagonal.

The passage $r(z) = \ \ \Omega/z \xrightarrow{\text{exponentiation}} R(u) = u\,I + \ \ P \xrightarrow{\text{RTT}}$ Yangian relations $\xrightarrow{\text{Gauss}}$ Drinfeld presentation is the ordered bar construction applied to $V_k(\mathfrak{sl}_2)$, producing $Y(\mathfrak{sl}_2)$ as the E_1-chiral Koszul dual.

Part B: the ordered chiral center. We compute $\mathrm{HH}^{*,\mathrm{ord}}_{\mathrm{ch}}(Y\ (\mathfrak{sl}_2),\ Y\ (\mathfrak{sl}_2))$ at low degrees on the formal disk D, and identify the kernel of the averaging map av.

Degree 0*: the center.* The degree-0 cohomology is the chiral center $Z(Y\ (\mathfrak{sl}_2))$: those elements a with $a_{(n)}b = 0$ for all b and $n \geq 0$. The quantum determinant qdet $T(u) = A(u)\,D(u - \ \ ) - B(u)\,C(u - \ \ )$ is central in $Y\ (\mathfrak{sl}_2)$ [37]. The center is the polynomial algebra generated by the coefficients of qdet $T(u)$:

$$\mathrm{HH}^{0,\mathrm{ord}}_{\mathrm{ch}} = Z(Y\ (\mathfrak{sl}_2)) = \mathbb{C}[\mathrm{qdet}\ T(u)].$$

The averaging map av_0 is an isomorphism at degree 0 (there is nothing to symmetrise), so $\ker(\mathrm{av}_0) = 0$.

Degree 2*: the KZ local system and* $\ker(\mathrm{av}_2) \neq 0$. This is the degree where the ordered theory diverges from the symmetric. The ordered chiral Hochschild complex at degree 2 involves the de Rham cohomology of $\mathrm{Conf}^{\mathrm{ord}}_2(\mathbb{C})$ with coefficients in the KZ local system. At degree 2, the leading-order KZ connection $\nabla = d - \quad \Omega\, d(z_1 - z_2)/(z_1 - z_2)$ captures the classical r-matrix data $r(z) = \quad \Omega/z$. The full quantum R-matrix $R(u) = u\,I + \quad P$ contributes higher-order corrections visible at degree ≥ 3 through the Drinfeld associator; at degree 2 the connection is:

$$\nabla_{\mathrm{KZ}} = d - \frac{\Omega}{z_1 - z_2}\, d(z_1 - z_2),$$

with coefficients in $\mathrm{End}(Y^{\otimes 2})$. Setting $w = z_1 - z_2$, the flat sections on $\mathbb{C}^\times$ satisfy $\partial\Phi/\partial w = (\quad \Omega/w)\,\Phi$.

For the Yangian on the fundamental representation $V \otimes V$, the Casimir decomposes along the irreducible $\mathfrak{sl}_2$-components:

$$V \otimes V = \mathrm{Sym}^2 V \oplus \bigwedge^2 V, \qquad \Omega\big|_{\mathrm{Sym}^2 V} = +\tfrac{1}{2}, \quad \Omega\big|_{\wedge^2 V} = -\tfrac{3}{2}.$$

The monodromy of ∇_{KZ} around $w = 0$ on each isospin component is

$$M_{\mathrm{sym}} = e^{2\pi i \quad /2} = e^{\pi i/(k+2)}, \qquad M_{\mathrm{alt}} = e^{-3\pi i/(k+2)}.$$

The averaging map at degree 2 is $\mathrm{av}_2 : H^*_{\mathrm{dR}}(\mathrm{Conf}^{\mathrm{ord}}_2, \nabla_{\mathrm{KZ}}) \to H^*_{\mathrm{dR}}(\mathrm{Conf}_2, \nabla_{\mathrm{KZ}})^{\Sigma_2}$, where Σ_2 acts by the flip $w \mapsto -w$ composed with the R-matrix.

For an E_∞-chiral algebra, $R = \mathrm{id}$ and the flip acts by the Koszul sign; av_2 is an isomorphism and $\ker(\mathrm{av}_2) = 0$. For the Yangian, $R(u) \neq \mathrm{id}$: the R-matrix mixes the $\mathrm{Sym}^2 V$ and $\wedge^2 V$ channels nontrivially, and the Σ_2-action on the ordered de Rham cohomology is the composition of the geometric flip with R.

The kernel is spanned by those flat sections that are antisymmetric under this twisted Σ_2-action:

$$\ker(\mathrm{av}_2) \cong \bigwedge^2 V = \mathbb{C}. \tag{2.5}$$

The generator is the flat section $\Phi_{\mathrm{alt}}(w) = w^{-3 \quad /2}$ on the antisymmetric channel of $V \otimes V$. The full R-matrix $R(u) = u\,I + \quad P$ evaluates on the antisymmetric component as $(u - \quad)$ id; the Σ_2-flip sends $\Phi_{\mathrm{alt}}(w)$ to $R_{21}(-w)\,\Phi_{\mathrm{alt}}(-w) = -\Phi_{\mathrm{alt}}(w)$ (the R-matrix eigenvalue -1 on $\wedge^2 V$ at the leading order cancels the geometric sign). This class is in the ordered cohomology but maps to zero under av_2: it is invisible to the symmetric chiral homology.

The scalar shadow surviving av_2 is $\kappa = 3(k+2)/4$ (equation (9.1)).

Degree 3*: the Drinfeld associator in* $\ker(\mathrm{av}_3)$. At degree 3, the KZ system on $\mathrm{Conf}^{\mathrm{ord}}_3(\mathbb{C})$ is

$$d\Phi = \left(\frac{\Omega_{12}}{z_1 - z_2}\, d(z_1 - z_2) + \frac{\Omega_{13}}{z_1 - z_3}\, d(z_1 - z_3) + \frac{\Omega_{23}}{z_2 - z_3}\, d(z_2 - z_3)\right)\Phi.$$

The fundamental group $\pi_1(\mathrm{Conf}_3^{\mathrm{ord}}(\mathbb{C})) = PB_3$ (pure braid group on three strands). The KZ monodromy gives a representation $\rho_{\mathrm{KZ}} \colon PB_3 \to \mathrm{GL}(V^{\otimes 3})$.

The Drinfeld associator $\Phi_{\mathrm{KZ}} \in \exp(\widehat{\mathrm{Lie}}(t_{12}, t_{23}))$ is the regularised holonomy of this connection along the real path from the boundary stratum $|z_1 - z_2| \ll |z_2 - z_3|$ to the stratum $|z_2 - z_3| \ll |z_1 - z_2|$ in $\overline{\mathrm{FM}}_3^{\mathrm{ord}}(\mathbb{C})$.

The averaging map av_3 sends ordered data to Σ_3-coinvariants. At degree 3, the relevant representation is the adjoint $\mathfrak{g} = \mathfrak{sl}_2$ (dimension 3), not the fundamental $V = \mathbb{C}^2$ (dimension 2) used at degree 2: the convolution algebra acts on $\mathfrak{g}^{\otimes n}$ via the adjoint action (the Casimir operators Ω_{ij} act through the adjoint representation on each tensor factor), and the Drinfeld associator lives in $\Lambda^3(\mathfrak{g})$. The kernel of av_3 contains the associator:

$$\ker(\mathrm{av}_3) \cong \Lambda^3(\mathfrak{g}) = \mathbb{C},$$

one-dimensional, carrying the associator class. This computation follows from the structure of the cohomology of $\mathrm{Conf}_3^{\mathrm{ord}}(\mathbb{C})$: the top-degree Arnold class $\omega_{12} \wedge \omega_{23} \in H^2(\mathrm{Conf}_3^{\mathrm{ord}})$ maps to the alternating representation of Σ_3, and the KZ-twisted version (with coefficients in $\mathfrak{g}^{\otimes 3}$) gives $\ker(\mathrm{av}_3) \cong \Lambda^3(\mathfrak{g})$. For $\mathfrak{g} = \mathfrak{sl}_2$: $\dim \Lambda^3(\mathfrak{g}) = \binom{3}{3} = 1$.

For E_∞-chiral algebras, the formality bridge forces $\ker(\mathrm{av}_3) = 0$: the associator is in the image of a Σ_3-equivariant quasi-isomorphism and carries no independent ordered data. For the Yangian, the $\mathcal{D}$-module is not Σ_3-equivariant (because $R \neq \mathrm{id}$), and the associator survives as a non-trivial element of the ordered chiral center invisible to any symmetric theory.

Explicit computation in $\ker(\mathrm{av}_3)$. On the fundamental representation $V = \mathbb{C}^2$ of $\mathfrak{sl}_2$, the KZ system at three points reduces to a 2×2 Fuchsian system on the multiplicity space $M = \mathrm{Hom}_{\mathfrak{sl}_2}(V_1, V^{\otimes 3})$, where V_j is the spin-$j/2$ irreducible (so $V_1 = \mathbb{C}^2$ is the fundamental). Since $V^{\otimes 3} \cong V_3 \oplus V_1^{\oplus 2}$, M is 2-dimensional: its two basis vectors correspond to the (12)-pair coupled to spin 0 (the $\bigwedge^2 V$ channel) and spin 1 (the $\mathrm{Sym}^2 V$ channel). The Casimir Ω_{12} acts on M as $A = \mathrm{diag}(-\frac{3}{2}, \frac{1}{2})$ (eigenvalue $-\frac{3}{2}$ on the spin-0 channel, $\frac{1}{2}$ on spin-1). The representation of Σ_3 on M has character $(2, 0, 0, 0, -1, -1)$: it is the standard representation with no trivial summand. The Reynolds operator $\frac{1}{6}\sum_{\sigma \in \Sigma_3} \sigma$ therefore vanishes identically on M, so the *entire* 2-dimensional multiplicity space lies in $\ker(\mathrm{av}_3)$. The Drinfeld associator acts on M as a pure E_1 invariant with no symmetric shadow.

At level $k = 2$ ($\kappa = 3(k+2)/4 = 3$), direct numerical integration of the Fuchsian system gives

$$\Phi_{\mathrm{KZ}}\big|_M = \begin{pmatrix} 0.884 & -0.204 \\ 0.306 & 1.061 \end{pmatrix}, \qquad \det \Phi_{\mathrm{KZ}}\big|_M = 1,$$

with eigenvalues on the unit circle (the connection is unitary on M). The off-diagonal entry $|\Phi_{12}| = 0.204$ measures the channel-mixing amplitude between the $j = 0$ and $j = 1$ sectors; it decays as $2.849/\kappa^2$ at large κ, consistent with the semiclassical expansion

$$\Phi_{\mathrm{KZ}}\big|_M = I - \frac{\zeta(2)}{\kappa^2}\,[t_{12}, t_{23}] + O(1/\kappa^3),$$

where $t_{ij} = \Omega_{ij}/(k + h^\vee)$ and $[t_{12}, t_{23}] \neq 0$ on M (reflecting the non-abelian Lie bracket). This is an explicit non-trivial class in $\ker(\mathrm{av}_3)$: an ordered invariant of the Yangian that is invisible to every symmetric chiral-homological observable.

Degree n: braid group monodromy. At degree n, the ordered chiral center carries the full pure braid group monodromy of the KZ system with the quantum R-matrix:

$$\rho_{\mathrm{KZ}}^{(n)} \colon PB_n \to \mathrm{GL}(V^{\otimes n}).$$

The symmetric shadow at degree n retains only the scalar invariants: the traces of the monodromy matrices and the Markov-trace data. The kernel $\ker(\mathrm{av}_n)$ carries the full matrix-valued monodromy, from which knot invariants (the Jones polynomial and its coloured generalisations) are extracted.

The Poincaré series of the ordered chiral centre, evaluated on the fundamental representation $V = \mathbb{C}^2$, admits a closed-form generating function. At degree n, the ordered datum is $V^{\otimes n}$ (dimension 2^n) and the symmetric shadow is $\mathrm{Sym}^n(V)$ (dimension $n+1$), so

$$P_V^{\mathrm{ord}}(t) = \sum_{n\geq 0} 2^n\, t^n = \frac{1}{1-2t}, \qquad P_V^{\Sigma}(t) = \sum_{n\geq 0} (n+1)\, t^n = \frac{1}{(1-t)^2}.$$

The kernel series recording the genuinely E_1 data at each degree is

$$\Delta P_V(t) \;=\; P_V^{\mathrm{ord}}(t) - P_V^{\Sigma}(t) \;=\; \frac{1}{1-2t} - \frac{1}{(1-t)^2} \;=\; \frac{t^2}{(1-2t)(1-t)^2},$$

with coefficients $\dim\ker(\mathrm{av}_n) = 2^n - (n+1)$ for $n \geq 0$. The first ten values are 0, 0, 1, 4, 11, 26, 57, 120, 247, 502: the kernel vanishes at degrees 0 and 1, starts at degree 2 with the antisymmetric R-matrix block $\bigwedge^2 V = \mathbb{C}$, and grows exponentially thereafter. The symmetric shadow retains $O(n)$ dimensions out of $O(2^n)$; almost all ordered data is invisible to the Beilinson–Drinfeld theory at large degree.

Why this is new. This is the first instance where the formality bridge (Theorem 7.4) fails: the $\mathcal{D}$-module is not Σ_n-equivariant because $R(u) \neq \mathrm{id}$, so the ordered chiral homology carries strictly more data than the symmetric theory.

n	$\dim\ker(\mathrm{av}_n)$	Ordered datum ($V = \mathbb{C}^2$)	Symmetric shadow
0	0	$Z(Y) = \mathbb{C}[\mathrm{qdet}]$	$Z(Y)$
1	0	$V = \mathbb{C}^2$	V
2	1	$V^{\otimes 2}$, dim 4; R-matrix	$\mathrm{Sym}^2 V$, $\kappa = 3(k{+}2)/4$
3	4	$V^{\otimes 3}$, dim 8; $\supseteq$ associator	$\mathrm{Sym}^3 V$, dim 4
4	11	$V^{\otimes 4}$, dim 16	$\mathrm{Sym}^4 V$, dim 5
n	$2^n{-}(n{+}1)$	$V^{\otimes n}$, dim 2^n; PB_n-monodromy	$\mathrm{Sym}^n V$, dim $n{+}1$

At degree 3, the 4-dimensional kernel contains the Drinfeld associator class $\Lambda^3(\mathfrak{g}) = \mathbb{C}$ as a distinguished one-dimensional subspace; the remaining 3 dimensions carry the mixed-symmetry Schur–Weyl components of $V^{\otimes 3}$ under Σ_3.

The computations in Example 2.3 determine the ordered chiral centre at low degrees. We now state the expected global structure.

Conjecture 2.4 (Ordered derived chiral centre of the chiral Yangian). *Let $\mathfrak{g} = \mathfrak{sl}_2$ and let $Y(\mathfrak{g})^{\mathrm{ch}}$ be the chiral Yangian on a smooth curve X.*

(i) (Infinite-dimensionality.) *The ordered derived chiral centre*

$$Z_{\mathrm{ch}}^{\mathrm{der,ord}}\big(Y(\mathfrak{sl}_2)^{\mathrm{ch}}\big)$$

is infinite-dimensional as a graded vector space. This stands in contrast to the derived chiral centre of any E_∞-chiral algebra, which has finite-dimensional cohomology concentrated in degrees $\{0, 1, 2\}$ (Theorem H).

(ii) (Poincaré series – fundamental representation.) *On the fundamental representation $V = \mathbb{C}^2$, the ordered and symmetric Poincaré series are*

$$P_V^{\mathrm{ord}}(t) = \frac{1}{1-2t}, \qquad P_V^{\Sigma}(t) = \frac{1}{(1-t)^2},$$

and the kernel generating function is

$$\Delta P_V(t) \;=\; \frac{1}{1-2t} - \frac{1}{(1-t)^2} \;=\; \frac{t^2}{(1-2t)(1-t)^2},$$

with n-th coefficient $\dim\ker(\mathrm{av}_n) = 2^n - (n+1)$ $(= 0, 0, 1, 4, 11, 26, 57, 120, 247, 502, \ldots)$. *At degree 2:* $\dim\ker(\mathrm{av}_2) = 1$, *the antisymmetric R-matrix block* $\bigwedge^2 V$ (Example 2.3). *At degree 3:* $\dim\ker(\mathrm{av}_3) = 4$, *containing the Drinfeld associator class* $\Lambda^3(\mathfrak{g}) = \mathbb{C}$ *and three mixed-symmetry Schur–Weyl dimensions.*

(iii) (Poincaré series – adjoint representation.) *On the adjoint representation* $\mathfrak{g} = \mathfrak{sl}_2$ $(\dim\mathfrak{g} = 3)$, *the kernel generating function is*

$$\Delta P_{\mathfrak{g}}(t) \;=\; \frac{1}{1-3t} - \frac{1}{(1-t)^3} \;=\; \frac{t^2(3-t)}{(1-3t)(1-t)^3},$$

with n-th coefficient $\dim\ker(\mathrm{av}_n) = 3^n - \binom{n+2}{2}$ $(= 0, 0, 3, 17, 66, 222, 701, \ldots)$. *For a representation of dimension d, the general formula is*

$$\Delta P_d(t) \;=\; \frac{1}{1-dt} - \frac{1}{(1-t)^d}, \qquad \dim\ker(\mathrm{av}_n) = d^n - \binom{n+d-1}{n}.$$

The ordered data grows exponentially (d^n) while the symmetric shadow grows polynomially ($O(n^{d-1})$); at large degree, the kernel carries asymptotically all of the ordered content. At each degree n, the PB_n-monodromy of the Knizhnik–Zamolodchikov system furnishes $\ker(\mathrm{av}_n)$ *with a faithful representation of the pure braid group.*

(iv) (E_2-algebra structure.) *The ordered derived chiral centre carries an E_2-algebra structure via the classical Deligne conjecture applied to the ordered bar coalgebra $B^{\mathrm{ord}}(Y^{\mathrm{ch}})$. It does* not *carry an E_3-structure: the $\mathcal{D}$-module underlying Y^{ch} does not descend to the symmetric power $X^{(n)}$, so the symmetric bar $\overline{B}^{\Sigma}$ does not exist, and the higher Deligne conjecture (which produces E_3 from the E_2-coalgebra $\overline{B}^{\Sigma}$) is inapplicable. The E_2-structure is the maximal algebraic structure visible to the ordered theory.*

(v) (Gerstenhaber bracket recovers $\mathfrak{g}$.) *The degree-1 component of the ordered derived chiral centre, equipped with the E_2 Gerstenhaber bracket, satisfies*

$$\mathrm{HH}^1_{\mathrm{ord}}\big(Y\ (\mathfrak{sl}_2)^{\mathrm{ch}}\big) \;\cong\; \mathfrak{sl}_2$$

as a Lie algebra. The bracket arises from the classical r-matrix $r(z) = \ \ \Omega/z$ (equation (2.1); on $V \otimes V$, $\Omega = P - \frac{1}{2}I$): at first order in , the collision residue of $r(z)$ recovers the Lie bracket $[X, Y]$.

(vi) (Symmetric comparison.) *The averaging map* $\mathrm{av}\colon Z_{\mathrm{ch}}^{\mathrm{der,ord}} \to Z_{\mathrm{ch}}^{\mathrm{der},\Sigma}$ *is surjective. Its kernel*

$$\ker(\mathrm{av}) \;=\; \bigoplus_{n\ge 2} \ker(\mathrm{av}_n)$$

carries the full pure braid group monodromy data of the KZ system; this is the genuinely E_1 content of the ordered theory, invisible to every symmetric chiral-homological observable.

Remark 2.5 (Categorical Drinfeld centre and the passage to E_3). The conjecture above describes the *algebraic* ordered derived centre, which is an E_2-algebra. To recover an E_3-structure, one must pass through the categorical Drinfeld centre: replacing the chiral Yangian $Y\ (\mathfrak{g})^{\mathrm{ch}}$ by its *category of modules* $Y\ (\mathfrak{g})^{\mathrm{ch}}$-mod and computing the Drinfeld centre $\mathcal{Z}(Y(\mathfrak{g})^{\mathrm{ch}}\text{-mod})$ produces a braided monoidal category whose deformation-class is an E_3-algebra. This is the content of the categorified averaging step in the E_n operadic circle ($\mathsf{E}_1 \to \mathsf{E}_2 \to \mathsf{E}_3$): the Drinfeld centre categorifies the Reynolds averaging map, and the passage from E_2 to E_3 requires working at the categorical rather than algebraic level. Concretely, the Drinfeld centre $\mathcal{Z}(Y(\mathfrak{sl}_2)^{\mathrm{ch}}\text{-mod})$ should recover the quantum group category $\mathrm{Rep}_q(\mathfrak{sl}_2)$ at $q = e^{2\pi i/(k+2)}$. This identification is the subject of the DK-5 programme; see [34, §6] for the current status in type A.

The computations in Example 2.3 and Conjecture 2.4(v) concern the kernel of the averaging map on the ordered chiral centre, a cohomological object. The following proposition establishes the underlying representation-theoretic input: the dimension of $\ker(\mathrm{av}_n)$ on the bare tensor power $V^{\otimes n}$, before passing to bar-complex cohomology.

Proposition 2.6 (Kernel of the Reynolds projector: general simple Lie algebras). *Let $\mathfrak{g}$ be any simple Lie algebra over $\mathbb{C}$, let V be any finite-dimensional representation of $\mathfrak{g}$ with $d = \dim V$, and let $\mathrm{av}_n : V^{\otimes n} \to (V^{\otimes n})_{\Sigma_n}$ denote the Reynolds averaging map (projection onto the Σ_n-coinvariant quotient). Then*

$$\dim \ker(\mathrm{av}_n\big|_{V^{\otimes n}}) \;=\; d^n - \binom{n+d-1}{d-1}.$$

The formula depends only on $d = \dim V$, not on $\mathfrak{g}$ or on the $\mathfrak{g}$-module structure of V.

Special cases. For $\mathfrak{g} = \mathfrak{sl}_2$ with the fundamental representation $V = \mathbb{C}^2$: $\dim \ker(\mathrm{av}_n) = 2^n - (n+1)$. *For any $\mathfrak{g}$ with d-dimensional representation V, at $n = 2$:* $\dim \ker(\mathrm{av}_2) = d^2 - d(d+1)/2 = d(d-1)/2 = \dim \bigwedge^2 V$.

Proof. The Reynolds projector $\mathrm{av}_n = \frac{1}{n!}\sum_{\sigma\in\Sigma_n}\sigma$ acts on $V^{\otimes n}$ by permuting tensor factors. Its image is the Σ_n-invariant subspace $\mathrm{Sym}^n(V)$, regardless of any Lie algebra structure on V. The dimension $\dim \mathrm{Sym}^n(V) = \binom{n+d-1}{d-1}$ (the number of degree-n monomials in d variables) depends only on $d = \dim V$, as the stars-and-bars argument uses no algebraic structure beyond the vector space. Since $\dim V^{\otimes n} = d^n$, the kernel has dimension $d^n - \binom{n+d-1}{d-1}$.

When $\mathfrak{g} = \mathfrak{gl}_r$ with $V = \mathbb{C}^r$, Schur–Weyl duality refines this to an isotypic decomposition. The tensor power decomposes into $(\Sigma_n, \mathfrak{gl}_r)$-bimodules:

$$V^{\otimes n} \;\cong\; \bigoplus_{\lambda \vdash n,\; \ell(\lambda)\le r} S_\lambda \otimes V_\lambda(\mathfrak{gl}_r),$$

where λ runs over partitions of n with at most r rows, S_λ is the corresponding Specht module of Σ_n, and $V_\lambda(\mathfrak{gl}_r)$ is the irreducible $\mathfrak{gl}_r$-representation of highest weight λ. The projector av_n selects the trivial isotypic component $S_{(n)} = \mathbb{C}$ and annihilates every S_λ with $\lambda \neq (n)$. The image is

$$\mathrm{im}(\mathrm{av}_n) \;\cong\; S_{(n)} \otimes V_{(n)}(\mathfrak{gl}_r) \;=\; \mathrm{Sym}^n(\mathbb{C}^r),$$

and the kernel decomposes by isotypic components:

$$\ker(\mathrm{av}_n) \;\cong\; \bigoplus_{\substack{\lambda\vdash n\\ \ell(\lambda)\le r,\; \lambda\neq(n)}} S_\lambda \otimes V_\lambda(\mathfrak{gl}_r).$$

For a general simple $\mathfrak{g}$ acting on V, the Schur–Weyl decomposition involves the commutant algebra $\mathrm{End}_{\mathfrak{g}}(V^{\otimes n})$ rather than the group algebra of $\mathrm{GL}(V)$, and the individual summands carry different $\mathfrak{g}$-representations with different dimensions. The total kernel dimension, however, depends only on $\dim \mathrm{Sym}^n(V) = \binom{n+d-1}{d-1}$, which is determined by d alone. □

Remark 2.7 (Explicit values and Conjecture 2.4). Table 1 records the kernel dimensions at low degrees for the fundamental representations of $\mathfrak{sl}_2$ through $\mathfrak{sl}_5$ (of dimensions $d = 2, 3, 4, 5$), together with the fundamental representations of $\mathfrak{so}_5$ ($d = 5$) and G_2 ($d = 7$). Since the formula depends only on

$d = \dim V$, the rows for $\mathfrak{sl}_5$ and $\mathfrak{so}_5$ are identical. These are representation-theoretic upper bounds on the dimensions of the corresponding components of the ordered chiral centre in Conjecture 2.4(v): $\ker(\mathrm{av}_n)$ on the ordered chiral centre is a subquotient of $\ker(\mathrm{av}_n|_{V^{\otimes n}})$, obtained by passing to bar-complex cohomology (only the cocycles in the non-trivial Σ_n-isotypics survive). At degree 2 for $\mathfrak{sl}_2$: the full 1-dimensional kernel $\bigwedge^2 V = \mathbb{C}$ survives (Example 2.3, equation (2.5)). At degree $n \geq 3$, the kernel on the chiral centre is in general strictly smaller than the representation-theoretic bound $d^n - \binom{n+d-1}{d-1}$.

	$n=2$	$n=3$	$n=4$	$n=5$	$n=6$
$\mathfrak{sl}_2$ (d=2)	1	4	11	26	57
$\mathfrak{sl}_3$ (d=3)	3	17	66	222	701
$\mathfrak{sl}_4$ (d=4)	6	44	221	968	4,012
$\mathfrak{sl}_5$, $\mathfrak{so}_5$ (d=5)	10	90	555	2,999	15,415
G_2 *fund.* (d=7)	21	259	2,191	16,345	116,725
Formula			$d^n - \binom{n+d-1}{d-1}$		

TABLE 1. Dimension of $\ker(\mathrm{av}_n|_{V^{\otimes n}})$ for a d-dimensional representation V at degrees $n = 2, \dots, 6$ (Proposition 2.6). The formula depends only on $d = \dim V$.

3. Background: the $\mathsf{E}_1/\mathsf{E}_\infty$ hierarchy

We summarise the structural framework; detailed treatments appear in [34, §I–II].

Definition 3.1 (Chirality hierarchy)**.** An E_∞*-chiral algebra* has symmetric OPE; its factorisation product descends to $X^{(n)} = X^n/\Sigma_n$. An E_1*-chiral algebra* has braided OPE: the vertex operators satisfy S-locality

$$(z-w)^N Y(a,z)Y(b,w) = (z-w)^N Y(b,w)Y(a,z)\cdot S(z-w)$$

where $S(z)$ is a vertex R-matrix satisfying the QYBE, unitarity, the shift condition, and the hexagon axiom [18]. The factorisation product lives on ordered configurations X^n, not on $X^{(n)}$.

Warning 3.2 (Multiple notions of E_1-chiral)**.** Five notions coexist: (A) strict ChirAss-algebra, (B) A_∞ in the chiral endomorphism operad, (C) EK quantum vertex algebra, (D) double A_∞, and (E) factorisation $\mathcal{D}$-module on $\mathrm{Ran}^{\mathrm{ord}}(X)$. On the Koszul locus, (B) and (C) determine each other via a Drinfeld associator; this paper uses (C) algebraically and (E) geometrically. See [34, Warning 2.21].

Definition 3.3 (Koszul locus)**.** An E_1-chiral algebra $\mathcal{A}$ lies on the *Koszul locus* if the bar-cobar counit is a quasi-isomorphism: $\Omega(\overline{B}^{\mathrm{ord}}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$. Standard vertex algebras at generic level and Yangians $Y(\mathfrak{g})$ lie on the Koszul locus.

Definition 3.4 (Ordered and symmetric Ran spaces)**.** The *ordered Ran space* $\mathrm{Ran}^{\mathrm{ord}}(X) = \coprod_{n\geq 1} U_n(X)$, $U_n(X) = X^n \setminus \bigcup_{i<j}\Delta_{ij}$, parametrises ordered tuples of distinct points, with concatenation maps $\mathrm{union}_{k,n-k}\colon \mathrm{Ran}^{\mathrm{ord}}(X)_k \times_{\mathrm{disj}} \mathrm{Ran}^{\mathrm{ord}}(X)_{n-k} \to \mathrm{Ran}^{\mathrm{ord}}(X)_n$. The Beilinson–Drinfeld Ran space is the coinvariant quotient $\mathrm{Ran}(X) = \mathrm{Ran}^{\mathrm{ord}}(X)/\Sigma_\bullet$.

Warning 3.5 (Bar-cobar gives E_1; the curve gives E_2)**.** The abstract bar-cobar adjunction produces an E_1-coalgebra with the deconcatenation coproduct. The symmetric bar $\overline{B}^{\Sigma}(\mathcal{A})$ of a chiral algebra on a curve carries the coshuffle coproduct, which is $\mathsf{E}_2/\mathsf{E}_\infty$: the Σ_n-symmetry of unordered configuration spaces $\mathrm{Conf}_k(X)$ provides the $\mathrm{FM}_2 \simeq \mathsf{E}_2$ operad. The coinvariant quotient $\overline{B}^{\mathrm{ord}} \to \overline{B}^{\Sigma}$ introduces the curve geometry. Without the curve, bar-cobar gives E_1 and the classical Deligne conjecture gives E_2 on the centre. Only affine topologization with conformal vector raises BRST cohomology to $\mathsf{E}_3^{\mathrm{top}}$.

Definition 3.6 (Shadow classification $G/L/C/M$)**.** The shadow classification [34, §II] partitions E_∞-chiral algebras by the pole structure of $r(z)$: G (Heisenberg, pole ≤ 1, depth r=2), L (affine KM, pole 1 non-abelian, r=3), C ($\beta\gamma/bc$, nilpotent r-matrix, r=4), M (Virasoro/$\mathcal{W}_N$, pole ≥ 2, r=∞). The shadow depth is the smallest r with $\ell_k = 0$ for $k \geq r{+}1$ in the symmetric convolution algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$.

Theorem 3.7 (Bridge; [34, Theorem 1.1(ii)])**.** *The chiral bar-cobar adjunction on a complex curve X is the holomorphic refinement of n=2 E_n Koszul duality: $\overline{B}^{\Sigma}(\mathcal{A})$ is an E_2-coalgebra from the coshuffle coproduct on $\mathrm{Conf}_k(X) \cong \mathrm{Conf}_k(\mathbb{R}^2)$. The E_2 structure comes from the curve, not from bar-cobar.*

Remark 3.8 (The bridge at the level of operads)*.* The real FM compactification $\overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{R}) \cong K_n$ (associahedron; Sinha [42]). The inclusion $\mathbb{R} \hookrightarrow \mathbb{C}$ gives

$$K_n = \overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{R}) \hookrightarrow \overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{C}). \tag{3.1}$$

An E_1-chiral algebra structure is an operadic map $A_\infty \to \mathrm{End}^{\mathrm{ch,ord}}_{\mathcal{A}}$ factoring through (3.1); K_n is real-codimension n in $\overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{C})$.

4. $\mathcal{D}$-modules on ordered configuration spaces

The integration theory for E_∞-chiral algebras lives on symmetric products $X^{(n)} = X^n/\Sigma_n$, but the E_1 theory requires ordered configurations X^n. What are the $\mathcal{D}$-modules on X^n that encode the OPE, and what happens to them at the diagonals? The answer is the ordered factorisation $\mathcal{D}$-module: a holonomic $\mathcal{D}$-module on X^n whose singularities along the diagonals are precisely the OPE poles.

4.1. Ordered configuration spaces and their compactifications.

The configuration space $U_n(X) = X^n \setminus \bigcup \Delta_{ij}$ and its Fulton–MacPherson compactification are the geometric substrate for ordered chiral homology. The compactification resolves the diagonals by blowup, producing boundary strata indexed by ordered collision patterns: rooted planar trees with n labelled leaves.

Definition 4.1 (Ordered configuration space)**.** Let X be a smooth algebraic curve over $\mathbb{C}$. For $n \geq 0$, the *ordered configuration space* is the smooth quasi-projective variety

$$U_n = U_n(X) := X^n \setminus \bigcup_{1\leq i<j\leq n} \Delta_{ij}$$

where $\Delta_{ij} = \{(x_1, \dots, x_n) \in X^n : x_i = x_j\}$ is the (i,j)-diagonal. For $n = 0$, $U_0 = \{\varnothing\} = \mathrm{Spec}\ \mathbb{C}$. For $n = 1$, $U_1 = X$.

Definition 4.2 (Ordered FM compactification)**.** The *ordered Fulton–MacPherson compactification* $\overline{\mathrm{FM}}_n^{\mathrm{ord}}(X)$ is the real oriented blowup of X^n along all diagonals Δ_{ij}: a smooth manifold with corners whose boundary strata are indexed by ordered collision patterns (rooted planar trees with n labelled leaves). Each boundary stratum ∂_T factors as a product of local FM spaces on tangent spaces times a residual FM space:

$$\partial_T \overline{\mathrm{FM}}_n^{\mathrm{ord}}(X) \cong \textstyle\prod_{i=1}^r \overline{\mathrm{FM}}_{k_i}^{\mathrm{ord}}(T_{x_i}X) \times \overline{\mathrm{FM}}_{n'}^{\mathrm{ord}}(X).$$

Standard references: Kashiwara [27], Hotta–Takeuchi–Tanisaki [23].

4.2. **The ordered factorisation $\mathcal{D}$-module.** Given the configuration spaces and their compactifications, one extends the external product $\mathcal{A}^{\boxtimes n}$ meromorphically across the diagonals. The resulting $*$-extension $j_* j^*(\mathcal{A}^{\boxtimes n})$ is the ordered factorisation $\mathcal{D}$-module: a holonomic $\mathcal{D}$-module on X^n whose singularity type along each diagonal Δ_{ij} is dictated by the OPE pole order of $\mathcal{A}$.

Construction 4.3 (Ordered factorisation $\mathcal{D}$-module)**.** Let $\mathcal{A}$ be a chiral algebra on X. The *external product* $\mathcal{A}^{\boxtimes n}$ is the $\mathcal{D}$-module on X^n defined by

$$\mathcal{A}^{\boxtimes n} := \mathcal{A} \boxtimes \mathcal{A} \boxtimes \cdots \boxtimes \mathcal{A} = \mathrm{pr}_1^* \mathcal{A} \otimes_{\mathcal{O}_{X^n}} \mathrm{pr}_2^* \mathcal{A} \otimes_{\mathcal{O}_{X^n}} \cdots \otimes_{\mathcal{O}_{X^n}} \mathrm{pr}_n^* \mathcal{A}$$

where $\mathrm{pr}_i \colon X^n \to X$ is the i-th projection.

The *ordered factorisation $\mathcal{D}$-module* at degree n is the $*$-extension:

$$\mathcal{F}_n^{\mathrm{ord}}(\mathcal{A}) := j_* j^*(\mathcal{A}^{\boxtimes n}) \in \mathcal{D}\text{-mod}(X^n).$$

Concretely, sections of $\mathcal{F}_n^{\mathrm{ord}}$ over an open $V \subset X^n$ are sections of $\mathcal{A}^{\boxtimes n}$ over $V \cap U_n$ that extend meromorphically across the diagonals. The poles along Δ_{ij} are the OPE poles: if the OPE $Y(a,z)b = \sum_{k \geq -N} a_{(k)} b\, z^{-k-1}$ has poles of order at most N for all $a, b \in \mathcal{A}$, then sections of $\mathcal{F}_n^{\mathrm{ord}}$ have poles of order at most N along each Δ_{ij}.

Definition 4.4 (Ordered factorisation $\mathcal{D}$-module on $\mathrm{Ran}^{\mathrm{ord}}(X)$)**.** For a chiral algebra $\mathcal{A}$ on X, the *ordered factorisation $\mathcal{D}$-module* is the $\mathcal{D}$-module on $\mathrm{Ran}^{\mathrm{ord}}(X)$ (Definition 3.4) whose restriction to the n-th stratum $U_n(X)$ is

$$\mathcal{F}^{\mathrm{ord}}(\mathcal{A})\big|_{U_n} = \mathcal{A}^{\boxtimes n}\big|_{U_n}$$

with the factorisation product: for the merging map $\mathrm{union}_{k,n-k}$, the factorisation isomorphism

$$\mathrm{union}_{k,n-k}^* \mathcal{F}^{\mathrm{ord}}\big|_{U_n} \;\simeq\; \mathcal{F}^{\mathrm{ord}}\big|_{U_k} \boxtimes \mathcal{F}^{\mathrm{ord}}\big|_{U_{n-k}}$$

is the OPE (the meromorphic extension across the diagonals). The KZ connection endows $\mathcal{F}^{\mathrm{ord}}$ with a flat connection on each stratum, compatible with the factorisation maps. The degree-by-degree $\mathcal{D}$-modules $\mathcal{F}_n^{\mathrm{ord}}(\mathcal{A})$ of Construction 4.3 are the restrictions of this single object to the strata of $\mathrm{Ran}^{\mathrm{ord}}(X)$.

Proposition 4.5 (Properties of the ordered factorisation $\mathcal{D}$-module)**.** *Let $\mathcal{A}$ be a chiral algebra on a smooth proper curve X.*

(i) (Holonomicity and singularity type.) $\mathcal{F}_n^{\mathrm{ord}}(\mathcal{A})$ *is a holonomic $\mathcal{D}$-module on X^n. The singularity type along $\bigcup \Delta_{ij}$ depends on the pole order of the r-matrix:*

- Class G (Heisenberg): $r(z) = k/z$, *pole order* 1. *Regular singularities.*
- Class L (affine KM): $r(z) = k\Omega/z$ (*trace-form*), *pole order* 1. *Regular singularities.*
- Class C ($\beta\gamma$): *r is a constant nilpotent endomorphism* (*d-log absorbs the simple OPE pole*)*. Regular singularities.*
- Class M (Virasoro, $\mathcal{W}_N$): $r^{\mathrm{Vir}}(z) = (c/2)/z^3 + 2T/z$, *pole order* 3. Irregular *singularities of Poincaré rank* 2.

(ii) (The KZ connection.) *The flat connection on $\mathcal{F}_n^{\mathrm{ord}}(\mathcal{A})$ restricts on U_n to the standard product connection on $\mathcal{A}^{\boxtimes n}|_{U_n}$ plus the Knizhnik–Zamolodchikov correction:*

$$\nabla_n = \nabla_{\mathcal{A}}^{\boxtimes n} - \sum_{i<j} r_{ij}(z_i - z_j)\, dz_{ij}$$

where $r_{ij}(z) \in \mathrm{End}(\mathcal{A} \otimes \mathcal{A})((z))$ is the classical r-matrix (degree-2 projection of $\Theta_{\mathcal{A}}$) acting on the (i,j) tensor factors, and $dz_{ij} = d(z_i - z_j)$ is the connection 1-form.

For affine Kac–Moody at level k, two equivalent conventions coexist. In the trace-form convention, with Ω_{tr} *the Casimir built from the normalised trace form* $(X, Y)_{\mathrm{tr}} = \mathrm{tr}(\mathrm{ad}_X \circ \mathrm{ad}_Y)/(2h^\vee)$*:*

$$\nabla_n^{\mathrm{KM}} = \nabla^{\boxtimes n} - \sum_{i<j} \frac{k\,(\Omega_{\mathrm{tr}})_{ij}}{z_i - z_j}\, d(z_i - z_j). \tag{4.1}$$

In the KZ normalisation, with Ω *the inverse Killing form Casimir:*

$$\nabla_n^{\mathrm{KM}} = \nabla^{\boxtimes n} - \sum_{i<j} \frac{\Omega_{ij}}{(k + h^\vee)(z_i - z_j)}\, d(z_i - z_j).$$

The bridge identity $k\,(\Omega_{\mathrm{tr}})_{ij} = \Omega_{ij}/(k + h^\vee)$ *equates the two at generic k (see* [34, §III.2]*). Both have a* simple pole *along* Δ_{ij}*: the connection is Fuchsian.*

(iii) (Flatness.) $\nabla_n^2 = 0$ *on* U_n*, following from the classical Yang–Baxter equation for* $r(z)$*.*

(iv) (Equivariant descent.) *For* E_∞*-chiral* $\mathcal{A}$ *(where* $S(z) = \mathrm{id}$ *up to Koszul signs):* $\mathcal{F}_n^{\mathrm{ord}}$ *is* Σ_n*-equivariant, and* $(\mathcal{F}_n^{\mathrm{ord}})^{\Sigma_n} \simeq \mathcal{F}_n^{\Sigma}$ *is the Beilinson–Drinfeld factorisation* $\mathcal{D}$*-module on* $X^{(n)}$*.*

(v) (Non-equivariance for E_1.) *For* E_1*-chiral* $\mathcal{A}$*:* $\mathcal{F}_n^{\mathrm{ord}}$ *is* not Σ_n*-equivariant. The braiding* $S(z)$ *on the* (i, j) *factor introduces monodromy as the i-th and j-th points circle each other.*

Proof. (i) Holonomicity from Kashiwara's theorem [27]. Singularity type from pole structure: class G/L (simple pole) regular; class C (nilpotent after d-log) regular; class M (cubic pole) irregular, Poincaré rank 2. Finite-dim de Rham regardless (Mebkhout [40]). (ii) KZ connection from $\Theta_{\mathcal{A}}|_{\deg 2}$; the 1-form is $r_{ij}\,dz_{ij}$, distinct from the Arnold forms $\omega_{ij} = d\log(z_i - z_j)$. (iii) Flatness from the CYBE. (iv) E_∞: $S = \mathrm{id}$, monodromy trivial, descent to $X^{(n)}$. (v) E_1: $S \neq \mathrm{id}$ (Drinfeld–Kohno), no descent. □

Warning 4.6 (Regularity dichotomy)**.** Class $G/L/C$: regular singularities (Fuchsian; Deligne Riemann–Hilbert applies; 3d gauge theories). Class M: irregular singularities (Poincaré rank ≥ 2; D'Agnolo–Kashiwara [16] required; 3d gravity [13]; Stokes data). The regularity type is the r-matrix pole order: ≤ 1 for $G/L/C$, ≥ 2 for M.

5. Ordered chiral homology

The ordered factorisation $\mathcal{D}$-module $\mathcal{F}^{\mathrm{ord}}(\mathcal{A})$ is a holonomic $\mathcal{D}$-module with flat connection. Its de Rham cohomology is a finite-dimensional graded vector space at each degree. Assembling these across degrees, with the chiral differential from FM boundary strata, produces the ordered chiral homology of $\mathcal{A}$ over X.

5.1. Definition.

The de Rham cohomology of a holonomic $\mathcal{D}$-module on a proper variety is finite-dimensional (Mebkhout). Applied to $\mathcal{F}^{\mathrm{ord}}(\mathcal{A})$ on $\mathrm{Ran}^{\mathrm{ord}}(X)$, this gives a graded vector space at each degree; the cross-degree chiral differential assembles these into a chain complex.

Definition 5.1 (Ordered chiral homology via $\mathrm{Ran}^{\mathrm{ord}}(X)$)**.** Let $\mathcal{A}$ be a chiral algebra on a smooth proper curve X, and let $\mathcal{F}^{\mathrm{ord}}(\mathcal{A})$ be the ordered factorisation $\mathcal{D}$-module on $\mathrm{Ran}^{\mathrm{ord}}(X)$ (Definition 4.4). The *ordered chiral homology* is the derived pushforward to the point:

$$\int_X^{\mathrm{ord}} \mathcal{A} \;:=\; R\Gamma_{\mathrm{dR}}\big(\mathrm{Ran}^{\mathrm{ord}}(X),\ \mathcal{F}^{\mathrm{ord}}(\mathcal{A})\big) \;=\; \pi_*\mathcal{F}^{\mathrm{ord}}(\mathcal{A})$$

where $\pi\colon \mathrm{Ran}^{\mathrm{ord}}(X) \to \mathrm{pt}$ is the projection. The degree-by-degree decomposition

$$\int_X^{\mathrm{ord}} \mathcal{A} \;=\; \Big(\bigoplus_{n\geq 0} R\Gamma_{\mathrm{dR}}\big(X^n,\ \mathcal{F}_n^{\mathrm{ord}}(\mathcal{A})\big),\ \ d_{\mathrm{ch}}\Big)$$

is the computation of this pushforward stratum by stratum, where $\mathrm{dR}(\mathcal{F}) = \mathcal{F} \otimes_{\mathcal{D}} \Omega^\bullet_{X^n}$ is the de Rham complex of the $\mathcal{D}$-module $\mathcal{F}$ and d_{ch} is the *chiral differential*: the alternating sum of the restriction maps to the codimension-1 boundary strata of $\overline{\mathrm{FM}}_n^{\mathrm{ord}}(X)$, corresponding to pairwise collisions of adjacent points. At each degree, the *ordered chiral homology at degree n* is

$$\int_{X,n}^{\mathrm{ord}} \mathcal{A} \;:=\; R\Gamma_{\mathrm{dR}}\big(X^n,\ \mathcal{F}_n^{\mathrm{ord}}(\mathcal{A})\big).$$

5.2. **Symmetric descent.** The quotient map $q\colon \mathrm{Ran}^{\mathrm{ord}}(X) \to \mathrm{Ran}(X)$ (Definition 3.4) induces the natural map from ordered to symmetric chiral homology.

Proposition 5.2 (Symmetric descent recovers Beilinson–Drinfeld). *Let $\mathcal{A}$ be an E_∞-chiral algebra on a smooth proper curve X. The quotient $q\colon \mathrm{Ran}^{\mathrm{ord}}(X) \to \mathrm{Ran}(X)$ induces the averaging map*

$$\int_X^{\mathrm{ord}} \mathcal{A} = R\Gamma\big(\mathrm{Ran}^{\mathrm{ord}}(X),\ \mathcal{F}^{\mathrm{ord}}\big) \xrightarrow{\ \mathrm{av}\ } R\Gamma\big(\mathrm{Ran}(X),\ \mathcal{F}^{\Sigma}\big) = \int_X \mathcal{A},$$

which is a quasi-isomorphism. At each degree, the Σ_n-coinvariants of the ordered chiral homology recover the Beilinson–Drinfeld chiral homology:

$$\Big(\int_{X,n}^{\mathrm{ord}} \mathcal{A}\Big)_{\Sigma_n} \;\simeq\; R\Gamma_{\mathrm{dR}}\big(X^{(n)},\ \mathcal{F}_n^{\Sigma}(\mathcal{A})\big) \;=:\; \int_{X,n} \mathcal{A}.$$

Proof. For E_∞-chiral $\mathcal{A}$, $\mathcal{F}_n^{\mathrm{ord}}$ is Σ_n-equivariant (Proposition 4.5(iv)). The equivariant descent for $\mathcal{D}$-modules works through the quotient stack $[X^n/\Sigma_n]$: the category of Σ_n-equivariant $\mathcal{D}$-modules on X^n is equivalent to the category of $\mathcal{D}$-modules on $[X^n/\Sigma_n]$. Since Σ_n is a finite group acting in characteristic zero, Maschke's theorem ensures that representations of Σ_n are semisimple; the quotient stack $[X^n/\Sigma_n]$ and the coarse moduli space $X^{(n)} = X^n/\Sigma_n$ have equivalent $\mathcal{D}$-module categories. The quotient $\pi\colon X^n \to X^{(n)}$ is ramified along the diagonals, but equivariant descent is not affected by ramification in characteristic zero.

The de Rham functor commutes with Σ_n-coinvariants (which are exact for the symmetric group acting on vector spaces over $\mathbb{C}$ of characteristic zero), giving: $R\Gamma_{\mathrm{dR}}(X^n, \mathcal{F}_n^{\mathrm{ord}})_{\Sigma_n} \simeq R\Gamma_{\mathrm{dR}}(X^{(n)}, (\mathcal{F}_n^{\mathrm{ord}})^{\Sigma_n}) \simeq R\Gamma_{\mathrm{dR}}(X^{(n)}, \mathcal{F}_n^{\Sigma})$. The right side is the Beilinson–Drinfeld chiral homology at degree n.

The cross-degree chiral differential d_{ch} is Σ-equivariant for E_∞-chiral algebras because the factorisation maps (collision/OPE) are symmetric: the OPE $Y(a,z)b$ and $Y(b,z)a$ agree up to signs. This ensures that the total complex, not just the fixed-degree pieces, descends to the symmetric quotient. □

Proposition 5.3 (Lossy descent for E_1-chiral algebras). *For an E_1-chiral algebra $\mathcal{A}$, the natural map*

$$\mathrm{av}_n\colon \int_{X,n}^{\mathrm{ord}} \mathcal{A} \longrightarrow \Big(\int_{X,n}^{\mathrm{ord}} \mathcal{A}\Big)_{\Sigma_n}$$

is surjective but not injective for $n \geq 2$. The kernel $\ker(\mathrm{av}_n)$ *carries ordered data invisible to the Beilinson–Drinfeld theory:*

(i) *At $n = 2$: the full spectral r-matrix $r(z)$ versus the scalar $\kappa(\mathcal{A})$ (for abelian algebras, $\kappa = \mathrm{av}(r(z))$; for non-abelian KM, $\kappa = \mathrm{av}(r(z)) + \dim(\mathfrak{g})/2$, the Sugawara shift).*

(ii) *At $n = 3$: the Drinfeld associator Φ appears in* $\ker(\mathrm{av}_3)$. *The space of sections of* av_3 *compatible with the Maurer–Cartan equation is a* GRT_1*-torsor.*

(iii) *At $n \geq 4$: the kernel carries higher-degree ordered shadow data from the E_1 obstruction tower.*

Proof. By [34, the non-splitting proposition], the short exact sequence $0 \to \ker(\mathrm{av}) \to \mathfrak{g}^{\mathsf{E}_1}_{\mathcal{A}} \to \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}} \to 0$ does not split as dg Lie algebras. At $n = 2$: for a genuinely E_1-chiral algebra, $R(z) \neq \mathrm{id}$ and the Σ_2-action on the ordered de Rham cohomology is twisted by R; the antisymmetric eigenspace of this twisted action gives $\ker(\mathrm{av}_2) \neq 0$ (see Example 2.3 for the Yangian case). At $n \geq 3$: the cross-degree obstruction (the bar differential is Σ_{n-2}-equivariant but not Σ_n-equivariant) forces non-trivial elements of $\ker(\mathrm{av}_n)$. The GRT_1-torsor structure at $n = 3$ follows from Drinfeld's theorem on associators combined with the Etingof–Kazhdan quantisation theorem. □

5.3. **Genus $g \geq 1$ curvature.** At genus zero the bar differential squares to zero and ordered chiral homology is an ordinary chain complex. At genus $g \geq 1$, this fails: the fiberwise bar differential acquires curvature proportional to the Koszul invariant $\kappa(\mathcal{A})$ (the scalar $\kappa(\mathcal{A}) = \mathrm{av}(r(z))$ in the abelian and scalar families, while non-abelian affine Kac–Moody has $\kappa(V_k(\mathfrak{g})) = \mathrm{av}(r(z)) + \dim(\mathfrak{g})/2$, the degree-2 average of the classical r-matrix; computed explicitly in §9) times the Hodge class $\omega_g = c_1(\lambda)$ on $\overline{\mathcal{M}}_g$. The curvature is a *base* class (on moduli), not a *fiber* form (on the curve).

Remark 5.4 (Curvature at higher genus). At genus $g \geq 1$, the fiberwise bar differential acquires curvature:

$$d^2_{\mathrm{fib}} = \kappa(\mathcal{A}) \cdot \omega_g \qquad \text{(UNIFORM-WEIGHT)} \tag{5.1}$$

where $\kappa(\mathcal{A})$ is the Koszul invariant of $\mathcal{A}$ and $\omega_g = c_1(\lambda)$ is the first Chern class of the Hodge line bundle on $\overline{\mathcal{M}}_g$; ω_g is a class on the *moduli space* $\overline{\mathcal{M}}_g$, not a form on the curve. The scalar form of the curvature holds in the uniform-weight sector; at multi-weight, a cross-channel correction $\delta d^2_{\mathrm{cross}}$ must be included.

Three points require emphasis.

(a) The $\mathcal{D}$-module connection at *fixed* degree n remains flat: $\nabla_n^2 = 0$ on U_n. The curvature (5.1) appears in the *cross-degree* chiral differential d_{ch}, which connects different degrees.
(b) The curvature $\kappa(\mathcal{A}) \cdot \omega_g$ is a *scalar* (it acts on all tensor factors uniformly), so it is Σ_n-invariant: it survives descent to the symmetric quotient unchanged.
(c) The period-corrected differential $D^{(g)}$ of [34, §II.3] restores $d^2 = 0$ at the total level by incorporating the Hodge correction.

6. Chiral quantum groups: R-matrices, A_∞-structures, and chiral coproducts

The classical theory of quantum groups rests on an equivalence between three descriptions of the same structure: a quasi-triangular Hopf algebra (H, Δ, R) can be specified by its R-matrix, by its coproduct, or by an operadic A_∞-algebra map. The goal of this section is to establish the chiral analogue: an equivalence between three descriptions of an E_1-chiral algebra equipped with quantum-group-type structure. The primitive datum is the E_1 ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with its deconcatenation coproduct, coassociative over $(\mathrm{ChirAss})^!$; the R-matrix and the chiral coproduct are both recoverable from this datum.

6.1. **Three structures on an E_1-chiral algebra.** An E_1-chiral algebra admits three equivalent descriptions: by a vertex R-matrix $S(z)$ (braiding data), by chiral A_∞ structure maps m_k^{ch} (higher associativity data), and by a chiral coproduct Δ^{ch} (coalgebra data). Each description captures a different projection of the ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$.

Definition 6.1 (E_1-chiral algebra with vertex R-matrix)**.** An E_1-chiral algebra with vertex R-matrix is a pair $(\mathcal{A}, S(z))$ where $\mathcal{A}$ is a graded vertex algebra and $S(z) \in \mathrm{End}(\mathcal{A} \otimes \mathcal{A})[[z, z^{-1}]]$ is a vertex R-matrix satisfying:

(i) (S-locality.) For all $a, b \in \mathcal{A}$ and sufficiently large N:

$$(z-w)^N Y(a,z)\,Y(b,w) \;=\; (z-w)^N Y(b,w)\,Y(a,z)\cdot S(z-w). \tag{6.1}$$

(ii) (Quantum Yang–Baxter equation.) In $\mathrm{End}(\mathcal{A}^{\otimes 3})$:

$$S_{12}(z_1-z_2)\,S_{13}(z_1-z_3)\,S_{23}(z_2-z_3) \;=\; S_{23}(z_2-z_3)\,S_{13}(z_1-z_3)\,S_{12}(z_1-z_2). \tag{6.2}$$

(iii) (Unitarity.) $S_{12}(z)\,S_{21}(-z) = \mathrm{id}$.

(iv) (Shift condition.) The R-matrix intertwines the translation operator T: $[T \otimes \mathrm{id} + \mathrm{id} \otimes T,\ S(z)] = \partial_z S(z)$.

(v) (Hexagon axiom.) On $\overline{\mathrm{FM}}_3^{\mathrm{ord}}(\mathbb{C})$, the braiding and associativity satisfy the coherence: for the two standard paths in $\overline{\mathrm{FM}}_3^{\mathrm{ord}}(\mathbb{C})$ connecting the same boundary strata,

$$S_{12}(z_{12})\,\Phi_{123}^{-1}\,S_{13}(z_{13})\,\Phi_{132} \;=\; \Phi_{213}^{-1}\,S_{23}(z_{23})\,\Phi_{123} \tag{6.3}$$

where Φ_{ijk} is the Drinfeld associator evaluated on the three-point KZ connection.

These are the axioms of Etingof–Kazhdan [18] for a quantum vertex operator algebra.

Definition 6.2 (A_∞-algebra in the chiral endomorphism operad)**.** An A_∞-algebra structure on $\mathcal{A}$ in the chiral endomorphism operad $\mathrm{End}^{\mathrm{ch}}_{\mathcal{A}}$ is an operadic map

$$A_\infty \;\longrightarrow\; \mathrm{End}^{\mathrm{ch}}_{\mathcal{A}}$$

where $\mathrm{End}^{\mathrm{ch}}_{\mathcal{A}}(n) = \mathrm{Hom}(\mathcal{A}^{\otimes n},\ \mathcal{A}((\lambda_1))\cdots((\lambda_{n-1})))$ is the spectral substitution operad ([34, §3]). Concretely, the data consist of chiral structure maps

$$m_k^{\mathrm{ch}}\colon \mathcal{A}^{\otimes k} \longrightarrow \mathcal{A}((\lambda_1))\cdots((\lambda_{k-1})), \qquad k \geq 1,$$

satisfying the Stasheff A_∞-relations in the chiral endomorphism operad: for each $n \geq 1$,

$$\sum_{\substack{i+j=n+1\\ 1\leq p\leq i}} (-1)^{\epsilon}\, m_i^{\mathrm{ch}} \circ_p m_j^{\mathrm{ch}} \;=\; 0, \tag{6.4}$$

where $\circ_p$ is the partial composition via spectral substitution (λ-variable nesting), and ϵ carries the standard Koszul sign from the suspended degree $\|m_k\| = k-2$. The structure maps m_k^{ch} are meromorphic in the spectral parameters $\lambda_1, \ldots, \lambda_{k-1}$. At $k=2$, the map m_2^{ch} is the OPE vertex operation $Y(-,z)$. At $k=1$, $m_1^{\mathrm{ch}} = d$ is the differential. The operations m_k^{ch} for $k \geq 3$ record the obstruction to strict associativity of the chiral product.

Definition 6.3 (E_1-chiral algebra with compatible chiral coproduct)**.** An E_1-chiral algebra with compatible chiral coproduct is a triple $(\mathcal{A}, Y, \Delta^{\mathrm{ch}})$ where $(\mathcal{A}, Y)$ is a chiral algebra and

$$\Delta^{\mathrm{ch}}\colon \mathcal{A} \longrightarrow (\mathcal{A} \hat{\otimes} \mathcal{A})((z))$$

is a chiral coproduct (a map to the completed tensor product with spectral parameter) satisfying:

(i) (Coassociativity up to associator.) The two iterated coproducts agree up to conjugation by the Drinfeld associator Φ:

$$(\Delta^{\mathrm{ch}} \otimes \mathrm{id}) \circ \Delta^{\mathrm{ch}} \;=\; \Phi \cdot (\mathrm{id} \otimes \Delta^{\mathrm{ch}}) \circ \Delta^{\mathrm{ch}} \cdot \Phi^{-1}.$$

(ii) (OPE compatibility.) The coproduct intertwines the vertex operation: for $a, b \in \mathcal{A}$,

$$\Delta^{\mathrm{ch}}(Y(a,z)\,b) \;=\; Y^{(2)}(\Delta^{\mathrm{ch}}(a), z) \cdot \Delta^{\mathrm{ch}}(b) \tag{6.5}$$

where $Y^{(2)}$ denotes the tensorwise vertex operation on $\mathcal{A} \hat{\otimes} \mathcal{A}$.

(iii) (Counit.) There exists a counit $\varepsilon\colon \mathcal{A} \to \mathbb{C}$ with $(\varepsilon \otimes \mathrm{id}) \circ \Delta^{\mathrm{ch}} = \mathrm{id} = (\mathrm{id} \otimes \varepsilon) \circ \Delta^{\mathrm{ch}}$, recovering the augmentation of $\mathcal{A}$.

The chiral coproduct determines a vertex R-matrix by

$$S(z) \;:=\; \sigma \circ \Delta^{\mathrm{ch}} \circ (\Delta^{\mathrm{ch,op}})^{-1}$$

where σ is the graded flip and $\Delta^{\mathrm{ch,op}} = \sigma \circ \Delta^{\mathrm{ch}}$ is the opposite coproduct. This is the chiral analogue of the classical formula $R = \Delta^{\mathrm{op}}(\Delta^{-1}(-))$ for a quasi-triangular Hopf algebra.

Remark 6.4 (The role of the Drinfeld associator). The Drinfeld associator $\Phi \in \exp(\widehat{\mathrm{Lie}}(t_{12}, t_{23}))$ controls the failure of strict coassociativity: the two parenthesizations traverse different paths in $\overline{\mathrm{FM}}_3^{\mathrm{ord}}(\mathbb{C})$, and Φ is the KZ parallel transport between them. It also appears in $\ker(\mathrm{av}_3)$ of Proposition 5.3, invisible to the symmetric shadow. For E_∞-chiral algebras, Φ acts trivially on coinvariants; for genuinely E_1-chiral algebras, it is essential structural data.

6.2. **The equivalence theorem.** The three descriptions of the preceding subsection are not independent: a vertex R-matrix determines the chiral A_∞ operations via the bar complex, the A_∞ operations determine the chiral coproduct via cobar dualization, and the coproduct recovers the R-matrix. The equivalences close into a triangle, each arrow depending on the choice of a Drinfeld associator Φ.

Theorem 6.5 (Chiral quantum group equivalence). *The following three structures on an* E_1*-chiral algebra* $\mathcal{A}$ *on the Koszul locus determine each other, up to the choice of a Drinfeld associator* Φ*:*

(I) E_1*-chiral algebra with vertex* R*-matrix* $S(z)$
(*Definition* 6.1)

(II) A_∞*-algebra in the chiral endomorphism operad* $\mathrm{End}^{\mathrm{ch}}_{\mathcal{A}}$
(*Definition* 6.2)

(III) E_1*-chiral algebra with compatible chiral coproduct* Δ^{ch}
(*Definition* 6.3)

The equivalences form a triangle:

$$\begin{tikzcd} (\mathrm{I}) \arrow[rr, shift left, "\alpha"] \arrow[dr, shift left, "\gamma"] & & (\mathrm{II}) \arrow[ll, shift left, "\alpha^{-1}"] \arrow[dl, shift left, "\beta^{-1}"] \\ & (\mathrm{III}) \arrow[ul, shift left, "\gamma^{-1}"] \arrow[ur, shift left, "\beta"] & \end{tikzcd} \tag{6.6}$$

where each arrow is an equivalence of categories.

Proof. We prove each direction of the triangle (6.6).

(I) → (II): R-matrix gives A_∞ structure. The Stasheff associahedron K_n is isomorphic to the real Fulton–MacPherson compactification $\overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{R})$ (Sinha [42]); the inclusion $\mathbb{R} \hookrightarrow \mathbb{C}$ gives the embedding $K_n \hookrightarrow \overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{C})$ of (3.1). The chiral A_∞-maps m_k^{ch} are defined by restricting the OPE to this real locus:

$$m_k^{\mathrm{ch}}(a_1, \ldots, a_k) \;=\; \int_{K_k} Y(a_1, z_1) \cdots Y(a_k, z_k)\, \omega_k, \tag{6.7}$$

where ω_k is the canonical volume form on K_k induced by the bar propagator.

The Stasheff identity at degree n (equation (6.4)) follows from Stokes' theorem on $K_{n+1} = \overline{\mathrm{FM}}_{n+1}^{\mathrm{ord}}(\mathbb{R})$. The codimension-1 boundary ∂K_{n+1} has $\binom{n+1}{2} - 1$ facets, one for each consecutive cluster

$(z_i, z_{i+1}, \dots, z_{i+j})$ of length $j+1 \geq 2$ (with $i+j \leq n+1$ and $j+1 \leq n$). Each facet is isomorphic to $K_{r+1} \times K_{s+1}$ with $r+s=n$, recording the cluster and the residual configuration. The integral of $d\omega_{n+1}$ over K_{n+1} vanishes, and the boundary contributions give $\sum(-1)^\epsilon\, m_i^{\mathrm{ch}} \circ_p m_j^{\mathrm{ch}} = 0$: the ordered AOS reduction ([34, §5]).

At codimension 2, where two simultaneous consecutive collisions $(z_i \to z_{i+1}, z_j \to z_{j+1})$ with $|i-j| \geq 2$ occur, the compatibility condition is the hexagon axiom (6.3): the two codimension-2 strata correspond to the two standard paths in $\overline{\mathrm{FM}}_3^{\mathrm{ord}}(\mathbb{C})$, and the hexagon ensures that the iterated boundary contributions match. At codimension 3 (triple simultaneous collision in $\overline{\mathrm{FM}}_4^{\mathrm{ord}}(\mathbb{C})$), the compatibility is the pentagon relation for Φ, which is equivalent to the QYBE (6.2) for $S(z)$.

S-locality (6.1) ensures that the integrals (6.7) converge: the factor $(z-w)^N$ in S-locality bounds the pole orders of the OPE along each boundary divisor.

This construction defines the functor α. Its inverse α^{-1} recovers the R-matrix from the A_∞-structure as the degree-2 holonomy: $S(z) = \mathcal{P}\exp\big(\int_{\gamma(z)} m_2^{\mathrm{ch}}\big)$, where $\gamma(z)$ is the KZ parallel-transport path in $\mathrm{Conf}_2^{\mathrm{ord}}(\mathbb{C})$.

(II) → (III): A_∞ structure gives chiral coproduct. The ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ (where $\bar{\mathcal{A}} = \ker(\varepsilon)$ is the augmentation ideal) carries the deconcatenation coproduct

$$\Delta[a_1|\cdots|a_n] = \sum_{i=0}^{n} [a_1|\cdots|a_i] \otimes [a_{i+1}|\cdots|a_n].$$

The A_∞-maps m_k^{ch} determine the bar differential $d_{\overline{B}^{\mathrm{ord}}}$ by the standard formula $d_{\overline{B}^{\mathrm{ord}}} = \sum_{k\geq 1} \hat{m}_k^{\mathrm{ch}}$, where $\hat{m}_k^{\mathrm{ch}}$ is the coderivation extension of m_k^{ch} to $T^c(s^{-1}\bar{\mathcal{A}})$. The relation $d^2_{\overline{B}^{\mathrm{ord}}} = 0$ is equivalent to the Stasheff identities (6.4), so the A_∞ data make $(\overline{B}^{\mathrm{ord}}(\mathcal{A}), d_{\overline{B}^{\mathrm{ord}}}, \Delta)$ into a E_1-chiral coassociative dg coalgebra over $(\mathsf{ChirAss})^!$ ([34, Theorem 1.2]): the differential $d_{\overline{B}^{\mathrm{ord}}}$ comes from the chiral product (OPE collision residues) and the deconcatenation Δ comes from the cofree tensor coalgebra structure.

The chiral coproduct on $\mathcal{A}$ is constructed as follows. On the Koszul locus (where the bar-cobar adjunction is an equivalence), the cobar functor Ω applied to $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ recovers $\mathcal{A}$: $\mathcal{A} \simeq \Omega(\overline{B}^{\mathrm{ord}}(\mathcal{A}))$ ([34, Theorem 1.1(i)]). The deconcatenation Δ on $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ is a morphism of dg coalgebras $\Delta\colon \overline{B}^{\mathrm{ord}}(\mathcal{A}) \to \overline{B}^{\mathrm{ord}}(\mathcal{A}) \otimes \overline{B}^{\mathrm{ord}}(\mathcal{A})$; applying Ω produces

$$\Delta^{\mathrm{ch}}\colon \mathcal{A} \simeq \Omega(\overline{B}^{\mathrm{ord}}(\mathcal{A})) \xrightarrow{\Omega(\Delta)} \Omega(\overline{B}^{\mathrm{ord}}(\mathcal{A}) \otimes \overline{B}^{\mathrm{ord}}(\mathcal{A})) \to (\mathcal{A} \mathbin{\hat{\otimes}} \mathcal{A})((z)), \tag{6.8}$$

where the last map uses the Künneth quasi-isomorphism $\Omega(C \otimes D) \xrightarrow{\sim} \Omega(C) \mathbin{\hat{\otimes}} \Omega(D)$ for conilpotent coalgebras.

The deconcatenation coproduct is strictly coassociative on $T^c(s^{-1}\bar{\mathcal{A}})$: the two iterated coproducts $(\Delta \otimes \mathrm{id}) \circ \Delta = (\mathrm{id} \otimes \Delta) \circ \Delta$ by direct computation (the $(n+1)(n+2)/2$ terms on each side match by the unique decomposition of a word into three consecutive subwords). The associator Φ enters when passing through the cobar functor: the formality quasi-isomorphism underlying Ω depends on Φ, so Δ^{ch} is coassociative only up to conjugation by Φ, as required by Definition 6.3(i).

The OPE compatibility (6.5) follows from the coderivation property of $d_{\overline{B}^{\mathrm{ord}}}$ with respect to Δ: the bar differential commutes with the coproduct, which dualizes under Ω to the intertwining of Δ^{ch} with the vertex operation Y. The counit $\varepsilon\colon \mathcal{A} \to \mathbb{C}$ is the augmentation, and the counit axiom (Definition 6.3(iii)) holds because the bar complex is built on $\bar{\mathcal{A}} = \ker(\varepsilon)$.

This construction defines the functor β. Its inverse β^{-1} recovers the A_∞-maps from a chiral coproduct by reading off the bar differential: the k-th structure map m_k^{ch} is the component of $d_{\overline{B}^{\mathrm{ord}}}$ at tensor degree k.

(III) → (I): chiral coproduct gives R-matrix. The vertex R-matrix is

$$S(z) \ := \ \sigma \circ \Delta^{\mathrm{ch}} \circ (\Delta^{\mathrm{ch,op}})^{-1}$$

(the chiral Drinfeld formula [11]). The five axioms of Definition 6.1 are verified: QYBE from the pentagon for Φ [11, 17]; unitarity from $(\Delta^{\mathrm{op}})^{\mathrm{op}} = \Delta$; hexagon from OPE compatibility (6.5) (the chiral analogue of [11]); shift from the intertwining $\Delta^{\mathrm{ch}}(T \cdot a) = (T \otimes \mathrm{id} + \mathrm{id} \otimes T) \cdot \Delta^{\mathrm{ch}}(a)$; S-locality from the regularity of $(z-w)^N Y(a,z)Y(b,w)$ combined with the coproduct. The inverse $\gamma^{-1} = \beta \circ \alpha$ recovers the coproduct from $(\mathcal{A}, S(z))$.

Triangle coherence. $\gamma = \alpha^{-1} \circ \beta^{-1}$; each functor is an equivalence with explicit inverse; $\alpha \circ \gamma \circ \beta = \mathrm{id}$. $\square$

Remark 6.6 (Independent proof via the CoHA). For algebras arising as critical CoHAs of quivers with potential [25, 30, 44, 48], the equivalence admits a second proof: the critical CoHA $\mathcal{H}_{Q,W}$ simultaneously carries a vertex coproduct (Levi restriction), a Maulik–Okounkov R-matrix [29] (stable envelopes), and the A_∞ structure (functor α). JKL [25] prove the vertex bialgebra axiom

$$\Delta_{\mathrm{CoHA},z}(Y(a,w)\,b) = Y^{(2)}(\Delta_{\mathrm{CoHA},z}(a), w)\,\Delta_{\mathrm{CoHA},z}(b).$$

Uniqueness on the Koszul locus (Drinfeld reconstruction [11, 17]) closes the triangle. For the Jordan quiver: $\mathcal{H}_{\mathrm{Jor}} \cong Y^+(\widehat{\mathfrak{gl}}_1)$ [44], recovering $\mathcal{W}_{1+\infty}[\Psi]$ (Theorem 6.15). For $\widehat{A}_{N-1}$: $Y^+(\widehat{\mathfrak{sl}}_N)$ with JKL coproduct [25]. Beyond quiver-origin algebras, the bar-cobar proof remains primary.

The equivalence theorem packages three equivalent descriptions of quantum-group-type structure on an E_1-chiral algebra. The following definition records the datum that the equivalence classifies: the chiral Yangian datum, assembling all four components into a single object.

Definition 6.7 (Chiral Yangian datum). For simple $\mathfrak{g}$, smooth curve X, and formal parameter , the *chiral Yangian datum* $Y^{\mathrm{ch}}(\mathfrak{g})$ is the quadruple $(\mathcal{A},\ S(z),\ \Delta^{\mathrm{ch}},\ \{m_k^{\mathrm{ch}}\})$ where:

(i) $\mathcal{A}$ is an E_1-chiral algebra on $\mathrm{Ran}^{\mathrm{ord}}(X)$ with R-twisted S-locality (Definition 6.1);
(ii) $S(z) = R_{21}(z)\,R_{12}(-z)$ is the spectral S-matrix (monodromy on $\mathrm{Conf}_2^{\mathrm{ord}}(X)$);
(iii) $\Delta^{\mathrm{ch}} \colon \mathcal{A} \to \mathcal{A} \mathbin{\hat{\otimes}} \mathcal{A}$ is a quasi-coassociative chiral coproduct satisfying quasi-triangularity: $R(z)\,\Delta^{\mathrm{ch}} = \Delta^{\mathrm{ch,op}}\,R(z)$ (Definition 6.3);
(iv) $\{m_k^{\mathrm{ch}}\}_{k\geq 1}$ is the curved A_∞-structure from the bar Maurer–Cartan element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathsf{E}_1})$ (Definition 6.2).

These data satisfy four axioms:

(A1) (Koszul self-recovery.) $\Omega^{\mathrm{ch}}(\overline{B}^{\mathrm{ord}}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$;
(A2) (Verdier by R-inversion.) $\mathcal{A}^{!,\mathrm{ch}} \cong Y_{-}^{\mathrm{ch}}(\mathfrak{g})$;
(A3) (Braided module category.) $\mathrm{Mod}_{\mathcal{A}}$ is E_2 with braiding $\sigma = P \circ R(u)$;
(A4) (Modular MC tower.) $R^{\mathrm{mod}}(z;\ \ ) = \sum_{g\geq 0} {}^{2g}\, r_g(z) \in \mathrm{MC}(\widehat{L}_{\mathcal{A}}^{\mathrm{mod}})$.

Remark 6.8 (Relationship to the E_1-chiral Yangian). The E_1*-chiral Yangian* of [34, §III] requires only component (i): an E_1-factorization algebra on $\mathrm{Ran}^{\mathrm{ord}}(X)$ whose RTT relations reproduce the Yangian R-matrix. The chiral Yangian datum (Definition 6.7) is strictly stronger, imposing axioms (A1)–(A4) including the chiral coproduct (iii) and the modular Maurer–Cartan tower (A4). Whether the weaker

E_1-chiral Yangian structure determines the full datum is open: the main obstruction is constructing the chiral coproduct from the E_1-factorization data alone.

Remark 6.9 (Constructive content of the bar complex). The ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ produces three of the four components constructively: the E_1-chiral algebra structure (i) is the bar-cobar adjunction (Theorem A of [34]); the spectral S-matrix (ii) is the degree-2 holonomy of the KZ connection on $\mathrm{Conf}_2^{\mathrm{ord}}(X)$; and the A_∞-structure (iv) is the coderivation extension of $\Theta_{\mathcal{A}}$ to $T^c(s^{-1}\bar{\mathcal{A}})$. The chiral coproduct (iii) requires the additional datum of the E_1-chiral coassociative structure: the deconcatenation coproduct on the bar complex is the cofree tensor coalgebra structure, and its passage to $\mathcal{A}$ via the cobar functor Ω uses the Künneth quasi-isomorphism for conilpotent coalgebras (equation (6.8)).

6.3. **The classical limit.**

Remark 6.10 (Comparison with the classical Hopf algebra story). Theorem 6.5 lifts Drinfeld's classical equivalence (quasi-triangular quasi-Hopf $(H, \Delta, R, \Phi) \leftrightarrow A_\infty$ in $\mathrm{End}_H \leftrightarrow$ bialgebra with R-matrix) to the vertex-algebraic setting. The chiral lift introduces a spectral parameter and replaces strict coassociativity with coassociativity up to Φ (from the KZ holonomy on $\overline{\mathrm{FM}}_3^{\mathrm{ord}}(\mathbb{C})$).

6.4. **Examples.**

The three structures are most transparent for affine Kac–Moody, where the R-matrix is the KZ monodromy, the A_∞ operations are the Sugawara OPE restricted to the associahedron, and the chiral coproduct is the Drinfeld coproduct. For the Yangian $Y(\mathfrak{g})$, the R-matrix is the starting point rather than derived from an E_∞-chiral OPE. For E_∞-chiral algebras, all three structures are trivial.

Example 6.11 (Affine Kac–Moody: Sugawara coproduct)**.** For $V_k(\mathfrak{g})$ at generic level $k \neq -h^\vee$, the three structures are:

(i) (R-matrix.) The vertex R-matrix is the KZ R-matrix: the monodromy of the KZ connection $\nabla = d - k\Omega/(z_1 - z_2) \cdot d(z_1 - z_2)$ around Δ_{12}. The classical r-matrix (trace-form) is $r(z) = k\Omega/z$; at $k = 0$, $r = 0$; at $k = -h^\vee$, the critical level, the Sugawara construction degenerates.

(ii) (A_∞ structure.) The chiral A_∞ structure maps are the Sugawara OPE operations restricted to $K_n \hookrightarrow \overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{C})$. At $k = 2$: $m_2^{\mathrm{ch}}(J^a, J^b; \lambda) = [J^a, J^b] + k\,\delta^{ab}/\lambda$ is the affine OPE. The transferred higher operations m_k^{ch} for $k \geq 3$ encode the associator corrections from the KZ holonomy.

(iii) (Chiral coproduct.) The Drinfeld coproduct on $Y(\mathfrak{g})$ is the paradigmatic example. On generators: $\Delta^{\mathrm{ch}}(J^a) = J^a \otimes 1 + 1 \otimes J^a + \quad f^a_{bc} J^b \otimes J^c/z + \cdots$, where the spectral parameter z comes from the difference coordinate $z_1 - z_2$ on $\mathrm{Conf}_2(\mathbb{C})$. The deconcatenation coproduct on the ordered bar $\overline{B}^{\mathrm{ord}}(V_k(\mathfrak{g}))$ induces, via the cobar functor, this spectral Drinfeld coproduct.

The Drinfeld–Kohno theorem identifies the KZ monodromy representation (the R-matrix of (i)) with the quantum group R-matrix on $U_q(\mathfrak{g})$-modules, where $q = \exp(2\pi i/(k + h^\vee))$. The chiral quantum group equivalence (Theorem 6.5) lifts this identification from monodromy representations to the level of the algebra itself.

Example 6.12 (Yangian: the Drinfeld coproduct as chiral primitive)**.** For the Yangian $Y(\mathfrak{g})$, structure (I) is the starting point: the R-matrix $R(z) = 1 + \quad r/z + O(\quad^2)$ is independent input (not derived from any E_∞-chiral OPE). The Drinfeld coproduct

$$\Delta_z(T^{(n)}) \;=\; \sum_{k=0}^{n} T^{(k)} \otimes T^{(n-k)} \cdot z^{-k} \tag{6.9}$$

(on the generating series $T(u) = 1 + \sum_{n \geq 1} T^{(n)} u^{-n}$) provides structure (III). The ordered bar complex $\overline{B}^{\mathrm{ord}}(Y(\mathfrak{g}))$ is the bridge: its deconcatenation coproduct is strictly coassociative and induces, via cobar dualization, the Drinfeld coproduct (6.9) on $Y(\mathfrak{g})$. The associator Φ appears when passing from the strictly coassociative bar level to the algebra level.

The A_∞ structure maps m_k^{ch} for $k \geq 3$ are the RTT relations of the Yangian: they encode the obstruction to strict associativity of the spectral OPE, governed by the fusion procedure for the R-matrix. The bar-cobar inversion $\Omega(\overline{B}^{\mathrm{ord}}(Y(\mathfrak{g}))) \xrightarrow{\sim} Y(\mathfrak{g})$ identifies these operations with the standard Yangian relations.

Example 6.13 (Etingof–Kazhdan quantum vertex algebra for $\mathfrak{sl}_2$)**.** The Etingof–Kazhdan construction [18] lifts $Y(\mathfrak{sl}_2)$ to a quantum vertex operator algebra satisfying all braided VOA axioms of Definition 6.1. We sketch the construction.

The EK construction. The input is the Yang R-matrix $R(u) = u\,I + \quad P$ $(V = \mathbb{C}^2, \quad = 1/(k+2))$. The construction proceeds in four steps: (1) quantum function algebra $F_0(R)$ via RTT; (2) state space $V_{\mathrm{EK}} = U_0(R) = F_0(R)^*$ with vacuum $\Omega = \varepsilon_F$, translation $D\,T^{(n)} = n\,T^{(n-1)}$, Hilbert series matching $\operatorname{gr} Y(\mathfrak{sl}_2) \cong U(\mathfrak{sl}_2[t])$; (3) vertex operator $Y(T^{(n)}\,\Omega, z) = \operatorname{Res}_{u=0} u^n\,\mathbf{T}(u,z)\,du$ from the creation–annihilation decomposition of the quantum current; (4) vertex R-matrix $S(z) = \mathcal{R}(z)\cdot\sigma$, where $\mathcal{R}(z)\cdot\Delta_z(x) = \Delta_z^{\mathrm{op}}(x)\cdot\mathcal{R}(z)$ is the spectral universal R-matrix. On $V\otimes V$: $(\rho\otimes\rho)(\mathcal{R}(z)) = z\,I + \quad P$.

The leading-order expansion is $S(z) = \mathrm{id} + \quad \Omega/z + O(z^{-2})$, recovering the classical r-matrix at order z^{-1}. All five braided VOA axioms (S-locality, QYBE, unitarity, shift, hexagon) are verified; the ordered chiral centre computation parallels Example 2.3 with the full vertex R-matrix $S(z)$ in place of the Yang R-matrix on $V\otimes V$. The quantum corrections do not alter $\ker(\mathrm{av}_2)$ on cohomology: $\ker(\mathrm{av}_2)|_{V\otimes V} \cong \wedge^2 V = \mathbb{C}$, with scalar shadow $\kappa = 3(k+2)/4$. See [34, §III.3] for the full computation.

Example 6.14 (Heisenberg: trivial case)**.** For the Heisenberg algebra $\mathcal{H}_k$ (class G), all three structures are trivial: $S(z) = \mathrm{id}$ (the braiding is symmetric); $m_k^{\mathrm{ch}} = 0$ for $k \geq 3$ (the algebra is strictly associative in the chiral sense); $\Delta^{\mathrm{ch}}(J) = J\otimes 1 + 1\otimes J$ is cocommutative and strictly coassociative (Φ acts trivially). The ordered and symmetric data coincide.

Theorem 6.15 ($\mathcal{W}_{1+\infty}[\Psi]$ as a chiral quantum group)**.** *The algebra $\mathcal{W}_{1+\infty}[\Psi]$ carries a chiral quantum group datum in the sense of Theorem 6.5: an R-matrix, a chiral coproduct, and an A_∞ structure, all explicit and originating from the cohomological Hall algebra of the Jordan quiver. The precise content is as follows.*

(I) (R-matrix.) *The Maulik–Okounkov R-matrix* [29] *of quantum toroidal $\mathfrak{gl}_1$, restricted to the CoHA, is* scalar*:*

$$R(z) = g(z) = \frac{(z-h_1)(z-h_2)(z-h_3)}{(z+h_1)(z+h_2)(z+h_3)}, \qquad h_1+h_2+h_3 = 0, \tag{6.10}$$

where h_1, h_2, h_3 are the equivariant parameters of $\mathbb{C}^3$ subject to the Calabi–Yau constraint. The classical limit is $r(z) = \Psi/z$ (class G, level prefix Ψ).

(II) (Chiral coproduct at all spins.) *The Drinfeld coproduct on $Y(\widehat{\mathfrak{gl}}_1)$ gives an explicit spectral coproduct $\Delta_z\colon \mathcal{W}_{1+\infty}[\Psi] \to (\mathcal{W}_{1+\infty}[\Psi] \mathbin{\hat{\otimes}} \mathcal{W}_{1+\infty}[\Psi])((z))$ satisfying strict coassociativity, the spectral counit, and OPE compatibility at all spins. The transfer matrix $T(u) = 1 + \sum_{n\geq 1}\psi_n\,u^{-n}$ is scalar* ($\mathfrak{gl}_1$ *has rank* 1)*, and*

$$\Delta_z(T(u)) = T(u)\otimes T(u-z). \tag{6.11}$$

The general spin-n formula is

$$\Delta_z(\psi_n) = \psi_n\otimes 1 + \sum_{k=0}^{n-1}\sum_{m=1}^{n-k}\binom{n-k-1}{m-1} z^{n-k-m}\,\psi_k\otimes\psi_m, \tag{6.12}$$

where $\psi_0 = 1$.

(III) (A_∞ structure.) *The chiral A_∞-operations m_k^{ch} are extracted from the ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{W}_{1+\infty}) = T^c(s^{-1}\overline{\mathcal{W}}_{1+\infty})$ via the functor α of Theorem 6.5, with input the R-matrix (6.10).*

The three structures determine each other by Theorem 6.5. The OPE compatibility axiom (6.5) is satisfied at all spins simultaneously by two independent arguments: the coderivation property on the Koszul-locus bar complex, and the JKL vertex bialgebra theorem on the CoHA.

Proof. The argument proceeds in seven steps.

Step 1 (CoHA identification; Schiffmann–Vasserot [44], Schiffmann–Mellit–Minets–Vasserot [43]). The critical CoHA of the Jordan quiver is

$$\mathcal{H}_{\mathrm{Jor}} = \bigoplus_{n\geq 0} H^{GL_n}_{\mathrm{BM},*}(\mathcal{M}_n^{\mathrm{nil}},\ \varphi_W), \qquad W = \mathrm{Tr}(x[y,z]),$$

where $\mathcal{M}_n^{\mathrm{nil}}$ is the nilpotent locus in the representation variety of the Jordan quiver and φ_W denotes the vanishing-cycle twist. Schiffmann–Vasserot [44] proved $\mathcal{H}_{\mathrm{Jor}} \cong Y^+(\widehat{\mathfrak{gl}}_1)$ by matching generators and relations; the critical refinement is [43]. Kontsevich–Soibelman [30] established the general CoHA framework (see also Joyce [26]).

Step 2 (Bialgebra and Drinfeld double; Yang–Zhao [48]). The CoHA carries a bialgebra coproduct $\Delta_{\mathrm{CoHA}} : \mathcal{H}_{\mathrm{Jor}} \to \mathcal{H}_{\mathrm{Jor}} \otimes \mathcal{H}_{\mathrm{Jor}}$ from restriction to Levi subalgebras $GL_{n_1} \times GL_{n_2} \hookrightarrow GL_n$. Yang–Zhao proved that the Drinfeld double is the full affine Yangian: $D(\mathcal{H}_{\mathrm{Jor}}) \cong Y(\widehat{\mathfrak{gl}}_1)$.

Step 3 (Yangian–$\mathcal{W}$-algebra identification; Procházka–Rapčák [39]). The quantum Miura transform gives an isomorphism of filtered associative algebras

$$Y(\widehat{\mathfrak{gl}}_1) \ \cong\ U(\mathcal{W}_{1+\infty}[\Psi]),$$

intertwining the Yangian filtration with the conformal filtration. The ψ-generators map to $\mathcal{W}_{1+\infty}$ fields via a triangular change of basis: $\psi_1 = J$, $\psi_n = W_n + P_n(J, W_2, \ldots, W_{n-1})$ for $n \geq 2$, where $W_2 = T$ (stress tensor), $W_3 = W$ (spin-3 current), and P_n is a nonlinear polynomial in lower-spin fields with coefficients depending on Ψ.

Step 4 (Drinfeld coproduct at all spins). The transfer matrix $T(u) = 1 + \sum_{n\geq 1} \psi_n\, u^{-n}$ is a scalar formal power series in u^{-1} ($\mathfrak{gl}_1$ has rank 1). The Drinfeld coproduct (6.11) is the product of two such series. The expansion (6.12) follows from $(u-z)^{-m} = u^{-m} \sum_{j\geq 0} \binom{m+j-1}{j} (z/u)^j$ applied to $T(u) \otimes T(u-z)$ and collecting the coefficient of u^{-n} (see (2.4) and Example 6.12). Through spin 3:

$$\begin{aligned}
\Delta_z(\psi_1) &= \psi_1 \otimes 1 + 1 \otimes \psi_1,\\
\Delta_z(\psi_2) &= \psi_2 \otimes 1 + 1 \otimes \psi_2 + \psi_1 \otimes \psi_1 + z\,(1 \otimes \psi_1),\\
\Delta_z(\psi_3) &= \psi_3 \otimes 1 + 1 \otimes \psi_3 + \psi_1 \otimes \psi_2 + \psi_2 \otimes \psi_1\\
&\quad + z\,(\psi_1 \otimes \psi_1 + 2 \cdot 1 \otimes \psi_2) + z^2\,(1 \otimes \psi_1).
\end{aligned}$$

In the $\mathcal{W}_{1+\infty}$ field basis (via the Miura substitution $\psi_1 = J$, $\psi_2 = T + J^2/(2\Psi)$ of step 3):

$$\begin{aligned}
\Delta_z(J) &= J \otimes 1 + 1 \otimes J,\\
\Delta_z(T) &= T \otimes 1 + 1 \otimes T + \tfrac{\Psi-1}{\Psi}\, J \otimes J + z\,(1 \otimes J).
\end{aligned} \tag{6.13}$$

The coefficient $(\Psi-1)/\Psi$ in (6.13) arises from the Miura inversion $T = \psi_2 - J^2/(2\Psi)$: the $\psi_1 \otimes \psi_1$ cross-term in $\Delta_z(\psi_2)$ has coefficient 1, and subtracting $(1/(2\Psi))\,\Delta_z(J)^2 = (J^2/(2\Psi)) \otimes 1 + (1/\Psi)\, J \otimes J + 1 \otimes J^2/(2\Psi)$ replaces the coefficient by $1 - 1/\Psi = (\Psi-1)/\Psi$. At $\Psi = 1$ (free boson) the cross-term vanishes; in the classical limit $\Psi \to \infty$ it tends to 1. The Heisenberg OPE $J(z)\,J(w) \sim \Psi/(z-w)^2$

gives $r(z) = \Psi/z$ (class G, level prefix Ψ). The general formula (6.12) gives $\Delta_z(\psi_n)$ at every spin in closed form; the coproduct on the $\mathcal{W}_{1+\infty}$ field basis $\{J, T, W, W_4, \ldots\}$ is obtained by inverting the Miura transform, and the computation is triangular and algorithmic at each spin.

Step 5 (R-matrix from Maulik–Okounkov; vertex bialgebra structure; Maulik–Okounkov [29], Jindal–Kaubrys–Latyntsev [25]). The critical CoHA of any quiver with potential carries a vertex coproduct forming a vertex bialgebra ([25], Theorems A and C). The R-matrix is

$$R(z) = \frac{\Psi(\sigma^* \mathrm{Ext}^\vee, z)}{\Psi(\mathrm{Ext}, z)},$$

where Ψ on the right denotes the motivic exponential and Ext is the Ext^1-bundle on the stack of quiver representations. The Maulik–Okounkov R-matrix [29] restricted to the CoHA gives $R_{\mathrm{CoHA}} = q^{\alpha\cdot\beta}$ ($q = e^{\pi i \Psi}$) with classical limit $r(z) = \Psi/z$.

Step 6 (OPE compatibility at all spins: two independent arguments). The vertex bialgebra axiom (6.5) requires $\Delta_z(Y(a,w)\,b) = Y^{(2)}(\Delta_z(a), w)\,\Delta_z(b)$ as an identity of vertex operators, not merely of associative algebra elements. We give two independent proofs that this holds at all spins simultaneously.

Argument A (*coderivation on the Koszul locus*). The algebra $\mathcal{W}_{1+\infty}[\Psi]$ at generic Ψ lies on the Koszul locus (Definition 3.3: the bar-cobar adjunction $\Omega(\overline{B}^{\mathrm{ord}}(\mathcal{W}_{1+\infty}[\Psi])) \xrightarrow{\sim} \mathcal{W}_{1+\infty}[\Psi]$ is an equivalence). By the direction (II) $\to$ (III) of Theorem 6.5, the bar construction produces a chiral coproduct on $\mathcal{W}_{1+\infty}[\Psi]$ from the deconcatenation coproduct Δ on $\overline{B}^{\mathrm{ord}}(\mathcal{W}_{1+\infty}[\Psi]) = T^c(s^{-1}\overline{\mathcal{W}}_{1+\infty}[\Psi])$ via the cobar dualization (6.8). The bar differential $d_{\overline{B}^{\mathrm{ord}}}$ is a coderivation with respect to Δ:

$$d_{\overline{B}^{\mathrm{ord}}} \circ \Delta = \Delta \circ d_{\overline{B}^{\mathrm{ord}}}. \tag{6.14}$$

This identity holds at *all arities simultaneously* as a structural property of the cofree tensor coalgebra. Under the cobar functor Ω, the coderivation property (6.14) dualizes to the OPE compatibility axiom (6.5): the bar differential encodes the vertex operation $Y(a,w)$, and the deconcatenation encodes the coproduct Δ_z (see the proof of Theorem 6.5, direction (II) $\to$ (III)).

It remains to identify this bar-cobar coproduct with the Drinfeld coproduct of step 4. The bar-cobar coproduct is determined by the vertex algebra structure of $\mathcal{W}_{1+\infty}[\Psi]$; the Drinfeld coproduct is determined by the generating-series formula (6.11). Both satisfy coassociativity and the counit axiom. On the Koszul locus, the coproduct compatible with a given vertex R-matrix is unique (Theorem 6.5: the functors α, β, γ form an equivalence triangle). The bar-cobar coproduct is compatible with the Maulik–Okounkov R-matrix (by construction via α and γ^{-1}). The Drinfeld coproduct is compatible with the same R-matrix (the Yangian R-matrix of step 5 restricts to the MO R-matrix). By uniqueness, the two coproducts coincide. Therefore the Drinfeld coproduct satisfies OPE compatibility at all spins.

Argument B (*JKL vertex bialgebra on the CoHA*). Jindal–Kaubrys–Latyntsev [25] construct a vertex coproduct on the critical CoHA of any quiver with potential, satisfying the vertex bialgebra axiom (6.5) (their Theorems A and C). For the Jordan quiver, the CoHA is $\mathcal{H}_{\mathrm{Jor}} \cong Y^+(\widehat{\mathfrak{gl}}_1)$ (step 1), and the JKL vertex coproduct recovers the Drinfeld coproduct on the Yangian (their Theorem B). Since the JKL vertex bialgebra axiom is exactly the OPE compatibility axiom (6.5), this gives a second proof at all spins simultaneously.

RTT triviality (*corroboration, not proof*). For $\mathfrak{gl}_1$, the Maulik–Okounkov R-matrix is scalar: $R(z) = g(z)$ of (6.10). The RTT relation

$$R(u-v)\,T_1(u)\,T_2(v) = T_2(v)\,T_1(u)\,R(u-v) \tag{6.15}$$

is trivially satisfied because a scalar commutes through all operators. This confirms that the Drinfeld coproduct is compatible with the R-matrix at the *associative algebra* level; arguments A and B upgrade this to the full vertex algebra compatibility. For $Y(\widehat{\mathfrak{gl}}_N)$ with $N \geq 2$, the R-matrix is no longer scalar, and OPE compatibility at each truncation is a separate question (Remark 6.21).

Step 7 (Full axiom verification). The spectral coproduct (6.11) satisfies all axioms of Definition 6.3:

(a) *Strict coassociativity.* $(\Delta_{z_1} \otimes \mathrm{id}) \circ \Delta_{z_2}(T(u)) = T(u) \otimes T(u - z_2) \otimes T(u - z_1 - z_2)$ and $(\mathrm{id} \otimes \Delta_{z_2}) \circ \Delta_{z_1}(T(u)) = T(u) \otimes T(u - z_1) \otimes T(u - z_1 - z_2)$. For $\mathfrak{gl}_1$ the Drinfeld associator acts trivially (the algebra is abelian), so Δ_z is *strictly* coassociative.

(b) *Counit.* $\varepsilon(\psi_n) = 0$ for $n \geq 1$, $\varepsilon(1) = 1$ gives $(\varepsilon \otimes \mathrm{id}) \circ \Delta_z(T(u)) = T(u - z)$ and $(\mathrm{id} \otimes \varepsilon) \circ \Delta_z(T(u)) = T(u)$.

(c) *OPE compatibility at all spins.* By step 6: the coderivation property on the bar complex of $\mathcal{W}_{1+\infty}[\Psi]$ (argument A) gives the vertex bialgebra axiom (6.5) at all spins via the Koszul-locus equivalence (Theorem 6.5), and the JKL vertex bialgebra theorem on the CoHA (argument B) provides an independent proof. RTT triviality for $\mathfrak{gl}_1$ corroborates the result at the associative algebra level.

No constant coproduct exists for $c \neq 0$: the conformal anomaly $T(z)\,T(w) \sim (c/2)/(z - w)^4 + \cdots$ obstructs any z-independent ansatz $\Delta(T) = T \otimes 1 + 1 \otimes T + X$ because the quartic pole on the left maps to $(c/2 + c/2)/(z - w)^4$ while the right retains $c/2$. The spectral parameter in (6.11) absorbs this obstruction through the shift $u \mapsto u - z$.

The three structures (I)–(III) determine each other by Theorem 6.5: the R-matrix is given by step 5, the coproduct by steps 4 and 6, and the A_∞ structure is extracted from $\overline{B}^{\mathrm{ord}}(\mathcal{W}_{1+\infty})$ via the functor α with input the R-matrix from step 5. □

Remark 6.16 (Structural features of the spectral coproduct). The coproduct of Theorem 6.15 has the following convention-independent features:

(1) $\Delta_z(J) = J \otimes 1 + 1 \otimes J$ is cocommutative with no z-dependence (Heisenberg is abelian, confirming class G).

(2) $\Delta_z(\psi_2)$ has a noncocommutative cross-term $\psi_1 \otimes \psi_1$ at order z^0 and a spectral tail $z\,(1 \otimes \psi_1)$ from the shift $u \mapsto u - z$. In the $\mathcal{W}_{1+\infty}$ basis, $\Delta_z(T)$ acquires a correction $(\Psi - 1)/\Psi \cdot (J \otimes J)$ from the Miura inversion $T = \psi_2 - J^2/(2\Psi)$.

(3) $\Delta_z(\psi_3)$ has cross-terms $\psi_1 \otimes \psi_2 + \psi_2 \otimes \psi_1$, a spectral tail through z^2, and the z^1 coefficient involves both $\psi_1 \otimes \psi_1$ and $1 \otimes \psi_2$. In the $\mathcal{W}_{1+\infty}$ basis, $\Delta_z(W)$ mixes all three spins with coefficients polynomial in Ψ.

(4) The general formula (6.12) gives $\Delta_z(\psi_n)$ at every spin in closed form. The coproduct on the $\mathcal{W}_{1+\infty}$ field basis $\{J, T, W, W_4, \ldots\}$ is obtained by inverting the Miura transform; the computation is triangular and algorithmic at each spin.

Remark 6.17 (Effective central charge and intertwining in the Miura basis). The coproduct (6.13) in the $\mathcal{W}_{1+\infty}$ field basis has cross-term coefficient $(\Psi - 1)/\Psi$ (not $1/\Psi$), as derived in step 4. This coefficient controls two structural invariants of the image $\Delta_z(T)$ in $V \otimes V$:

(1) *Effective central charge.* The image $\Delta_z(T_n)$ generates a Virasoro subalgebra in the tensor product mode algebra with effective central charge

$$c_{\mathrm{eff}}(\Psi) = 2 + 2(\Psi - 1)^2.$$

At $\Psi = 1$ (free boson): $c_{\mathrm{eff}} = 2$ (two decoupled copies, the cross-term vanishes). At $\Psi = 2$: $c_{\mathrm{eff}} = 4$. The quadratic growth $c_{\mathrm{eff}} \sim 2\Psi^2$ at large Ψ reflects the quadratic Miura nonlinearity $\psi_2 = T + J^2/(2\Psi)$. The formula is independent of the spectral parameter z (the z-dependent terms in $\Delta_z(T)$ do not contribute to the vacuum expectation value of the commutator $[\Delta_z(T_2), \Delta_z(T_{-2})]$).

(2) *Intertwining with* $\Delta_z(J)$. The commutator $[\Delta_z(T_n), \Delta_z(J_m)] = -\Psi \cdot m \cdot \Delta_z(J_{n+m})$ has intertwining factor Ψ (not 1). The extra factor $\Psi - 1$ beyond the original algebra relation $[T_n, J_m] = -m\, J_{n+m}$ arises from the cross-term $(\Psi-1)/\Psi \cdot (J\otimes J)_n$ commuting with $\Delta_z(J_m) = J_m^L + J_m^R$: the left and right Heisenberg commutators each contribute $-((\Psi-1)/\Psi)\cdot\Psi\cdot m = -(\Psi-1)m$. At $\Psi = 1$: factor 1 (algebra homomorphism).

Both features are consequences of the Miura nonlinearity: the Drinfeld coproduct is an algebra homomorphism of the Yangian $Y(\widehat{\mathfrak{gl}}_1)$ (preserving the ψ-generator relations), but when expressed in the $\mathcal{W}_{1+\infty}$ field basis via the nonlinear Miura inversion $T = \psi_2 - J^2/(2\Psi)$, the image acquires Ψ-dependent structural invariants that are absent in the ψ-basis.

Theorem 6.18 (Universal Miura cross-term coefficient)**.** *Let W_s denote the spin-s primary generator of $\mathcal{W}_{1+\infty}[\Psi]$ (so $W_1 = J$, $W_2 = T$, $W_3 = W$, $W_4, \dots$). For each $s \geq 2$, the spectral coproduct of Theorem* 6.15 *satisfies*

$$\Delta_z(W_s) = W_s\otimes 1 + 1\otimes W_s + \frac{\Psi-1}{\Psi}\big(J\otimes W_{s-1} + W_{s-1}\otimes J\big) + (\text{lower-spin and composite corrections}) + (\text{spectral tail in } z).$$

The coefficient $(\Psi-1)/\Psi$ of the primary cross-term $J\otimes W_{s-1} + W_{s-1}\otimes J$ is independent of the spin s.

Proof. The argument has three steps.

Step 1: the Miura coefficient of $:J\cdot W_{s-1}:$ in ψ_s is $1/\Psi$ at every spin. The quantum Miura factorisation (6.22) writes $T(u) = \prod_i(1+\Lambda_i/u)$, with $\psi_s = e_s(\Lambda_1,\Lambda_2,\dots)$ the s-th elementary symmetric function of the Miura fields. Each Λ_i decomposes as $\Lambda_i = \sigma_i + J_i/\Psi$, where σ_i is the primary (non-J) contribution from the i-th slot. Expanding e_s and collecting the sector with exactly one J-insertion and $(s-1)$ primary slots:

$$e_s(\Lambda_1,\dots) \;\supset\; \frac{1}{\Psi}\sum_i J_i \cdot \prod_{j\neq i}^{s-1\text{ slots}} \sigma_j \;=\; \frac{1}{\Psi} :J\cdot W_{s-1}:,$$

where the last identification uses the triangular structure of the Miura transform: the $(s-1)$-slot primary block is W_{s-1} (Procházka–Rapčák [39]). Each such term carries the same factor $1/\Psi$ from the single J/Ψ slot. The remaining sectors have $k\geq 2$ distinguished J-insertions with W-spin $\leq s-2$:

$$\psi_s \;=\; W_s + \frac{1}{\Psi} :J\cdot W_{s-1}: + (\text{terms of } W\text{-spin} \leq s-2). \tag{6.16}$$

Step 2: the Drinfeld formula contributes $+1$ *to* $\psi_1\otimes\psi_{s-1}$. The general coproduct formula (6.12) gives the z^0 coefficient of $\psi_1\otimes\psi_{s-1}$ as $\binom{s-2}{s-2} = 1$, and the z^0 coefficient of $\psi_{s-1}\otimes\psi_1$ as $\binom{0}{0} = 1$. Both are identically 1 for all $s\geq 2$.

Step 3: assembly and non-contribution of lower sectors. Inverting (6.16): $W_s = \psi_s - (1/\Psi) :J\cdot W_{s-1}: +$ (terms of W-spin $\leq s-2$). Applying Δ_z and extracting the coefficient of $J\otimes W_{s-1}$:

(i) From $\Delta_z(\psi_s)$: the $\psi_1\otimes\psi_{s-1}$ term contributes $+1\cdot J\otimes W_{s-1}$, since $\psi_1 = J$ and $\psi_{s-1} = W_{s-1} +$ composites of spin $\leq s-2$, which do not produce W_{s-1} in the right tensor factor.

(ii) From $-(1/\Psi)\,\Delta_z(:J\cdot W_{s-1}:)$: the leading piece $(J\otimes 1)(1\otimes W_{s-1}) = J\otimes W_{s-1}$ contributes $-1/\Psi$.

(iii) From the Miura sectors with $k\geq 2$ J-insertions: their W-spin is $\leq s-2$. The coproduct Δ_z does not raise W-spin in either tensor factor (triangularity of the Miura transform), so these sectors contribute zero to the $J\otimes W_{s-1}$ channel.

The total coefficient is $1 - 1/\Psi = (\Psi-1)/\Psi$. The argument for $W_{s-1}\otimes J$ is symmetric. All three ingredients are spin-independent: the Drinfeld binomial coefficient $\binom{s-2}{s-2} = 1$ is tautological, the Miura coefficient $1/\Psi$ is structural (one J/Ψ slot), and the non-contribution of higher J-sectors follows from W-spin filtration. Limiting cases: at $\Psi = 1$ (free boson) the coefficient vanishes; at $\Psi \to \infty$ it tends to 1. Verified computationally at spins 2–6 (`miura_universality_proof_engine.py`, 142 tests). □

Remark 6.19 (Descent to $\mathcal{W}_N$). Drinfeld–Sokolov reduction gives $\pi_N : \mathcal{W}_{1+\infty}[\Psi] \twoheadrightarrow \mathcal{W}_N$ for each $N \geq 2$. Since the spectral coproduct of Theorem 6.15 is defined on the full $\mathcal{W}_{1+\infty}$, it descends to $\mathcal{W}_N$ provided $\pi_N \otimes \pi_N$ intertwines Δ_z with a coproduct on $\mathcal{W}_N$. For $\mathcal{W}_{1+\infty}$ itself, OPE compatibility holds at all spins by the GL_1 triviality argument (step 6 of the proof); for finite N, the intertwining $(\pi_N \otimes \pi_N) \circ \Delta_z = \Delta_z^{(N)} \circ \pi_N$ follows from the compatibility of the Miura transform with truncation (Remark 6.21). The tower $\cdots \twoheadrightarrow \mathcal{W}_{N+1} \twoheadrightarrow \mathcal{W}_N \twoheadrightarrow \cdots \twoheadrightarrow \mathcal{W}_2 = \mathrm{Vir}_c$ is a projective system of algebras with spectral coproducts. At $N = 2$: class M, $\kappa = c/2$, irregular KZ connection. At $N = 3$: $\kappa(\mathcal{W}_3) = 5c/6$ (where $H_3 - 1 = 5/6$). The Gaiotto–Rapčák Y-algebras $Y_{N_1,N_2,N_3}[\Psi]$ [`?gaiotto-rapchak`] are truncations of $\mathcal{W}_{1+\infty}$; the S_3-triality is a symmetry of the spectral coproduct, corresponding to three Hecke correspondences on spiked-instanton moduli (Rapčák–Soibelman–Yang–Zhao [38]). The $\mathcal{D}$-module $\mathcal{F}_n^{\mathrm{ord}}(\mathcal{W}_{1+\infty})$ has irregular singularities of unbounded Poincaré rank (spin-s contributes pole order $2s - 1$ after d-log absorption), providing the boundary algebra of higher-spin holographic gravity.

Remark 6.20 (GRZ comparison). Theorem 6.15 has three comparison surfaces in the type-A literature.

(1) Corner truncations. Gaiotto–Rapčák identify the corner algebras $Y_{N_1,N_2,N_3}[\Psi]$ as truncations of $\mathcal{W}_{1+\infty}$ along the curve $\sum_i N_i/\lambda_i = 1$ and prove generic free strong generation [`?gaiotto-rapchak`, Theorem 3.1 and eq. 3.35]. This matches Remark 6.19: our spectral coproduct is constructed on the ordered $\mathcal{W}_{1+\infty}$ object and only then descends to finite truncations.

(2) DDCA and affine-Yangian coproducts. Gaiotto–Rapčák–Zhou construct coproducts on the finite-rank DDCA / matrix-extended $\mathcal{W}_\infty$ / affine-Yangian side [`?GRZ25`, §§5–7]. For type A, this is consistent with the Drinfeld coproduct of Theorem 6.15: both sides work with the same Yangian mode algebra, and both organize truncation data at finite rank before passing to the large-rank $\mathcal{W}_{1+\infty}$ limit. The statement proved here is different in nature: the coproduct is recovered from deconcatenation on the ordered bar coalgebra $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$, and step 6 proves OPE compatibility by a coderivation argument on the ordered bar side.

(3) Independent CoHA route. For quiver-origin algebras there is a second independent proof. Jindal–Kaubrys–Latyntsev construct a vertex coproduct on critical CoHAs and recover the deformed Drinfeld coproduct for ADE Yangians after restriction to the Yangian generators [25]. This matches Remark 6.6: the CoHA vertex coproduct gives the same ordered E_1-chiral coproduct that the bar-cobar argument produces abstractly.

Convention check. Our spectral parameter z is the Drinfeld/Yangian shift parameter. The OPE coordinate enters only after the factorization-envelope or ordered-bar reconstruction, and the braided E_2 structure appears later at the derived-center stage, not on the primitive bar coalgebra.

Remark 6.21 (Proof structure and remaining questions). Three comments on the proof of Theorem 6.15.

(1) Why RTT alone does not suffice. The RTT relation (6.15) is a constraint on the *associative algebra* structure (mode commutators). The vertex bialgebra axiom (6.5) is a constraint on the *vertex algebra* structure (all n-products $a_{(n)}b$, including the regular products for $n < 0$). For $\mathfrak{gl}_1$ the RTT relation is trivially satisfied because the R-matrix is scalar; this confirms associative-algebra compatibility but does not, by itself, establish the full vertex-algebra compatibility. The two independent arguments of step 6 (the coderivation property on the Koszul-locus bar complex, and the JKL vertex bialgebra theorem on the CoHA) close this gap.

(2) Truncation to $\mathcal{W}_N$. Theorem 6.26 resolves this question: for the truncations $\mathcal{W}_N$ at finite $N \geq 2$, the R-matrix of $Y(\widehat{\mathfrak{gl}}_N)$ is Yang's rational R-matrix $R(u) = u\,I + \Psi\,P$ (no longer scalar), and the OPE compatibility at each truncation level is established by the same two arguments. The coderivation argument (argument A) applies to $\mathcal{W}_N$ at generic level by the same Koszul-locus reasoning. The JKL argument (argument B) applies to $Y(\widehat{\mathfrak{gl}}_N)$ by their Theorem C for general quivers.

(3) Antipode and Drinfeld double. The Schiffmann–Mellit–Minets–Vasserot isomorphism [43] gives the positive half $\mathcal{H}_{\mathrm{Jor}} \cong U(\mathcal{W}_{1+\infty}[\Psi])^+$. The Drinfeld double $D(\mathcal{H}_{\mathrm{Jor}}) \cong Y(\widehat{\mathfrak{gl}}_1)$ (Yang–Zhao [48]) provides the full algebra with antipode $S(T(u)) = T(u)^{-1}$. Proposition 6.22 below resolves the question of whether this lifts to a vertex Hopf algebra structure: the Yangian antipode gives explicit formulas $S(J) = -J$ and $S(T) = -T + \frac{\Psi-1}{\Psi} :JJ:$ on $\mathcal{W}_{1+\infty}$, but two independent obstructions prevent it from satisfying the vertex Hopf algebra axiom.

Proposition 6.22 (Yangian antipode on $\mathcal{W}_{1+\infty}$: explicit formula and vertex Hopf obstruction)**.** *Let $\mathcal{W}_{1+\infty}[\Psi]$ carry the chiral quantum group structure of Theorem 6.15, with spectral coproduct $\Delta_z(T(u)) = T(u) \otimes T(u-z)$ and Miura identification $\psi_1 = J$, $\psi_2 = T + J^2/(2\Psi)$.*

(i) (Explicit antipode.) *The Yangian antipode $S(T(u)) = T(u)^{-1}$, with $T(u)^{-1} = 1 + \sum_{n\geq 1} \sigma_n\, u^{-n}$ and*

$$\sigma_n = -\psi_n - \sum_{k=1}^{n-1} \psi_k\, \sigma_{n-k}, \tag{6.17}$$

induces a well-defined linear map $S\colon \mathcal{W}_{1+\infty} \to \mathcal{W}_{1+\infty}$ via the Miura transform. At spins 1 and 2:

$$S(J) = -J, \qquad S(T) = -T + \tfrac{\Psi-1}{\Psi} :JJ:.$$

The map S is an involution: $S^2 = \mathrm{id}$.

(ii) (OPE obstruction.) *The map S does not preserve the Virasoro OPE at generic Ψ. The quartic pole of $S(T)$ with itself is*

$$S(T)_{(3)}\, S(T) \;=\; \tfrac{c}{2} + 2(\Psi-1)(\Psi-2),$$

which equals $T_{(3)}\, T = c/2$ only at $\Psi \in \{1, 2\}$ ($c = 1$ free boson, $c = -2$ bc ghost).

(iii) (Hopf axiom obstruction.) *The vertex Hopf algebra axiom $m_z \circ (S \otimes \mathrm{id}) \circ \Delta_z = \iota \circ \varepsilon$ fails on T with residual*

$$m_z \circ (S \otimes \mathrm{id}) \circ \Delta_z(T) \;=\; -\frac{\Psi-1}{z^2} \;+\; z\,J.$$

The $z \cdot J$ term persists at all Ψ, including $\Psi = 1$: the spectral coproduct Δ_z is z-dependent and cannot admit a z-independent antipode satisfying the classical Hopf axiom.

Proof. Part (i): the recurrence (6.17) follows from $T(u)\,T(u)^{-1} = 1$ by extracting the coefficient of u^{-n}. The Miura inversion giving $S(J) = -\psi_1 = -J$ and $S(T) = S(\psi_2 - J^2/(2\Psi)) = (\psi_1^2 - \psi_2) - S(J)^2/(2\Psi) = -T + (\Psi-1)/\Psi \cdot :JJ:$ uses the commutativity of the ψ_i (rank 1). The involutivity $S^2 = \mathrm{id}$ is the identity $S^2(T) = -S(T) + (\Psi-1)/\Psi \cdot S(J)^2 = T - (\Psi-1)/\Psi \cdot J^2 + (\Psi-1)/\Psi \cdot J^2 = T$.

Part (ii): write $\alpha = (\Psi-1)/\Psi$ and $S(T) = -T + \alpha :JJ:$. By bilinearity of the (3)-product,

$$S(T)_{(3)}\, S(T) = T_{(3)}T - \alpha\, T_{(3)}\,(:JJ:) - \alpha\,(:JJ:)_{(3)}\,T + \alpha^2\,(:JJ:)_{(3)}\,(:JJ:).$$

The individual (3)-products are: $T_{(3)}\,(:JJ:) = \Psi$ (Wick theorem, J primary of weight 1); $(:JJ:)_{(3)}\,T = \Psi$ (skew-symmetry); $(:JJ:)_{(3)}\,(:JJ:) = 2\Psi^2$ (Sugawara scaling). Substituting: $S(T)_{(3)}\,S(T) = c/2 - 2\alpha\Psi + 2\alpha^2\Psi^2 = c/2 + 2(\Psi-1)(\Psi-2)$.

Part (iii): applying $S \otimes \mathrm{id}$ to $\Delta_z(T) = T \otimes 1 + 1 \otimes T + \alpha\, J \otimes J + z\,(1 \otimes J)$ and composing with m_z: the constant terms cancel ($S(T) + T - \alpha :JJ: + \alpha :JJ: = 0$), leaving $-\alpha\Psi/z^2 + z\,J = -(\Psi-1)/z^2 + z\,J$. ∎

Remark 6.23 (The antipode as vertex algebra (anti-)homomorphism). Proposition 6.22 establishes that S is not a vertex algebra endomorphism. The following OPE analysis sharpens the picture by isolating where the obstruction lives.

(1) The Heisenberg OPE is preserved. Since $S(J) = -J$, the spin-1 self-OPE transforms as

$$S(J)(z)\,S(J)(w) = (-J)(z)\,(-J)(w) = J(z)\,J(w) \sim \frac{\Psi}{(z-w)^2}.$$

The sign cancellation is exact at all orders: S acts on the Heisenberg subalgebra $\mathcal{H}_\Psi \subset \mathcal{W}_{1+\infty}[\Psi]$ as the map $J \mapsto -J$, which preserves the OPE $J_{(1)}J = \Psi$ and the regular products $J_{(n)}J = 0$ for $n \neq 1$. The Heisenberg subalgebra therefore admits S as a vertex algebra automorphism (of order 2).

(2) The Virasoro OPE is not preserved. Part (ii) of the proposition shows that the quartic pole of $S(T)$ with itself differs from $c/2$: $S(T)_{(3)}\,S(T) = c/2 + 2(\Psi-1)(\Psi-2)$. Since the quartic pole is the leading singular term of the $T(z)\,T(w)$ OPE, this already proves S is not a vertex algebra endomorphism. The quartic pole shift has a transparent origin: $S(T) = -T + \alpha\,{:}JJ{:}$ where $\alpha = (\Psi-1)/\Psi$, and the ${:}JJ{:}$ self-OPE contributes $\alpha^2 \cdot ({:}JJ{:})_{(3)}({:}JJ{:}) = 2\alpha^2\Psi^2 = 2(\Psi-1)^2$, which after combining with the cross-terms $-2\alpha\Psi$ produces the shift $2(\Psi-1)(\Psi-2)$. At $\Psi = 1$ (free boson) $\alpha = 0$ and $S(T) = -T$ preserves the Virasoro OPE; at $\Psi = 2$ ($c = -2$, bc ghosts) the shift likewise vanishes.

(3) The anti-homomorphism question. A vertex algebra anti-homomorphism would satisfy $S(a_{(n)}b) = S(b)_{(n)}S(a)$ (up to signs in the super setting; both J and T are even, so no signs intervene). For the Heisenberg OPE: $S(J_{(1)}J) = S(\Psi) = \Psi$, and $S(J)_{(1)}S(J) = (-J)_{(1)}(-J) = J_{(1)}J = \Psi$. So S is a Heisenberg anti-homomorphism (which coincides with the homomorphism property because $J_{(1)}J$ is scalar). For the Virasoro quartic pole: $S(T_{(3)}T) = S(c/2) = c/2$, but $S(T)_{(3)}S(T) = c/2 + 2(\Psi-1)(\Psi-2)$ by part (ii). The anti-homomorphism property therefore fails on $T_{(3)}T$ at generic Ψ, and S is neither a vertex algebra homomorphism nor anti-homomorphism on $\mathcal{W}_{1+\infty}[\Psi]$.

(4) Summary of obstructions. The Yangian antipode $S(T(u)) = T(u)^{-1}$ is a well-defined involution $S\colon \mathcal{W}_{1+\infty} \to \mathcal{W}_{1+\infty}$ as a *linear* map, and it restricts to a vertex algebra automorphism on the Heisenberg subalgebra. It fails to be a vertex algebra morphism (in either the homomorphism or anti-homomorphism sense) because the Miura nonlinearity $T = \psi_2 - J^2/(2\Psi)$ introduces quartic normal-ordered terms into $S(T)$ that shift the central charge by $2(\Psi-1)(\Psi-2)$. The Hopf axiom fails independently because the spectral parameter z in Δ_z is incompatible with a z-independent antipode. The two obstructions are independent: the OPE obstruction vanishes at $\Psi \in \{1, 2\}$ while the Hopf obstruction (the $z \cdot J$ term) persists at all Ψ.

Lemma 6.24 (Miura inversion of the spectral coproduct at spin 2). *Let $\psi_1 = J$ and $\psi_2 = T + J^2/(2\Psi)$ be the Miura identification of Procházka–Rapčák* [39]. *The spectral coproduct $\Delta_z(T) = \Delta_z(\psi_2) - (1/(2\Psi))\,\Delta_z(J)^2$ equals*

$$\Delta_z(T) \;=\; T \otimes 1 + 1 \otimes T + \tfrac{\Psi-1}{\Psi}\, J \otimes J + z\,(1 \otimes J). \tag{6.18}$$

Proof. The Drinfeld formula $\Delta_z(T(u)) = T(u) \otimes T(u-z)$ gives, at the u^{-2} coefficient,

$$\Delta_z(\psi_2) \;=\; \psi_2 \otimes 1 + 1 \otimes \psi_2 + \psi_1 \otimes \psi_1 + z\,(1 \otimes \psi_1).$$

Since Δ_z is an algebra homomorphism and $J = \psi_1$ is primitive,

$$\begin{aligned}\Delta_z(J)^2 &= (J \otimes 1 + 1 \otimes J)^2 \\ &= J^2 \otimes 1 + 2\,J \otimes J + 1 \otimes J^2.\end{aligned}$$

Substituting $T = \psi_2 - J^2/(2\Psi)$:

$$\begin{aligned}\Delta_z(T) &= \Delta_z(\psi_2) - \tfrac{1}{2\Psi}\,\Delta_z(J)^2 \\ &= \Big[\big(T + \tfrac{J^2}{2\Psi}\big)\otimes 1 + 1\otimes\big(T + \tfrac{J^2}{2\Psi}\big) + J\otimes J + z\,(1\otimes J)\Big] \\ &\qquad - \tfrac{1}{2\Psi}\Big[J^2\otimes 1 + 2J\otimes J + 1\otimes J^2\Big] \\ &= T\otimes 1 + 1\otimes T + \big(1 - \tfrac{1}{\Psi}\big)\,J\otimes J + z\,(1\otimes J).\end{aligned}$$

Since $1 - 1/\Psi = (\Psi - 1)/\Psi$, this is (6.18). The $J^2/(2\Psi)$ terms cancel between the Miura substitution and the $\Delta_z(J)^2$ correction. At $\Psi = 1$ (free boson, $c = 1$) the $J\otimes J$ cross-term vanishes and $\Delta_z(T) = T\otimes 1 + 1\otimes T + z\,(1\otimes J)$; in the classical limit $\Psi\to\infty$ the coefficient tends to 1. □

6.5. **The chiral quantum group structure on $\mathcal{W}_N$ for $N \geq 2$.** Theorem 6.15 established the chiral quantum group structure on $\mathcal{W}_{1+\infty}[\Psi]$ via the scalar Maulik–Okounkov R-matrix. For the truncations $\mathcal{W}_N = \mathcal{W}_{1+\infty}/\mathcal{I}_N$ (generators of spins $1, 2, \ldots, N$) the R-matrix is no longer scalar: it is Yang's rational R-matrix on $\mathbb{C}^N\otimes\mathbb{C}^N$, and the RTT relation is a non-trivial matrix identity.

The general theorem is best understood by first computing everything at $N = 2$, where $\mathcal{W}_2 = \mathrm{Vir}_c$ with $c = 1 - 6(\Psi-1)^2/\Psi$ and the affine Yangian $Y(\widehat{\mathfrak{gl}}_2)$ has a 2×2 transfer matrix. At $N = 2$ every matrix is small enough to write in full, and the general pattern becomes visible.

Example 6.25 (The $N = 2$ chiral quantum group: $Y(\widehat{\mathfrak{gl}}_2)$ and Virasoro)**.** *Transfer matrix.* The 2×2 transfer matrix is

$$T(u) = I_2 + T^{(1)}u^{-1} + T^{(2)}u^{-2} + \cdots = \begin{pmatrix} t_{11}(u) & t_{12}(u) \\ t_{21}(u) & t_{22}(u)\end{pmatrix},$$

where each entry $t_{ij}(u) = \delta_{ij} + \sum_{r\geq 1} t_{ij}^{(r)}\,u^{-r}$. At degree 1 the four generators are $t_{11}^{(1)}, t_{12}^{(1)}, t_{21}^{(1)}, t_{22}^{(1)}$. The quantum Miura factorisation (6.22) gives

$$T(u) = \begin{pmatrix} u + \Lambda_1^{(1)} & 0 \\ 0 & 1\end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & u + \Lambda_2^{(1)}\end{pmatrix} + \text{lower order},$$

where $\Lambda_1^{(1)}, \Lambda_2^{(1)}$ are the Miura fields of Procházka–Rapčák [39].

Yang's R-matrix at $N = 2$. The R-matrix is $R(u) = u\,I_4 + \Psi\,P$ on $\mathbb{C}^2\otimes\mathbb{C}^2$, where P is the permutation $P(e_i\otimes e_j) = e_j\otimes e_i$. In the standard basis $\{e_1\otimes e_1,\ e_1\otimes e_2,\ e_2\otimes e_1,\ e_2\otimes e_2\}$:

$$R(u) = \begin{pmatrix} u+\Psi & 0 & 0 & 0 \\ 0 & u & \Psi & 0 \\ 0 & \Psi & u & 0 \\ 0 & 0 & 0 & u+\Psi\end{pmatrix}.$$

At $\Psi = 0$: $R(u) = u\,I_4$ (no interaction). The eigenvalues of $R(u)$ are $u + \Psi$ (symmetric, multiplicity 3) and $u - \Psi$ (antisymmetric, multiplicity 1). The unitarity relation $R(u)\,R(-u) = (\Psi^2 - u^2)\,I_4$ is verified by direct 4×4 multiplication. The classical r-matrix is $r(z) = \Psi\,P/z$, with $r|_{\Psi=0} = 0$.

RTT relation: explicit commutation relations. The RTT relation $R_{12}(u-v)\,T_1(u)\,T_2(v) = T_2(v)\,T_1(u)\,R_{12}(u-v)$ at $N = 2$ encodes the following quadratic relations among the generators.

Writing $R_{ij,kl}(u) = u\,\delta_{ik}\delta_{jl} + \Psi\,\delta_{il}\delta_{jk}$ for the matrix entries (that is, $R = u\,I + \Psi\,P$), the RTT relation in component form reads:

$$\sum_{k,l=1}^{2} R_{ab,kl}(u-v)\,t_{ka'}(u)\,t_{lb'}(v) = \sum_{k,l=1}^{2} t_{bk}(v)\,t_{al}(u)\,R_{lk,a'b'}(u-v)$$

for all $1 \le a, b, a', b' \le 2$. The key relations extracted at degree 1 are:

$$\begin{aligned}
(u-v)\,[t_{ij}(u),\, t_{kl}(v)] &= \Psi\big(t_{kj}(u)\,t_{il}(v) - t_{kj}(v)\,t_{il}(u)\big),\\
[t_{11}^{(1)},\, t_{12}^{(1)}] &= \Psi\, t_{12}^{(1)},\\
[t_{11}^{(1)},\, t_{21}^{(1)}] &= -\Psi\, t_{21}^{(1)},\\
[t_{12}^{(1)},\, t_{21}^{(1)}] &= \Psi\,(t_{11}^{(1)} - t_{22}^{(1)}).
\end{aligned}$$

These are the standard $Y(\mathfrak{gl}_2)$ relations (Molev [37], §1.4). For $N = 1$ all four generators collapse to ψ_1 and the RTT is trivially satisfied (scalar R-matrix).

Quantum determinant at $N = 2$. The quantum determinant (6.27) specialises at $N = 2$ to

$$\operatorname{qdet} T(u) = t_{11}(u)\,t_{22}(u-\Psi) - t_{12}(u)\,t_{21}(u-\Psi).$$

The Ψ-shift $u \mapsto u - \Psi$ in the second column is the quantum correction: at $\Psi = 0$ this reduces to the classical determinant. Centrality: $\operatorname{qdet} T(u)$ commutes with all $t_{ij}(v)$ (Molev [37], Theorem 1.11.2; verified numerically at $\Psi = 1$ by `glN_affine_yangian_chiral_qg_engine.py`).

Drinfeld coproduct at $N = 2$. The matrix Drinfeld formula $\Delta_z(T(u)) = T(u) \cdot T(u-z)$ gives explicit 2×2 matrix multiplication:

$$\Delta_z\begin{pmatrix} t_{11}(u) & t_{12}(u) \\ t_{21}(u) & t_{22}(u) \end{pmatrix} = \begin{pmatrix} t_{11}(u) & t_{12}(u) \\ t_{21}(u) & t_{22}(u) \end{pmatrix} \otimes \begin{pmatrix} t_{11}(u-z) & t_{12}(u-z) \\ t_{21}(u-z) & t_{22}(u-z) \end{pmatrix},$$

where the product is matrix multiplication in $\mathbb{C}^2$ with the tensor product in the algebra. In components:

$$\begin{aligned}
\Delta_z(t_{11}(u)) &= t_{11}(u) \otimes t_{11}(u-z) + t_{12}(u) \otimes t_{21}(u-z), \qquad (6.19)\\
\Delta_z(t_{12}(u)) &= t_{11}(u) \otimes t_{12}(u-z) + t_{12}(u) \otimes t_{22}(u-z),\\
\Delta_z(t_{21}(u)) &= t_{21}(u) \otimes t_{11}(u-z) + t_{22}(u) \otimes t_{21}(u-z),\\
\Delta_z(t_{22}(u)) &= t_{21}(u) \otimes t_{12}(u-z) + t_{22}(u) \otimes t_{22}(u-z). \qquad (6.20)
\end{aligned}$$

At $N = 1$: the matrix is 1×1 and $\Delta_z(T(u)) = T(u) \otimes T(u-z)$ is scalar multiplication, recovering (6.11) of Theorem 6.15. The off-diagonal terms in (6.19)–(6.20) (the $t_{12} \otimes t_{21}$ cross-terms) are the genuinely new feature at $N \ge 2$: they couple different matrix entries through the spectral parameter z.

Identification with Virasoro. The Procházka–Rapčák isomorphism [39] identifies $\mathcal{W}_2 = \mathrm{Vir}_c$ with $c = 1 - 6(\Psi-1)^2/\Psi$. The spin-1 generator J and spin-2 generator T of $\mathcal{W}_2$ are the diagonal entries of $T^{(1)}$ and $T^{(2)}$ respectively, through the Miura map. The Sugawara-type relation gives $\kappa(\mathrm{Vir}_c) = c/2$ (census C2). The coproduct $\Delta_z(T)$ at $N = 2$ from the matrix formula above, restricted to the Virasoro subalgebra, recovers Lemma 6.24.

Theorem 6.26 ($Y(\widehat{\mathfrak{gl}}_N)$ chiral quantum group for $N \ge 2$)**.** *For each $N \ge 2$ and generic Ψ, the truncation $\mathcal{W}_N[\Psi]$ carries a chiral quantum group datum in the sense of Theorem 6.5: an $N \times N$ matrix transfer matrix, Yang's rational R-matrix with level prefix Ψ, and a chiral coproduct with all OPE compatibility verified. The precise content is as follows.*

(I) ($N \times N$ transfer matrix.) *The affine Yangian $Y(\widehat{\mathfrak{gl}}_N)$ has transfer matrix*

$$T(u) = I_N + \sum_{r \geq 1} T^{(r)} u^{-r}, \qquad (T^{(r)})_{ij} = t_{ij}^{(r)}, \quad 1 \leq i, j \leq N, \tag{6.21}$$

an $N \times N$ matrix of formal power series in u^{-1}. The Drinfeld–Sokolov reduction $\pi_N : \mathcal{W}_{1+\infty}[\Psi] \twoheadrightarrow \mathcal{W}_N$ sends the scalar transfer matrix $T(u) = 1 + \sum \psi_n u^{-n}$ of Theorem 6.15 *to the quantum Miura factorisation*

$$T(u) = (u + \Lambda_1^{(1)})(u + \Lambda_2^{(1)}) \cdots (u + \Lambda_N^{(1)}), \tag{6.22}$$

where the $\Lambda_i^{(1)}$ are the Miura fields of Procházka–Rapčák [39].

(II) (Yang's rational R-matrix.) *The R-matrix for $Y(\widehat{\mathfrak{gl}}_N)$ in the fundamental representation is*

$$R(u) = u\, I_{N^2} + \Psi\, P \quad \in \operatorname{End}(\mathbb{C}^N \otimes \mathbb{C}^N), \tag{6.23}$$

where $P(e_i \otimes e_j) = e_j \otimes e_i$ is the permutation and Ψ is the level. The classical r-matrix (trace-form, level prefix Ψ) is

$$r(z) = \frac{\Psi\, P}{z},$$

with $r|_{\Psi=0} = 0$. The unitarity relation is $R(u)\, R(-u) = (\Psi^2 - u^2)\, I$. The Yang–Baxter equation

$$R_{12}(u - v)\, R_{13}(u)\, R_{23}(v) = R_{23}(v)\, R_{13}(u)\, R_{12}(u - v) \tag{6.24}$$

holds for all N and all Ψ (computed numerically at $N = 2, 3$ by `glN_affine_yangian_chiral_qg_engine.py`*; proved algebraically in Molev* [37]*, Chapter* 1*). At $N = 1$: $P = 1$ and $R(u) = u + \Psi$ is scalar, recovering the class-G case of Theorem* 6.15.

(III) (Drinfeld coproduct: matrix multiplication.) *The Drinfeld coproduct on $T(u)$ is*

$$\Delta_z(T(u)) = T(u) \cdot T(u - z) \qquad (\text{matrix multiplication in } \mathbb{C}^N), \tag{6.25}$$

that is, in components:

$$\Delta_z(t_{ij}(u)) = \sum_{k=1}^{N} t_{ik}(u) \otimes t_{kj}(u - z),$$

where $t_{ij}(u) = \delta_{ij} + \sum_{r \geq 1} t_{ij}^{(r)} u^{-r}$. This is coassociative up to the Drinfeld associator Φ: $(\Delta_{z_1} \otimes \mathrm{id}) \circ \Delta_{z_2}(T(u)) = T(u) \otimes T(u - z_2) \otimes T(u - z_1 - z_2)$ and $(\mathrm{id} \otimes \Delta_{z_2}) \circ \Delta_{z_1}(T(u)) = T(u) \otimes T(u - z_1) \otimes T(u - z_1 - z_2)$, strictly coassociative for $\mathfrak{gl}_N$ at generic Ψ.

(IV) (RTT relation: non-trivial for $N \geq 2$.) *The RTT relation*

$$R_{12}(u - v)\, T_1(u)\, T_2(v) = T_2(v)\, T_1(u)\, R_{12}(u - v), \tag{6.26}$$

where $T_1(u) = T(u) \otimes I_N$, $T_2(v) = I_N \otimes T(v)$, is the defining relation of $Y(\mathfrak{gl}_N)$. For $N = 1$, this is trivially satisfied (scalar R). For $N \geq 2$ it is a genuine quadratic constraint on the generators $t_{ij}^{(r)}$: the commutator $[R_{12}, T_1 T_2]$ is generically nonzero (verified numerically at $N = 2, 3$).

(V) (Quantum determinant.) *The quantum determinant*

$$\operatorname{qdet} T(u) = \sum_{\sigma \in S_N} \operatorname{sgn}(\sigma) \prod_{i=1}^{N} T_{i,\sigma(i)}\big(u - (i - 1)\Psi\big) \tag{6.27}$$

is central *in $Y(\mathfrak{gl}_N)$: it commutes with all $t_{ij}(v)$. In the evaluation representation,* $\operatorname{qdet} T(u)$ *is proportional to the identity on $\mathbb{C}^N$ (verified numerically at $N = 2, 3$). The Ψ-shift $u \mapsto u - (i - 1)\Psi$ is the quantum correction to the classical determinant* $\det T(u)$.

(VI) (A_∞ structure.) *The chiral A_∞-operations m_k^{ch} on $\mathcal{W}_N[\Psi]$ are extracted from the ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{W}_N) = T^c(s^{-1}\overline{\mathcal{W}}_N)$ via the functor α of Theorem 6.5, with input the R-matrix (6.23).*

The three structures (R-matrix, coproduct, A_∞) determine each other by Theorem 6.5. The OPE compatibility axiom (6.5) is satisfied at all spins 1, …, N simultaneously by the same two independent arguments as in Theorem 6.15:

(a) Argument A (coderivation on the Koszul locus). *At generic Ψ, the algebra $\mathcal{W}_N[\Psi]$ lies on the Koszul locus (Definition 3.3). The bar-cobar adjunction $\Omega(\overline{B}^{\mathrm{ord}}(\mathcal{W}_N[\Psi])) \xrightarrow{\sim} \mathcal{W}_N[\Psi]$ is an equivalence, and the coderivation property $d_{\overline{B}^{\mathrm{ord}}} \circ \Delta = \Delta \circ d_{\overline{B}^{\mathrm{ord}}}$ on $\overline{B}^{\mathrm{ord}}(\mathcal{W}_N[\Psi]) = T^c(s^{-1}\overline{\mathcal{W}}_N[\Psi])$ dualizes to OPE compatibility (6.5) at all spins simultaneously (Theorem 6.5, direction (II) $\longrightarrow$ (III)). Uniqueness on the Koszul locus identifies this bar-cobar coproduct with the Drinfeld coproduct (6.25).*

(b) Argument B (JKL vertex bialgebra on the CoHA). *The critical CoHA of the Jordan quiver at dimension vector $n \cdot \delta$ (δ the minimal imaginary root of $\widehat{\mathfrak{gl}}_N$) satisfies the vertex bialgebra axiom (6.5) by Jindal–Kaubrys–Latyntsev [25], Theorem C for general quivers. The Schiffmann–Vasserot identification $\mathcal{H}_{\mathrm{Jor},N} \cong Y^+(\widehat{\mathfrak{gl}}_N)$ transports this to the Yangian, giving a second proof at all spins simultaneously.*

Proof. The proof follows the seven-step structure of Theorem 6.15, with the scalar R-matrix replaced by Yang's matrix R-matrix.

Step 1 (CoHA identification; Schiffmann–Vasserot [44]). The critical CoHA of the framed Jordan quiver at rank N satisfies $\mathcal{H}_{\mathrm{Jor},N} \cong Y^+(\widehat{\mathfrak{gl}}_N)$. The $N = 1$ case is Schiffmann–Vasserot [44]; the general case uses the Drinfeld–Sokolov hierarchy for $\widehat{\mathfrak{gl}}_N$.

Step 2 (Yangian–$\mathcal{W}_N$ identification; Procházka–Rapčák [39]). The quantum Miura transform gives an isomorphism of filtered associative algebras $Y(\widehat{\mathfrak{gl}}_N) \cong U(\mathcal{W}_N[\Psi])$. The transfer matrix (6.21) maps to the Miura factorisation (6.22). At $N = 2$: $\mathcal{W}_2 = \mathrm{Vir}_c$ with $c = 1 - 6(\Psi - 1)^2/\Psi$ and $\kappa(\mathrm{Vir}_c) = c/2$. At $N = 3$: $\mathcal{W}_3$ has generators at spins 2, 3 and $\kappa(\mathcal{W}_3) = 5c/6$ (where $H_3 - 1 = 5/6$).

Step 3 (R-matrix: Yang's rational R-matrix). The R-matrix (6.23) in the fundamental representation of $\mathfrak{gl}_N$ is:

$$R(u) = u\, I_{N^2} + \Psi\, P \qquad \text{(additive convention).}$$

The level prefix Ψ is mandatory: at $\Psi = 0$, $R(u) = u\, I$ (no interaction). The classical r-matrix is $r(z) = \Psi\, P/z$, with $r|_{\Psi=0} = 0$. The key properties, verified numerically at $N = 2, 3$ and proved algebraically for all N (Molev [37]):

(i) Permutation involution: $P^2 = I$.
(ii) Unitarity: $R(u)\, R(-u) = (\Psi^2 - u^2)\, I$.
(iii) Yang–Baxter equation (6.24).

At $N = 1$: $P = 1$ on $\mathbb{C}^1 \otimes \mathbb{C}^1$ and $R(u) = u + \Psi$ is scalar, so the RTT relation (6.26) is trivially satisfied. For $N \geq 2$: P is a genuine $N^2 \times N^2$ permutation matrix with eigenvalues ± 1, and the RTT is non-trivial.

Step 4 (Drinfeld coproduct at all spins). The matrix Drinfeld formula (6.25) gives $\Delta_z(T(u)) = T(u) \cdot T(u - z)$. In components, the coefficient of u^{-n} is:

$$\Delta_z(T^{(n)})_{ij} = \sum_{\substack{a,b \geq 0 \\ a+b+c=n}} \binom{b+c-1}{c} z^c \sum_{k=1}^{N} t_{ik}^{(a)} \otimes t_{kj}^{(b)},$$

where $c = n - a - b \geq 0$ and $\binom{-1}{0} = 1$ by convention. The matrix structure sums over the auxiliary index k, distinguishing this from the scalar ($N = 1$) formula (6.12).

Step 5 (RTT verification). The RTT relation (6.26) is a consequence of the Yang–Baxter equation (6.24) by evaluating the universal R-matrix in the fundamental representation. In the evaluation representation $T(u) \mapsto R(u-a)$, the RTT becomes the YBE with shifted spectral parameters. The RTT at $N \geq 2$ encodes genuine quadratic relations among the generators $t_{ij}^{(r)}$: the commutator $[R_{12}(u-v),\ T_1(u)\,T_2(v)]$ is generically nonzero (verified at $N = 2$: commutator norm = 1.0; $N = 3$: commutator norm = 1.0; $N = 1$: commutator = 0).

Step 6 (OPE compatibility at all spins: two arguments). The two independent proofs of OPE compatibility extend from $\mathcal{W}_{1+\infty}$ to $\mathcal{W}_N$:

Argument A. The coderivation property $d_{\overline{B}^{\mathrm{ord}}} \circ \Delta = \Delta \circ d_{\overline{B}^{\mathrm{ord}}}$ on the bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{W}_N[\Psi]) = T^c(s^{-1}\overline{\mathcal{W}}_N[\Psi])$ is a structural property of the cofree tensor coalgebra, valid at each truncation N. At generic Ψ, $\mathcal{W}_N[\Psi]$ lies on the Koszul locus, and the bar-cobar adjunction identifies the bar-cobar coproduct with the Drinfeld coproduct by uniqueness (Theorem 6.5).

Argument B. The JKL vertex bialgebra theorem [25] applies to the CoHA of the Jordan quiver at arbitrary rank N (their Theorem C holds for general quivers with potential). The vertex coproduct on $\mathcal{H}_{\mathrm{Jor},N} \cong Y^+(\widehat{\mathfrak{gl}}_N)$ satisfies (6.5) and identifies with the Drinfeld coproduct.

Step 7 (Quantum determinant and full axiom verification). The quantum determinant (6.27) is central in $Y(\mathfrak{gl}_N)$ (Molev [37], Theorem 1.11.2), verified numerically to be proportional to I_N in the evaluation representation at $N = 2, 3$. The coproduct satisfies coassociativity, the counit axiom, and OPE compatibility at all spins $1, \ldots, N$ by step 6. Together with the Yang R-matrix and the A_∞ structure from the bar complex, this gives the complete chiral quantum group datum for $\mathcal{W}_N[\Psi]$. □

Remark 6.27 (The $N = 1 \to N \geq 2$ transition). The passage from $N = 1$ (Theorem 6.15) to $N \geq 2$ (Theorem 6.26) has the following structural features.

(1) The R-matrix transitions from scalar ($R(u) = u + \Psi$, trivialising the RTT) to matrix-valued ($R(u) = u\,I + \Psi\,P$, making the RTT a genuine constraint). This is the non-abelian threshold.
(2) The transfer matrix transitions from the scalar series $T(u) = 1 + \sum \psi_n u^{-n}$ to the $N \times N$ matrix $T(u) = I_N + \sum T^{(r)} u^{-r}$. The Drinfeld coproduct $\Delta_z(T(u)) = T(u) \cdot T(u-z)$ is matrix multiplication in $\mathbb{C}^N$, not scalar multiplication.
(3) The quantum determinant $\operatorname{qdet} T(u)$ is a genuinely new central element for $N \geq 2$ (at $N = 1$ it reduces to $T(u)$ itself).
(4) Both arguments for OPE compatibility (coderivation on the Koszul locus, JKL on the CoHA) apply uniformly for all N: argument A uses only the structural properties of the cofree tensor coalgebra $T^c(s^{-1}\overline{\mathcal{W}}_N)$ and the Koszul-locus equivalence; argument B uses JKL for the Jordan quiver at rank N.

6.6. **Structural consequences.**

Corollary 6.28 (The ordered bar complex encodes all three structures). *The ordered bar complex* $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$*, equipped with its bar differential* $d_{\overline{B}^{\mathrm{ord}}}$ *and deconcatenation coproduct* Δ*, is the universal object from which all three structures in Theorem* 6.5 *are recovered:*

(i) *The bar differential* $d_{\overline{B}^{\mathrm{ord}}}$ *encodes the* A_∞ *structure maps* m_k^{ch} *(by the standard bar-cobar correspondence: coderivations of* $T^c(s^{-1}\bar{\mathcal{A}})$ *biject with elements of* $\prod_k \mathrm{End}^{\mathrm{ch}}_{\mathcal{A}}(k)$*).*

(ii) *The deconcatenation coproduct Δ induces, via cobar dualization, the chiral coproduct Δ^{ch}.*

(iii) *The vertex R-matrix $S(z)$ is recovered as the monodromy of the KZ connection, which is itself the degree-2 projection of the Maurer–Cartan element $\Theta_{\mathcal{A}}$ in $\mathfrak{g}^{\mathsf{E}_1}_{\mathcal{A}}$.*

This is the structural content of the E_1 primacy principle: the ordered bar complex with deconcatenation is the primitive object; the R-matrix, the chiral coproduct, and the A_∞ operations are projections of this single datum.

Remark 6.29 (Relation to factorization quantum groups). When the hexagon holds strictly, the chiral quantum group $(\mathcal{A}, S(z), \Delta^{\mathrm{ch}})$ is a *factorization quantum group* (Latyntsev [31]): its ordered bar complex defines a factorization coalgebra on the curve. When the hexagon holds only up to Φ, one obtains a *quasi-factorization quantum group* with regularized holonomy data (R, Φ).

7. Topological Hochschild versus ordered chiral homology of $D^\times$

Two chain complexes compute invariants of a chiral algebra on a punctured disk: the cyclic bar complex (topological Hochschild, from the circle $S^1 = \partial D$) and the ordered chiral chain complex (from the punctured disk $D^\times$ itself, with its complex structure). For E_∞-chiral algebras, operad formality identifies the two. For E_1-chiral algebras, they diverge: the ordered complex carries holomorphic data from the spectral parameter that the cyclic bar complex cannot see.

7.1. Two computations.

Let $\mathcal{A}$ be a chiral algebra on a curve X. Fix a point $p \in X$ and let $D = \operatorname{Spec} \mathbb{C}[[z]]$ be the formal disk at p, with $D^\times = \operatorname{Spec} \mathbb{C}((z))$ the punctured formal disk. The two chain complexes are the following.

Definition 7.1 (Topological Hochschild). The *topological Hochschild homology* $\mathrm{HH}^{\mathrm{top}}_*(\mathcal{A}) := \int_{S^1} \mathcal{A}$ is computed by the cyclic bar complex with the topological Hochschild differential and A_∞ higher operations. Input: the E_1-topological structure (m_2, m_k). No z-dependence; no Arnold relations.

Definition 7.2 (Ordered chiral homology of $D^\times$). The *ordered chiral chain complex on $D^\times$ at degree n* is $C^{\mathrm{ord}}_n(D^\times, \mathcal{A}) := (\Omega^*_{\log}(\overline{\mathrm{FM}}^{\mathrm{ord}}_n(D^\times)) \otimes (s^{-1}\bar{\mathcal{A}})^{\otimes n},\ d_{\mathrm{dR}} + d_{\mathrm{bar}} - \nabla_{\mathrm{KZ}})$, where d_{bar} encodes the full OPE (extracting residues at FM boundary strata via the Arnold forms $\omega_{ij} = d\log(z_i - z_j)$, which are coefficients in d_{bar}, not in ∇_{KZ}) and $\nabla_{\mathrm{KZ}} = \sum_{i<j} r_{ij}(z_i - z_j)\, d(z_i - z_j)$. The *ordered chiral homology of $D^\times$* is the derived pushforward over $\mathrm{Ran}^{\mathrm{ord}}(D^\times)$; the degree-by-degree computation gives:

$$\int_{D^\times}^{\mathrm{ord}} \mathcal{A} = R\Gamma_{\mathrm{dR}}\big(\mathrm{Ran}^{\mathrm{ord}}(D^\times),\ \mathcal{F}^{\mathrm{ord}}(\mathcal{A})\big) = \bigoplus_{n\geq 0} H^*\big(C^{\mathrm{ord}}_n(D^\times, \mathcal{A})\big).$$

The input data: the full OPE $Y(a,z)b = \sum_k a_{(k)} b\, z^{-k-1}$, the Arnold forms on $\mathrm{Conf}^{\mathrm{ord}}_n(D^\times)$, the FM compactification, and the r-matrix. The output depends on the complex structure of $D^\times$ (the coordinate z).

The comparison map $\int^{\mathrm{ord}}_{D^\times} \mathcal{A} \to \mathrm{HH}^{\mathrm{top}}_*(\mathcal{A})$ is induced by the boundary retraction $D^\times \to S^1$, forgetting the complex structure. It is a quasi-isomorphism for E_∞-chiral algebras (Theorem 7.4) and fails for genuinely E_1-chiral algebras (Theorem 7.5).

7.2. Four levels of formality.

Definition 7.3 (Formality hierarchy). Four conditions, each strictly stronger, control the ordered/symmetric comparison: (a) operad formality (Kontsevich–Tamarkin; unconditional); (b) algebra formality ($m^{\mathrm{tr}}_k = 0$ for $k \geq 3$; class G only); (c) convolution formality ($\ell_k = 0$ for $k \geq 3$ in $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$; class G); (d) SC-formality ($\Leftrightarrow$ class G). Only (a) holds unconditionally for E_∞-chiral algebras; it guarantees that the ordered and symmetric complexes compute the same cohomology, but the transferred A_∞-structure may be nontrivial for class $L/C/M$.

Theorem 7.4 (Formality bridge for E_∞-chiral algebras)**.** *Let $\mathcal{A}$ be an E_∞-chiral algebra and Φ a Drinfeld associator. The Kontsevich–Tamarkin operad formality quasi-isomorphism induces a quasi-isomorphism of chain complexes:*

$$C_n^{\mathrm{ord}}(D^\times, \mathcal{A}) \xrightarrow[\Phi]{\sim} B_n^{\mathrm{cyc}}(\mathcal{A}). \tag{7.1}$$

In particular, $\int_{D^\times}^{\mathrm{ord}} \mathcal{A} \simeq \mathrm{HH}_*^{\mathrm{top}}(\mathcal{A})$ as graded vector spaces.

The transferred A_∞-structure on the cohomological model $H^*(\mathcal{A})$ may still carry non-trivial higher operations m_k^{tr} for $k \geq 3$. For class L (affine KM), class C ($\beta\gamma$), and class M (Virasoro): these transferred operations are non-trivial. What operad formality guarantees is that the ordered and symmetric complexes compute the same cohomology.

Proof sketch; full proof in [34, §I.3]. The argument has six steps. (1) Kontsevich–Tamarkin operad formality gives a Σ_n-equivariant quasi-isomorphism $\varphi_\Phi\colon C_*(\overline{\mathrm{FM}}_n(\mathbb{C})) \xrightarrow{\sim} H^*(\overline{\mathrm{FM}}_n(\mathbb{C}))$ depending on a Drinfeld associator Φ. (2) Tensoring with id on $(s^{-1}\bar{\mathcal{A}})^{\otimes n}$ gives a chain map between the ordered chiral complex and the Arnold model. (3) The transferred bar differential reduces to $k = 0$ alone: only the leading OPE mode $a_{(0)}b = m_2(a, b)$ pairs with the Arnold generator $\omega_{ij} \in H^1$; higher modes $a_{(k)}b$ ($k \geq 1$) pair with boundary forms exact in $H^*(\overline{\mathrm{FM}}_n^{\mathrm{ord}})$. (4) The result is the topological Hochschild differential d_{HH}, giving the quasi-isomorphism $C_n^{\mathrm{ord}}(D^\times, \mathcal{A}) \xrightarrow{\sim} B_n^{\mathrm{cyc}}(\mathcal{A})$. (5) The transferred A_∞-structure $\{m_k^{\mathrm{tr}}\}$ on $H^*(\mathcal{A})$ may be nontrivial for class $L/C/M$ but does not affect the chain-level comparison. (6) For genuinely E_1-chiral $\mathcal{A}$, $S(z) \neq \mathrm{id}$ breaks Σ_n-equivariance of d_{bar}, preventing the transfer. □

Theorem 7.5 (Formality failure for genuinely E_1-chiral algebras)**.** *Let $\mathcal{A}$ be a genuinely E_1-chiral algebra* (*tier (c): the R-matrix $S(z)$ is not derived from any E_∞-chiral OPE; $S(z) \neq \mathrm{id}$*)*. The chain-level formality quasi-isomorphism* (7.1) *does* not *exist.*

- (i) (Operad formality holds.) *The Kontsevich–Tamarkin formality of $C_*(\overline{\mathrm{FM}}_n(\mathbb{C}))$ as an operad is unconditional: it depends only on the operad structure, not on the specific algebra.*
- (ii) (Equivariance fails.) *The $\mathcal{D}$-module $\mathcal{F}_n^{\mathrm{ord}}$ is not Σ_n-equivariant* (*because $S(z) \neq \mathrm{id}$*)*, so the ordered complex does* not *descend to the symmetric quotient. The ordered chiral homology is strictly richer than topological Hochschild: the ordered complex carries holomorphic data from the spectral parameter dependence of $S(z)$ that has no counterpart in the cyclic bar complex.*
- (iii) (Dimension count.) $\dim C_n^{\mathrm{ord}}(D^\times, \mathcal{A}) = (\dim \bar{\mathcal{A}})^n \cdot n!$ *(Arnold's Poincaré polynomial $\prod_{j=0}^{n-1}(1 + jt)$ evaluates to $n!$ at $t = 1$), while* $\dim B_n^{\mathrm{cyc}} = (\dim \mathcal{A})^n$.

8. The E_3 structure and Chern–Simons theory

The symmetric bar $\overline{B}^{\Sigma}(\mathcal{A})$ carries the E_2 structure of unordered configurations. Its chiral Hochschild cochains form the derived chiral centre $Z_{\mathrm{ch}}^{\mathrm{der}}(\mathcal{A})$; together with the boundary algebra $\mathcal{A}$ this is the universal $\mathrm{SC}^{\mathrm{ch,top}}$ pair. For affine KM at non-critical level, the Sugawara element topologises BRST cohomology to $\mathsf{E}_3^{\mathrm{top}}$. The full development appears in [34, §I.4, §IV]; we state the results needed for this paper.

8.1. The derived chiral centre.

The ordered and symmetric chiral Hochschild cochain complexes differ by the presence or absence of Σ_k-coinvariants. The derived chiral centre is the cohomology of the symmetric version; its ordered refinement carries the full E_1 data.

Definition 8.1 (Symmetric and ordered chiral chiral Hochschild complexes)**.** Let $\mathcal{A}$ be a chiral algebra on a curve X.

(i) The *symmetric chiral Hochschild cochain complex* is:

$$\mathrm{HH}^{*,\Sigma}_{\mathrm{ch}}(\mathcal{A},\mathcal{A}) = \bigoplus_{k\geq 0} \Omega^*(\overline{\mathrm{FM}}_k(\mathbb{C})) \otimes_{\Sigma_k} \mathrm{Hom}(\mathcal{A}^{\otimes k},\mathcal{A}),$$

with the chiral Hochschild differential. The derived chiral centre is $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) := \mathrm{HH}^{*,\Sigma}_{\mathrm{ch}}(\mathcal{A},\mathcal{A})$.

(ii) The *ordered chiral Hochschild cochain complex* is:

$$\mathrm{HH}^{*,\mathrm{ord}}_{\mathrm{ch}}(\mathcal{A},\mathcal{A}) = \bigoplus_{k\geq 0} \Omega^*(\overline{\mathrm{FM}}^{\mathrm{ord}}_k(\mathbb{C})) \otimes \mathrm{Hom}(\mathcal{A}^{\otimes k},\mathcal{A}),$$

without Σ_k-coinvariants, with the ordered chiral chiral Hochschild differential.

Proposition 8.2 (Derived centre, Swiss-cheese package, and affine topologization)**.** *Let $\mathcal{A}$ be a chiral algebra on a smooth algebraic curve X.*

(i) (Source of the E_2 structure.) *The curve geometry provides the E_2-coalgebra structure on $\overline{B}^{\Sigma}(\mathcal{A})$ (Warning* 3.5*).*

(ii) (Derived centre and open/closed package.) *Dualising the E_2-coalgebra $\overline{B}^{\Sigma}(\mathcal{A})$ to an algebra (which requires the finiteness conditions satisfied by standard chiral algebras, see* [34, Thm A]*), the derived centre*

$$Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = \mathrm{HH}^*(\overline{B}^{\Sigma}(\mathcal{A}), \overline{B}^{\Sigma}(\mathcal{A})) \quad \textit{is the universal closed-sector } \mathsf{E}_2\textit{-algebra.}$$

Together with the boundary algebra $\mathcal{A}$, it forms the universal $\mathrm{SC}^{\mathrm{ch,top}}$ *pair; a physical bulk is obtained only after an open–closed comparison with a field theory.*

(iii) (Scope and topologization boundary.) *For E_∞-chiral algebras, the symmetric bar $\overline{B}^{\Sigma}(\mathcal{A})$ exists as a well-defined E_2-coalgebra and the derived-centre package of* (i)–(ii) *is available. If, in addition, $\mathcal{A}$ admits an inner conformal-vector datum whose antighost contraction makes holomorphic translations Q-exact, the affine topologization theorem upgrades BRST cohomology of the derived centre to $\mathsf{E}_3^{\mathrm{top}}$. For genuinely E_1-chiral algebras (Yangians, EK quantum vertex algebras), the factorisation $\mathcal{D}$-module does not descend to $X^{(n)}$ (Theorem* 7.5*), so $\overline{B}^{\Sigma}(\mathcal{A})$ does not exist. The ordered bar $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ carries an E_1-coalgebra structure, giving only E_2 on $\mathrm{HH}^*(\overline{B}^{\mathrm{ord}}(\mathcal{A}), \overline{B}^{\mathrm{ord}}(\mathcal{A}))$ via the classical Deligne conjecture ($\mathsf{E}_1 + 1 = \mathsf{E}_2$). The passage from boundary data to bulk for E_1-chiral algebras requires the Drinfeld centre construction (see* [36, conj:drinfeld-center-equals-bulk]*).*

Remark 8.3 (Bulk E_2 versus affine $\mathsf{E}_3^{\mathrm{top}}$). Two different structures appear on the affine derived centre at non-critical level.

(a) *Algebraic bulk* E_2 (Proposition 8.2): the chiral Hochschild cohomology of the E_2-coalgebra $\overline{B}^{\Sigma}(V_k(\mathfrak{g}))$, via the centre formalism. This requires dualisation of the coalgebra to an algebra, which uses the finiteness of $V_k(\mathfrak{g})$ (finite-type as a graded vector space in each conformal-weight stratum). No conformal vector is needed.

(b) *Topological* $\mathsf{E}_3^{\mathrm{top}}$ (Proposition 8.11): the Sugawara Virasoro element at non-critical level topologises the holomorphic direction of the affine bulk, producing a topological E_3-algebra on $\mathbb{R}^3 \cong X \times \mathbb{R}$ after passing to BRST cohomology, with an unconditional chain-level model on a quasi-isomorphic complex. This requires the conformal vector and fails at $k = -h^\vee$.

Structure (a) is algebraic and holomorphic; structure (b) is geometric-topological. Theorem 8.6 compares their deformation parameter spaces and first nontrivial operations; full family identification remains open.

Warning 8.4 (The bar complex itself is not E_3)**.** The symmetric bar $\overline{B}^{\Sigma}(\mathcal{A})$ does *not* carry an E_3-algebra structure. The obstruction is topological: E_3 requires $\mathrm{Conf}_k(\mathbb{R}^3)$, but a curve provides only $\mathrm{Conf}_k(\mathbb{R}^2)$. For $\widehat{\mathfrak{sl}}_2$, the obstruction space $\mathrm{HH}^3_{\mathsf{E}_2}(\overline{B}^{\Sigma}, \overline{B}^{\Sigma})$ is one-dimensional and the obstruction class is non-zero. The E_3-algebra is the *derived centre*, not the bar complex.

8.2. **The CFG comparison.** Costello, Francis, and Gwilliam construct a filtered E_3-algebra $\mathcal{A}^{\lambda}$ from BV quantisation of Chern–Simons theory on $\mathbb{R}^3$. The derived chiral centre of $V_k(\mathfrak{g})$ and the CFG algebra both deform $C^*(\mathfrak{g})$ over $(k+h^\vee)H^3(\mathfrak{g})[[k+h^\vee]]$. The question is whether these deformation families are isomorphic as E_3-algebras.

Theorem 8.5 (Costello–Francis–Gwilliam [12])**.** *Let $\mathfrak{g}$ be a simple finite-dimensional Lie algebra. BV quantisation of Chern–Simons theory on $\mathbb{R}^3$ with gauge algebra $\mathfrak{g}$ and invariant pairing λ yields a filtered E_3-algebra $\mathcal{A}^{\lambda}$ on the Chevalley–Eilenberg cochains $C^*(\mathfrak{g})$.*

(i) *The classical observables are $C^*(\mathfrak{g}) = \mathrm{Sym}(\mathfrak{g}^\vee[-1])$ with the CE differential: a commutative (E_∞) algebra.*

(ii) *The quantum observables $\mathcal{A}^{\lambda}$ are a filtered E_3-algebra deforming $C^*(\mathfrak{g})$, with deformation space $(k+h^\vee)H^3(\mathfrak{g})[[k+h^\vee]]$.*

(iii) *For any E_3-algebra $\mathcal{A}$, $\mathrm{Mod}_{\mathcal{A}}$ is E_2-monoidal (braided monoidal). Filtered Koszul duality gives $\mathrm{Perf}_{C^*(\mathfrak{g})} \simeq \mathrm{Rep}_{\mathrm{fin}}(\mathfrak{g})^{\mathrm{dg}}$, so deforming $C^*(\mathfrak{g})$ as E_3 deforms $\mathrm{Rep}_{\mathrm{fin}}(\mathfrak{g})$ as braided monoidal: this recovers the Drinfeld–Jimbo quantum group.*

(iv) (Theorem 1.4 of [12].) *Canonical bijection between: perturbative CS quantisations, filtered E_3-algebras deforming $C^*(\mathfrak{g})$, braided monoidal deformations of $\mathrm{Rep}_{\mathrm{fin}}(\mathfrak{g})$, and quasi-triangular quasi-Hopf algebras deforming $U(\mathfrak{g})$. All non-canonically isomorphic to $(k+h^\vee)H^3(\mathfrak{g})[[k+h^\vee]]$.*

Theorem 8.6 (Matching deformation spaces and first nontrivial bulk data)**.** *Let $\mathfrak{g}$ be a simple finite-dimensional Lie algebra and $V_k(\mathfrak{g})$ the affine Kac–Moody vertex algebra at level k. Writing $\lambda = k + h^\vee$ for the departure from critical level, the affine derived-centre package and the CFG bulk package live over the same formal base*

$$\lambda H^3(\mathfrak{g})[[\lambda]].$$

(i) (Matching deformation spaces.) *The affine topologized bulk and the Costello–Francis–Gwilliam bulk are both governed by $(k+h^\vee)H^3(\mathfrak{g})[[k+h^\vee]]$: for the affine side the deformation parameter is the departure from critical level, while for the CFG side it is the Chern–Simons coupling.*

(ii) (Base point mismatch.) *At the critical level $k=-h^\vee$, the affine derived centre is related to the Feigin–Frenkel centre $\mathrm{Fun}(\mathrm{Op}_{\mathfrak{g}^\vee}(D))$, not directly to $C^*(\mathfrak{g})$. Any comparison with the CFG package is therefore perturbative, in the formal neighbourhood of critical level.*

(iii) (Formal-disk comparison.) *By Theorem* 8.10*, formal disk restriction matches the associated graded and the P_3 bracket of the affine bulk package with those of the CFG bulk package.*

(iv) (Strongest proved identification.) *For simple $\mathfrak{g}$, $H^3(\mathfrak{g}) \cong \mathbb{C}$ is one-dimensional, so part* (iii) *identifies the first nontrivial deformation direction at each order. What remains open is a chain-level equivalence of the full deformation families as $\mathsf{E}_3^{\mathrm{top}}$-algebras; that step requires a compatibility theorem beyond the present associated-graded and P_3 comparison.*

Proof. Part (i) is Theorem 1.3(iii) on the affine side and Theorem 8.5(ii), (iv) on the CFG side. Part (ii) is the critical level boundary already recorded in Theorem 1.3(v). Part (iii) is exactly the content of Theorem 8.10.

For part (iv), the one-dimensionality of $H^3(\mathfrak{g})$ for simple $\mathfrak{g}$ shows that the formal moduli problem has a single tangent direction at each order. The associated-graded and P_3 comparison therefore pins down the first nontrivial bulk data, but it does not by itself construct an $\mathsf{E}_3^{\mathrm{top}}$ equivalence of the full families. That remaining step is the open chain-level identification problem. □

8.3. **Explicit E_3 operations, topologization, and Chern–Simons comparison.** For $\mathfrak{g} = \mathfrak{sl}_2$ at generic level $k \neq -2$, the derived chiral centre is the graded vector space

$$Z^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{sl}_2)) \;=\; \underbrace{\mathbb{C}}_{=\mathrm{HH}^0} \;\oplus\; \underbrace{\mathfrak{sl}_2[-1]}_{=\mathrm{HH}^1} \;\oplus\; \underbrace{\mathbb{C}[-2]}_{=\mathrm{HH}^2},$$

with total dimension 5. Write $\mathbf{1}$ for the generator of $\mathrm{HH}^0 = \mathbb{C}$, $\{e, f, h\}$ for the standard basis of $\mathrm{HH}^1 = \mathfrak{sl}_2$ (outer derivations: infinitesimal level-preserving deformations parametrised by $\mathfrak{g}$ itself), and η for the generator of $\mathrm{HH}^2 = \mathbb{C}$ (the central charge deformation cocycle). The $\mathfrak{sl}_2$-module structure is: HH^0 trivial, HH^1 adjoint, HH^2 trivial.

The KZ coupling constant is $h_{\mathrm{KZ}} = 1/(k + h^\vee) = 1/(k + 2)$.

Remark 8.7 (Convention for). Throughout this paper, $= h_{\mathrm{KZ}} = 1/(k + h^\vee)$ is the KZ coupling constant: $\to 0$ at weak coupling ($k \to \infty$), diverges at critical level $k = -h^\vee$. The quantum group parameter is $q = e^{2\pi i}$.

An E_3-algebra carries three levels of operations: the graded-commutative product μ (E_∞), the Gerstenhaber bracket $[-,-]$ of degree -1 (E_2, from $S^1 = \mathrm{Conf}_2(\mathbb{R}^2)/\mathbb{R}^+$), and the P_3 bracket $\{-,-\}$ of degree -2 (E_3, from $S^2 = \mathrm{Conf}_2(\mathbb{R}^3)/\mathbb{R}^+$).

Proposition 8.8 (Explicit E_3 operations on $Z^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{sl}_2))$)**.** *Let $k \neq -2$. The E_3-algebra structure on $Z^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{sl}_2))$ is determined by the following data.*

(I) Cup product *(E_∞ level).*

(a) $\mathbf{1}$ *is the unit:* $\mathbf{1} \cdot f = f \cdot \mathbf{1} = f$ *for all* f.
(b) $\mu(X, Y) = 0$ *for all* $X, Y \in \mathrm{HH}^1 = \mathfrak{sl}_2$.
(c) $\mu(X, \eta) = 0$ *for all* $X \in \mathrm{HH}^1$, *and* $\eta \cdot \eta = 0$.

(II) Gerstenhaber bracket *(E_2 level, degree -1).*

$$[\mathbf{1}, -] = 0, \qquad [X, Y] = 0 \quad \textit{for all } X, Y \in \mathrm{HH}^1 = \mathfrak{sl}_2,$$
$$[X, \eta] = 0 \quad \textit{for all } X \in \mathrm{HH}^1, \qquad [\eta, \eta] = 0.$$

(III) The P_3 bracket *(E_3 level, degree -2).*

$$\begin{aligned} \{X, Y\} &= h_{\mathrm{KZ}} \cdot (X, Y) = \frac{1}{k + h^\vee}\,(X, Y) \quad \textit{for } X, Y \in \mathrm{HH}^1, \\ \{X, \eta\} &= 0 \quad \textit{for } X \in \mathrm{HH}^1, \\ \{\mathbf{1}, -\} &= 0, \\ \{\eta, \eta\} &= 0, \end{aligned}$$

where (X, Y) *denotes the normalised Killing form and* $h_{\mathrm{KZ}} = 1/(k + h^\vee) = 1/(k + 2)$ *is the KZ coupling constant* (*Remark* 8.7).

Proof sketch. The cup product vanishes on $\mathrm{HH}^1 \otimes \mathrm{HH}^1$ because $\Lambda^2(\mathfrak{sl}_2)^{\mathfrak{sl}_2} = 0$ (the exterior square $\Lambda^2(\mathrm{ad}) \cong \mathrm{ad}$ has no trivial summand). The BV operator $\Delta = 0$ on the entire derived centre (no equivariant maps between trivial and adjoint), so the BV relation

$$[f, g] = \Delta(f \cdot g) - (\Delta f) \cdot g - (-1)^{|f|} f \cdot (\Delta g).$$

forces $[X, Y] = 0$ for all $X, Y \in \mathrm{HH}^1$. The P_3 bracket is determined by $\mathfrak{sl}_2$-equivariance: $\{X, Y\} \in \mathrm{HH}^0 = \mathbb{C}$ must be an invariant symmetric form on the adjoint, hence proportional to the Killing form; the coefficient is fixed by the r-matrix residue $\mathrm{Res}_{z=0}[r(z)] = \Omega/(k + h^\vee)$ (KZ normalisation), giving $\alpha = h_{\mathrm{KZ}}$. See [34, §IV.3] for the full proof. □

The quantum corrections from the full vertex R-matrix $S(z)$ of the EK quantum VOA (Example 6.13) do not alter the P_3 bracket on cohomology: the quantum-corrected flat sections shift subleading terms but preserve the $\ker(\mathrm{av}_2)$ dimension and the scalar shadow $\kappa = 3(k+2)/4$.

The chiral P_3 bracket and the filtered affine bulk package. The PVA λ-bracket on $V_k(\mathfrak{g})$, $\{J^a\ {}_\lambda\ J^b\} = f^{ab}{}_c J^c + k\,(a, b)\,\lambda$, induces a degree-$(-2)$ bracket on the chiral Chevalley–Eilenberg complex $\mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_k) = (\mathrm{Sym}^c(\mathfrak{g}^*[-1]) \otimes \omega_X,\ d_{\mathrm{CE}},\ m_0 = \frac{k+h^\vee}{2h^\vee}\,\kappa)$:

$$\{\phi_a, \phi_b\}^{\mathrm{ch}} \;=\; f^{ab}{}_c\,\phi_c \;+\; k\,(a, b)\,\partial.$$

This equips $\mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_k)$ with the structure of a P_3-algebra in $\mathcal{D}$-modules on X (see [34, §IV.4]).

Definition 8.9 (The filtered affine bulk package)**.** The *affine bulk package* $\mathrm{CE}^{\mathrm{ch}}_k(\mathfrak{g}) := (\mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_k),\ \{-,-\}^{\mathrm{ch}},\ \mathcal{D}\text{-module structure})$ carries a $(k+h^\vee)$-adic filtration with associated graded the uncurved complex $\mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_{-h^\vee})$. It is a filtered factorisation $\mathcal{D}$-module with P_3 operation on the curve; $\mathsf{E}_3^{\mathrm{top}}$ enters only after affine topologization of the derived centre.

Theorem 8.10 (Formal disk restriction recovers CFG)**.** *Restricting* $\mathrm{CE}^{\mathrm{ch}}_k(\mathfrak{g})$ *to the formal disk* $D = \mathrm{Spec}\ \mathbb{C}[[z]]$ *gives a map of filtered* E_3*-algebras* $\Gamma(D,\ \mathrm{CE}^{\mathrm{ch}}_k(\mathfrak{g})) \to \mathcal{A}^\lambda$ *to the CFG* E_3*-algebra (Theorem* 8.5*), with* $\lambda = k + h^\vee$*. This map is an isomorphism on associated graded algebras and matches the* P_3 *brackets. For simple* $\mathfrak{g}$ *(where* $H^3(\mathfrak{g}) \cong \mathbb{C}$*), order-by-order uniqueness identifies the first nontrivial deformation direction at each order; full chain-level* $\mathsf{E}_3^{\mathrm{top}}$ *equivalence of the deformation families remains open.*

Proof. On the associated graded, both sides reduce to $C^*(\mathfrak{g}) = \mathrm{Sym}(\mathfrak{g}^\vee[-1])$ with the CE differential. The P_3 brackets match: the chiral bracket $\{\phi_a, \phi_b\}^{\mathrm{ch}} = f^{ab}{}_c\,\phi_c + k(a, b)\,\partial$ restricts on D to the CFG bracket with the derivation ∂ becoming the zero-mode δ_0, giving $\{X, Y\} = h_{\mathrm{KZ}}\,(X, Y)$ on cohomology. The KZ connection $\nabla = d + \sum k\,\Omega_{ij}/(z_i - z_j)\,dz_{ij}$ trivialises on D, so the $\mathcal{D}$-module becomes a vector space. The one-dimensionality of $H^3(\mathfrak{g})$ for simple $\mathfrak{g}$ forces order-by-order agreement by the deformation theory of E_3-algebras. □

Proposition 8.11 (Topological enhancement via Sugawara)**.** *At non-critical level* $k \neq -h^\vee$*, the Sugawara element* $T_{\mathrm{Sug}} = \frac{1}{2(k+h^\vee)} \sum_a :J^a J_a:$ *has an antighost contraction making holomorphic translations BRST-exact:* $[Q, b] = L_{-1}$*. By Khan–Zeng [28], this upgrades the holomorphic-topological symmetry to fully topological* $\mathsf{E}_3^{\mathrm{top}}$*. At* $k = -h^\vee$*, the Sugawara element is undefined and the enhancement fails; perturbatively it holds order-by-order in* $(k + h^\vee)$*.*

Proof. The Sugawara construction (Frenkel–Ben-Zvi [20]) produces an exact Virasoro element at $k \neq -h^\vee$ satisfying $L_{-1} \cdot J^a = \partial J^a$. The BRST relation $[Q, b] = L_{-1}$ gives BRST-exactness of holomorphic translation, the datum Khan–Zeng [28] requires. □

Remark 8.12 (Two E_3 structures). The algebraic bulk E_2 (chiral Hochschild cohomology of the E_2-coalgebra $\overline{B}^{\Sigma}$, no conformal vector needed) and the topological $\mathsf{E}_3^{\mathrm{top}}$ (Sugawara topologization, conformal vector at non-critical level) are distinct; Theorem 8.6 compares their deformation parameter spaces. Full family identification remains open.

Warning 8.13 (The bar complex itself is not E_3)**.** The symmetric bar $\overline{B}^{\Sigma}(\mathcal{A})$ does not carry an E_3 structure: E_3 requires $\mathrm{Conf}_k(\mathbb{R}^3)$, but a curve provides only $\mathrm{Conf}_k(\mathbb{R}^2)$. The E_3-algebra is the derived centre, not the bar complex.

The remaining subsections of the E_3 analysis (quantum corrections from the vertex R-matrix, the chiral P_3 bracket from the PVA λ-bracket, the filtered affine bulk package, the $\mathcal{D}$-module structure, and the Khan–Zeng topological enhancement) are developed in full in [34, §IV.3–IV.7].

9. Examples

The four shadow classes G, L, C, M produce qualitatively different ordered chiral homology theories. Class G (Heisenberg) has regular singularities, finite shadow tower, and trivial kernel $\ker(\mathrm{av})$. Class L (affine Kac–Moody) has regular singularities but a non-trivial R-matrix kernel. Class C ($\beta\gamma$ system) has regular singularities with nilpotent monodromy (the r-matrix is a constant nilpotent after d log-absorption; pole order 0). Class M (Virasoro) has irregular singularities of unbounded rank and an infinite shadow tower.

Example 9.1 (Heisenberg: class G, $r = 2$)**.** The Heisenberg algebra $\mathcal{H}_k$: $J(z)J(w) \sim k/(z-w)^2$, $r(z) = k/z$, $\kappa = k$. Class G, E_∞-chiral; formality bridge gives ordered = symmetric. The ordered chiral Hochschild cohomology on D:

$$\mathrm{HH}^{n,\mathrm{ord}}_{\mathrm{ch}}(\mathcal{H}_k, \mathcal{H}_k) = \begin{cases} \mathbb{C} & n = 0 \text{ (vacuum)}, \\ \mathbb{C} & n = 1 \text{ (outer derivation)}, \\ \mathbb{C} & n = 2 \text{ (level deformation)}, \\ 0 & n \geq 3 \text{ (Koszul length 2)}, \end{cases}$$

with Poincaré polynomial $1 + t + t^2$, total dimension 3, amplitude $[0, 2]$. The KZ connection $\nabla = d - k\,dw/w$ has scalar monodromy $M = e^{2\pi i k}$; regular singularities. The E_3 obstruction vanishes ($\Lambda^3(\mathbb{C}^1) = 0$): trivial E_3 structure (abelian CS). Ordered = topological Hochschild by formality.

Example 9.2 (Affine Kac–Moody: class L, $r = 3$; explicit computation for $\widehat{\mathfrak{sl}}_2$)**.** We carry out the ordered chiral Hochschild computation for $V_k(\mathfrak{sl}_2)$ on the formal disk, the simplest non-abelian Kac–Moody algebra.

Data. $\mathfrak{g} = \mathfrak{sl}_2$, $h^\vee = 2$, $\dim(\mathfrak{g}) = 3$. Generators $\{e, f, h\}$ with brackets $[e, f] = h$, $[h, e] = 2e$, $[h, f] = -2f$. Normalised invariant form: $(e, f) = (f, e) = 1$, $(h, h) = 2$. Inverse Casimir: $\Omega = e \otimes f + f \otimes e + \frac{1}{2} h \otimes h$. The Koszul invariant is

$$\kappa = 3(k+2)/4. \tag{9.1}$$

The r-matrix in trace-form convention is $r(z) = k\Omega/z$ (at $k = 0$, $r = 0$); in KZ normalisation, $r(z) = \Omega/((k+2)z)$. Class L (shadow depth $r = 3$): ℓ_1, ℓ_2, ℓ_3 non-trivial, $\ell_k = 0$ for $k \geq 4$.

Degree 2: KZ conformal blocks. The ordered chiral Hochschild at degree 2 is the de Rham cohomology of $\mathrm{Conf}_2(\mathbb{C})$ with coefficients in the KZ local system. The KZ equation at degree 2, setting $z = z_1 - z_2$, is

$$\frac{\partial \Phi}{\partial z} = \frac{\Omega}{(k+2)\,z}\,\Phi. \tag{9.2}$$

Since $V_k(\mathfrak{sl}_2)$ is E_∞-chiral, the coefficients live in $V_k(\mathfrak{sl}_2)^{\otimes 2}$, which decomposes under $\mathfrak{sl}_2$ by Clebsch–Gordan. For the adjoint representation acting on itself ($j = 1$): $V_1 \otimes V_1 = V_0 \oplus V_1 \oplus V_2$. The Casimir eigenvalue of Ω on the isospin-J component is

$$\lambda_J = \frac{C_2(J) - C_2(1) - C_2(1)}{2} = \frac{J(J+1) - 4}{2},$$

where $C_2(j) = j(j+1)$ in the normalisation $(\theta|\theta) = 2$. The eigenvalues are:

$$\lambda_0 = -2, \quad \lambda_1 = -1, \quad \lambda_2 = 1.$$

On each isospin component, the KZ equation (9.2) has the explicit solution $\Phi_J(z) = z^{\lambda_J/(k+2)}$. The monodromy around $z = 0$ (the diagonal $\Delta_{12} \subset X^2$) is

$$M_J = \exp\bigl(2\pi i\, \lambda_J/(k+2)\bigr).$$

The de Rham cohomology $H^*_{\mathrm{dR}}(\mathrm{Conf}_2(\mathbb{C}), \nabla_{\mathrm{KZ}})$ decomposes as:

- $H^0 = \ker(M - \mathrm{id})$: the subspace on which the monodromy is trivial. At generic k, this is zero (no isospin channel has $\lambda_J/(k+2) \in \mathbb{Z}$).
- $H^1 = \mathrm{coker}(M - \mathrm{id})$: the space of KZ conformal blocks at degree 2. At generic k, $H^1 \cong V_0 \oplus V_1 \oplus V_2$ (three-dimensional over the isospin decomposition: one block per channel).

The averaging map $\mathrm{av}_2 \colon H^1_{\mathrm{ord}} \to H^1_{\Sigma}$ projects each conformal block to its Σ_2-coinvariant. Under the flip $z \mapsto -z$, the monodromy on V_J picks up a sign $(-1)^J$ (from the permutation of the two adjoint insertions); at generic k, av_2 is surjective with $\ker(\mathrm{av}_2)$ carrying the odd-J blocks. The scalar $\kappa = 3(k+2)/4$ (equation (9.1)) is the complete information surviving the averaging.

Degree 3: the Drinfeld associator. At degree 3, the KZ system on $\mathrm{Conf}_3(\mathbb{C})$ is

$$d\Phi = \frac{1}{k+2}\left(\frac{\Omega_{12}}{z_1 - z_2}\, d(z_1 - z_2) + \frac{\Omega_{13}}{z_1 - z_3}\, d(z_1 - z_3) + \frac{\Omega_{23}}{z_2 - z_3}\, d(z_2 - z_3)\right)\Phi.$$

The fundamental group $\pi_1(\mathrm{Conf}_3(\mathbb{C})) = PB_3$ (the pure braid group on three strands) has generators $\sigma_{12}^2, \sigma_{23}^2$ (Dehn twists around the diagonals). The KZ monodromy gives a representation $PB_3 \to \mathrm{GL}(V_k(\mathfrak{g})^{\otimes 3})$. The Drinfeld associator $\Phi_{\mathrm{KZ}} \in \exp(\widehat{\mathrm{Lie}}(t_{12}, t_{23}))$ is the regularised holonomy along the path from the boundary stratum $|z_1 - z_2| \ll |z_2 - z_3|$ to the stratum $|z_2 - z_3| \ll |z_1 - z_2|$ in the real locus of $\overline{\mathrm{FM}}_3^{\mathrm{ord}}(\mathbb{C})$.

The kernel of the averaging map av_3 is the space of ordered data invisible to $\overline{B}^{\Sigma}$. For $\mathfrak{g} = \mathfrak{sl}_2$, this kernel is one-dimensional: $\ker(\mathrm{av}_3) \cong \Lambda^3(\mathbb{C}^3) = \mathbb{C}$, carrying the Drinfeld associator. The associator Φ_{KZ} is the unique (up to GRT_1-action) non-trivial element of this kernel.

For general $\mathfrak{g}$, the ordered data at degree 3 is controlled by $\Lambda^3(\mathfrak{g}) = \Lambda^3(\mathbb{C}^{\dim \mathfrak{g}})$, with $\dim \Lambda^3(\mathfrak{g}) = \binom{\dim \mathfrak{g}}{3}$. At $\mathfrak{g} = \mathfrak{sl}_2$: $\binom{3}{3} = 1$.

E_3 **obstruction.** The E_2-coalgebra $\overline{B}^{\Sigma}(V_k(\mathfrak{sl}_2))$ (Warning 3.5) has obstruction to E_3-promotion in

$$\mathrm{HH}^3_{\mathsf{E}_2}(\overline{B}^{\Sigma}, \overline{B}^{\Sigma}) \;\cong\; H^3(\mathfrak{sl}_2) \;\cong\; \Lambda^3(\mathfrak{sl}_2^*)^{\mathfrak{sl}_2} \;=\; \mathbb{C},$$

which is one-dimensional and carries the non-trivial obstruction class (Warning 8.13). The E_3-algebra is the derived chiral centre $Z^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{sl}_2))$, not $\overline{B}^{\Sigma}$ itself.

Identification with CFG. The E_3-deformation space is $(k+2)H^3(\mathfrak{sl}_2)[[k+2]] \cong (k+2)\mathbb{C}[[k+2]]$: one-dimensional at each order in $(k+2)$. For $\mathfrak{sl}_2$, $H^3(\mathfrak{sl}_2) = \mathbb{C}$ (generated by the three-form $\omega_3 = (e^* \wedge f^* \wedge h^*) \circ [\cdot,\cdot]$, the Cartan form). The deformation parameter is the departure from critical level $k + 2 = k + h^\vee$. By Theorem 8.10, the formal disk restriction $\Gamma(D, \mathrm{CE}^{\mathrm{ch}}_k(\mathfrak{sl}_2))$ recovers the CFG filtered E_3-algebra $C^*_{k+2}(\mathfrak{sl}_2)$ of perturbative Chern–Simons quantisations; the full identification of the deformation families is Theorem 8.6. At the associative level, the Drinfeld–Kohno theorem identifies the KZ monodromy with the R-matrix of $U_q(\mathfrak{sl}_2)$ at $q = \exp(2\pi i/(k+2))$: the chiral quantum group equivalence (Theorem 6.5) lifts this from monodromy representations to the algebra.

Summary of ordered chiral Hochschild for $\widehat{\mathfrak{sl}}_2$.

Degree n	Ordered datum	Symmetric shadow
1	$V_k(\mathfrak{sl}_2)$	$V_k(\mathfrak{sl}_2)$
2	KZ local system; conformal blocks $V_0 \oplus V_1 \oplus V_2$	scalar $\kappa = 3(k+2)/4$
3	KZ associator Φ_{KZ}; $\ker(\mathrm{av}_3) = \mathbb{C}$	trivial (associator lost)
≥ 4	higher KZ holonomy	trivial

The KZ connection (4.1) has simple poles: regular singularities. The 3d theory is Chern–Simons gauge theory with gauge group G.

Example 9.3 ($\beta\gamma$ system: class C, $r = 4$)**.** The $\beta\gamma$ system: $\beta(z)\gamma(w) \sim 1/(z-w)$, $c_{\beta\gamma}(\lambda) = 2(6\lambda^2 - 6\lambda + 1)$, $\kappa = c_{\beta\gamma}/2 = 6\lambda^2 - 6\lambda + 1$ (at $\lambda = 1/2$: $c = -1$; at $\lambda = 2$: $c = 26$). Class C, E_∞-chiral, depth $r = 4$. The formality bridge gives ordered = symmetric.

The collision residue after d-log absorption is a constant nilpotent $\Theta \in \mathrm{End}(V \otimes V)$ with $\Theta^2 = 0$ ($V = \mathbb{C}\beta \oplus \mathbb{C}\gamma$); the R-matrix $R = I + 2\pi i\,\Theta$ is unipotent (the exponential terminates at linear order). The KZ connection has logarithmic singularities (regular) with unipotent monodromy. The chiral Hochschild cohomology: $\mathrm{HH}^0 = \mathbb{C}$, $\mathrm{HH}^1 = 0$ (all derivations inner, from the simple-pole OPE), $\mathrm{HH}^2 = \mathbb{C}$ (OPE residue deformation), $\mathrm{HH}^n = 0$ for $n \geq 3$ (Koszul length 2). Poincaré polynomial $1 + t^2$, total dimension 2, amplitude $[0, 2]$. The Koszul dual bc system satisfies $c_{bc}(\lambda) = 1 - 3(2\lambda - 1)^2$ with $\kappa(\beta\gamma) + \kappa(bc) = 0$ (complementarity conductor $K = 0$, same family as Heisenberg; verification: $c_{\beta\gamma}(2) = 26$, $c_{bc}(2) = -26$). Trivial E_3 structure ($\Lambda^3(\mathbb{C}^2) = 0$). See [34, §III.4] for the full computation.

Example 9.4 (Virasoro: class M, $r = \infty$)**.** Vir_c: $T(z)T(w) \sim (c/2)/(z-w)^4 + 2T(w)/(z-w)^2 + \partial T(w)/(z-w)$, $r^{\mathrm{Vir}}(z) = (c/2)/z^3 + 2T/z$ (cubic pole), $\kappa = c/2$. Class M, E_∞-chiral, depth $r = \infty$. The formality bridge gives ordered = symmetric; the infinite shadow tower is a symmetric phenomenon.

The KZ connection at degree 2 has a pole of order 3: *irregular* singularities of Poincaré rank 2. The classical Riemann–Hilbert correspondence does not apply; one needs D'Agnolo–Kashiwara [16]. The formal solution has exponential growth $\Phi(z) \sim z^{2T} \exp(-(c/4)z^{-2})$ near $z = 0$, with 4 Stokes rays at $\arg(z) = 0, \pi/2, \pi, 3\pi/2$. The Stokes multipliers carry the irregular monodromy data invisible to the classical Riemann–Hilbert correspondence.

For $\mathcal{W}_N$, the Stokes ray count is $4N - 4$ ($\mathcal{W}_2$: 4; $\mathcal{W}_3$: 8), reflecting the Poincaré rank $2N - 2$ of the highest-spin channel.

The 3d theory is holomorphic-topological gravity (Beltrami parametrisation; Costello–Gwilliam [13]), in contrast to gauge theory for class $G/L/C$.

Remark 9.5 (The conformal anomaly forces the spectral parameter). The quartic pole obstruction $c/2 = \kappa(\mathrm{Vir}_c)$ prevents any z-independent coproduct for $c \neq 0$; the spectral parameter in $\Delta_z(T(u)) = T(u) \otimes T(u-z)$ absorbs the mismatch through the shift $u \mapsto u - z$.

10. Ordered chiral homology on elliptic curves: $\int_{E_\tau}^{\mathrm{ord}} Y(\mathfrak{sl}_2)$

The preceding sections develop ordered chiral homology at genus 0: the base curve is $\mathbb{P}^1$ (or the formal disk D), the bar propagator is the Arnold form $d\log(z_1 - z_2)$, and the flat transport is governed separately by the KZ connection. The five explicit examples (§9) treat chiral algebras from classes G, L, C, M and the genuinely E_1 Yangian on $D^{\times}$.

The purpose of this section is to construct the ordered chiral homology of the Yangian $Y(\mathfrak{sl}_2)$ on an elliptic curve $E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$. This is a genuinely new invariant: (i) the base curve has genus 1, so the bar propagator acquires the elliptic function $\wp_1 = \theta_1'/\theta_1$ in place of $1/z$; (ii) the algebra is genuinely E_1, so the ordered theory differs from the symmetric; (iii) the KZ connection is replaced by the KZB connection, and the B-cycle monodromy produces the quantum group parameter $q = e^{2\pi i \ }$.

The construction proceeds degree by degree.

10.1. **Setup: the ordered chiral chain complex on E_τ.** Fix $\tau \in \mathbb{H} = \{\mathrm{Im}(\tau) > 0\}$ and let $E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$ with coordinate z. The ordered configuration space at degree n is

$$\mathrm{Conf}_n^{\mathrm{ord}}(E_\tau) = \{(z_1, \dots, z_n) \in E_\tau^n : z_i \neq z_j \text{ for } i \neq j\}.$$

No ordering on E_τ itself is needed: "ordered" means the labelling of points is retained (no Σ_n-quotient).

The bar propagator on E_τ is

$$\eta_{ij}^{(1)} = d \log E(z_i, z_j) = \wp_1(z_i - z_j, \tau)\, d(z_i - z_j),$$

where $E(z, w)$ is the prime form on E_τ and $\wp_1(z, \tau) = \theta_1'(z|\tau)/\theta_1(z|\tau)$ is the logarithmic derivative of the Jacobi theta function. This is weight 1 (: the bar propagator is always weight 1).

The ordered chiral chain complex at degree n on E_τ is

$$C_n^{\mathrm{ord}}(E_\tau, \mathcal{A}) = \Big(\Omega^*_{\log}\big(\overline{\mathrm{FM}}_n^{\mathrm{ord}}(E_\tau)\big) \otimes (s^{-1}\bar{\mathcal{A}})^{\otimes n},\ \ d_{\mathrm{total}}\Big)$$

with

$$d_{\mathrm{total}} = d_{\mathrm{dR}} + d_{\mathrm{bar}} - \nabla_{\mathrm{KZB}}.$$

The decisive change from genus 0 to genus 1: the KZ connection $\nabla_{\mathrm{KZ}} = d - \sum r_{ij}\, dz_{ij}$ is replaced by the KZB connection

$$\nabla_{\mathrm{KZB}} = d - \frac{1}{k + h^\vee} \sum_{i<j} \Omega_{ij}\, \wp_1(z_{ij}, \tau)\, dz_{ij} - \frac{1}{2\pi i(k + h^\vee)} \sum_{i<j} \wp(z_{ij}, \tau)\, \Omega_{ij}\, d\tau, \qquad (10.1)$$

where $\ \ = 1/(k + h^\vee) = 1/(k + 2)$ for $\mathfrak{sl}_2$ (the Yangian convention $\ \ = h_{\mathrm{KZ}}$; see Remark 8.7 for the relationship to the departure-from-critical convention used in §8). The $d\tau$ term is absent at genus 0; it encodes the curvature $\kappa(\mathcal{A}) \cdot \omega_1$ from the Hodge class $\omega_1 = c_1(\lambda)$ on $\overline{\mathcal{M}}_{1,n}$.

Definition 10.1 (Ordered chiral homology on E_τ)**.** The *ordered chiral homology of $\mathcal{A}$ on E_τ* is

$$\int_{E_\tau}^{\mathrm{ord}} \mathcal{A} \ := \ R\Gamma_{\mathrm{dR}}\big(\mathrm{Ran}^{\mathrm{ord}}(E_\tau),\ \mathcal{F}^{\mathrm{ord}}(\mathcal{A})\big) \ = \ \bigoplus_{n\geq 0} H^*\big(C_n^{\mathrm{ord}}(E_\tau, \mathcal{A}),\ d_{\mathrm{total}}\big).$$

10.2. **Degree 0: the center.** At degree 0 there are no marked points and no configuration space; the ordered chiral homology reduces to the chiral center of $\mathcal{A}$.

Proposition 10.2 (Degree-0 ordered chiral homology on E_τ)**.** *For $\mathcal{A} = Y(\mathfrak{sl}_2)$:*

$$H^*\big(C_0^{\mathrm{ord}}(E_\tau, Y(\mathfrak{sl}_2))\big) = Z(Y(\mathfrak{sl}_2)) = \mathbb{C}[\mathrm{qdet}\ T(u)].$$

The quantum determinant qdet $T(u) = A(u)\, D(u - \ \ ) - B(u)\, C(u - \ \ )$ *generates the center as a polynomial algebra* (*Molev*)*. The averaging map* av_0 *is an isomorphism* (*there is nothing to symmetrise at degree* 0)*, so* $\ker(\mathrm{av}_0) = 0$.

Remark 10.3. The center is independent of the base curve: it depends only on the algebra $\mathcal{A}$ and not on E_τ. This reflects the degree-0 factorisation $\mathcal{D}$-module being a skyscraper at the empty configuration.

10.3. **Degree 1: de Rham cohomology with Yangian coefficients.** At degree 1, $\mathrm{Conf}_1^{\mathrm{ord}}(E_\tau) = E_\tau$ and the ordered chiral chain complex is

$$C_1^{\mathrm{ord}}(E_\tau, Y) = \Big(\Omega^*(E_\tau) \otimes s^{-1}\bar{Y}, \quad d_{\mathrm{dR}} + d_{\mathrm{bar}}\Big),$$

where $\bar{Y} = \ker(\varepsilon)$ is the augmentation ideal of $Y(\mathfrak{sl}_2)$ and $|s^{-1}v| = |v| - 1$. At degree 1 the KZB connection is trivial (no pairs $i < j$), so $\nabla_{\mathrm{KZB}} = d$, absorbed into d_{dR}. The bar differential d_{bar} acts on each factor of $s^{-1}\bar{Y}$ via the A_∞ structure maps m_k of the chiral algebra.

The de Rham complex of E_τ has cohomology

$$H^0(E_\tau) = \mathbb{C}, \qquad H^1(E_\tau) = \mathbb{C}^2, \qquad H^2(E_\tau) = \mathbb{C},$$

generated respectively by 1; the A-cycle form dz and the B-cycle form $d\bar{z}/\mathrm{Im}(\tau)$; and the volume form $dz \wedge d\bar{z}/(2i\,\mathrm{Im}(\tau))$. By the Künneth theorem:

$$H^*\big(C_1^{\mathrm{ord}}\big) = H^*(E_\tau) \otimes H^*(s^{-1}\bar{Y}, d_{\mathrm{bar}}).$$

The bar cohomology $H^*(s^{-1}\bar{Y}, d_{\mathrm{bar}})$ is the degree-1 bar cohomology of the chiral algebra. For $Y(\mathfrak{sl}_2)$ viewed through its RTT presentation, the bar differential at degree 1 is the internal differential of the Yangian (the differential inherited from the dg structure of the ordered bar complex of $V_k(\mathfrak{sl}_2)$ at the first page of the bar spectral sequence). Since $Y(\mathfrak{sl}_2)$ is the Koszul dual of $V_k(\mathfrak{sl}_2)$ and the affine algebra is Koszul (class L), the bar cohomology at degree 1 is concentrated in degree 1:

$$H^*(s^{-1}\bar{Y}, d_{\mathrm{bar}}) = s^{-1}\mathfrak{sl}_2 \quad \text{(concentrated in cohomological degree 1).} \tag{10.2}$$

This is the desuspended Lie algebra: the three generators $\mathcal{E}^{(0)}, \mathcal{F}^{(0)}, H^{(0)}$ in mode zero, shifted by s^{-1}.

Proposition 10.4 (Degree-1 ordered chiral homology on E_τ)**.**

$$H^*\big(C_1^{\mathrm{ord}}(E_\tau, Y(\mathfrak{sl}_2))\big) \cong H^*(E_\tau) \otimes s^{-1}\mathfrak{sl}_2 \cong (s^{-1}\mathfrak{sl}_2)^{\oplus 4},$$

with the four copies sitting in cohomological degrees 1, 2, 2, 3 (*from the tensor product of* $H^{0,1,1,2}(E_\tau)$ *with the degree-*1 *bar cohomology* $s^{-1}\mathfrak{sl}_2$)*. The total dimension is* $4 \cdot 3 = 12$*.*

Remark 10.5. At degree 1, $\Sigma_1 = \{e\}$, so the averaging map is the identity and $\ker(\mathrm{av}_1) = 0$. The ordered and symmetric theories agree at degree 1: the non-trivial ordered data enters only at degree ≥ 2.

Remark 10.6 (Comparison with genus 0). At genus 0, $H^*(D^\times)$ contributes only in degree 0 (and degree 1 from the circle at infinity), so the degree-1 cohomology is 3-dimensional. The passage to genus 1 multiplies by $\dim H^*(E_\tau) = 4$: the new B-cycle produces the extra copy.

10.4. **Degree 2: the elliptic KZB local system.** At degree 2, the configuration space is $\mathrm{Conf}_2^{\mathrm{ord}}(E_\tau) = E_\tau^2 \setminus \Delta$, and the KZB connection acts non-trivially. The ordered chiral chain complex is

$$C_2^{\mathrm{ord}}(E_\tau, Y) = \Big(\Omega^*_{\log}\big(\overline{\mathrm{FM}}_2^{\mathrm{ord}}(E_\tau)\big) \otimes (s^{-1}\bar{Y})^{\otimes 2}, \quad d_{\mathrm{dR}} + d_{\mathrm{bar}} - \nabla_{\mathrm{KZB}}\Big).$$

The key new ingredient is the elliptic collision residue: the degree-2 classical r-matrix on E_τ is the Belavin elliptic r-matrix

$$r^{\mathrm{ell}}(z,\tau) = k \cdot r^{\mathrm{Belavin}}(z,\tau) = k\Big(\frac{\wp_1(z,\tau)}{2}\, H \otimes H + \phi_+(z,\tau)\, E \otimes F + \phi_-(z,\tau)\, F \otimes E\Big),$$

where $\phi_\pm(z,\tau) = \theta_1'(0|\tau)\,\theta_1(z \pm \frac{1}{2}|\tau)/(\theta_1(z|\tau)\,\theta_1(\pm\frac{1}{2}|\tau))$ are Belavin's ϕ-functions. At $k = 0$, $r^{\mathrm{ell}} = 0$ (the algebra becomes abelian). The degeneration $\tau \to i\infty$ sends $\wp_1(z,\tau) \to \pi\cot(\pi z) \to 1/z + O(z)$ and $\phi_\pm \to 1/z + O(1)$, recovering $r^{\mathrm{rat}}(z) = k\,\Omega/z$.

The KZB connection at degree 2 on the relative coordinate $w = z_1 - z_2$ is (Bernard [5]; Calaque–Enriquez–Etingof [6])

$$\nabla^{(2)}_{\mathrm{KZB}} = d - \frac{1}{k+h^\vee}\,\Omega\,\wp_1(w,\tau)\,dw - \frac{1}{2\pi i(k+h^\vee)}\,\wp(w,\tau)\,\Omega\,d\tau.$$

The flat sections satisfy

$$\begin{aligned}\frac{\partial\Phi}{\partial w} &= \frac{1}{k+h^\vee}\,\wp_1(w,\tau)\,\Omega\,\Phi,\\ \frac{\partial\Phi}{\partial\tau} &= \frac{1}{2\pi i(k+h^\vee)}\,\wp(w,\tau)\,\Omega\,\Phi.\end{aligned} \tag{10.3}$$

The flatness of the KZB system follows from the heat equation for theta functions: $\partial_\tau\,\wp_1(z,\tau) = \frac{1}{2\pi i}\,\partial_z\,\wp(z,\tau)$.

Decomposition by $\mathfrak{sl}_2$-isospin. On the fundamental representation $V = \mathbb{C}^2$, the Casimir decomposes as before:

$$V\otimes V = \mathrm{Sym}^2 V \oplus \bigwedge\nolimits^2 V, \qquad \Omega\big|_{\mathrm{Sym}^2} = +\tfrac{1}{2}, \quad \Omega\big|_{\wedge^2} = -\tfrac{3}{2}.$$

The KZB system decomposes into scalar equations on each isospin channel. On the symmetric channel:

$$\frac{\partial\Phi_{\mathrm{sym}}}{\partial w} = \frac{1}{2(k+h^\vee)}\,\wp_1(w,\tau)\,\Phi_{\mathrm{sym}}.$$

Monodromy. The function $\wp_1(w,\tau) = \theta_1'(w|\tau)/\theta_1(w|\tau)$ is periodic under $w\mapsto w+1$ and quasi-periodic under $w\mapsto w+\tau$ with increment $-2\pi i$. The puncture monodromy (small loop around $w=0$) captures the residue:

$$M_\gamma = \exp\!\big(2\pi i\quad\Omega\big)\quad\text{where}\quad = \frac{1}{k+h^\vee}.$$

On each channel: $M_\gamma^{\mathrm{sym}} = e^{\pi i/(k+2)}$, $M_\gamma^{\mathrm{alt}} = e^{-3\pi i/(k+2)}$.

The B-cycle monodromy ($w\mapsto w+\tau$) is the genuinely new genus-1 datum. The quasi-periodicity $\wp_1(w+\tau) = \wp_1(w) - 2\pi i$ produces:

$$M_B = \exp\!\big(-2\pi i\quad\Omega\big) = M_\gamma^{-1}. \tag{10.4}$$

Remark 10.7 (Dynamical parameter). The B-cycle monodromy (10.4) is the leading-order result from the quasi-periodicity of $\wp_1$. The full genus-1 monodromy representation involves the *dynamical parameter* $\lambda\in\mathfrak{h}^*$ (Felder [32]; Felder–Varchenko [33]): the B-cycle monodromy operator depends on the basepoint through a shift $\lambda\mapsto\lambda - h^{(i)}$, producing the dynamical Yang–Baxter equation rather than the ordinary YBE (Etingof–Varchenko [19]; see also [34, Thm 5.6]). The gauge-invariant content is the constraint $M_\gamma\cdot[M_A, M_B] = \mathrm{id}$ from $\pi_1(E_\tau\setminus\{0\})$.

On each isospin channel:

$$M_B^{\mathrm{sym}} = e^{-\pi i/(k+2)}, \qquad M_B^{\mathrm{alt}} = e^{3\pi i/(k+2)}.$$

Proposition 10.8 (Degree-2 ordered chiral homology on E_τ). *The degree-2 ordered chiral homology $H^*(C_2^{\mathrm{ord}}(E_\tau, Y\,))$ is the de Rham cohomology of the KZB local system on* $\mathrm{Conf}_2^{\mathrm{ord}}(E_\tau)$ *with coefficients in $(s^{-1}\bar{Y}\,)^{\otimes 2}$. On evaluation modules $V_a\otimes V_b$:*

(i) *The KZB local system has puncture monodromy $M_\gamma = q^\Omega$ and B-cycle monodromy $M_B = q^{-\Omega}$, where $q = e^{2\pi i}$.*

(ii) *The de Rham cohomology of the KZB local system on* $E_\tau \setminus \{0\}$ *(equivalently,* $\mathrm{Conf}_2^{\mathrm{ord}}(E_\tau)$ *in the relative coordinate* $w = z_1 - z_2$*) is:*

$$H^p_{\mathrm{dR}}(E_\tau \setminus \{0\},\ \nabla_{\mathrm{KZB}}) = \begin{cases} \ker(M_\gamma - I) \cap \ker(M_B - I) = \ker(q^\Omega - I) & p = 0, \\ (\textit{see } (10.5) \textit{ below}) & p = 1. \end{cases}$$

(iii) *For generic (equivalently, generic level* k*),* q^Ω *has distinct eigenvalues on each isospin channel, and* $H^0_{\mathrm{dR}} = 0$*: there are no global flat sections.*

(iv) *The* H^1 *is computed by the Euler characteristic of the local system. By the Riemann–Hilbert correspondence and the Euler characteristic formula for a rank-*d^2 *local system on a genus-*1 *curve with one puncture:*

$$\dim H^1_{\mathrm{dR}}(E_\tau \setminus \{0\},\ \nabla_{\mathrm{KZB}}) = d^2 \cdot (2 \cdot 1 - 2 + 1) = d^2, \tag{10.5}$$

where $d = \dim V = 2$*, so* $\dim H^1 = 4$.

Proof. The Euler characteristic of a local system of rank r on a surface of genus g with s punctures is $\chi = r(2 - 2g - s)$. For $g = 1, s = 1, r = d^2 = 4$: $\chi = 4(2 - 2 - 1) = -4$. Since $H^0 = 0$ for generic (the monodromy has no invariant vectors) and $H^2 = 0$ (the puncture forces vanishing), we have $\dim H^1 = 4$. The four-dimensional space decomposes as $3 + 1$ along the isospin channels $\mathrm{Sym}^2 V$ (dimension 3) and $\bigwedge^2 V$ (dimension 1). □

Remark 10.9 (The symmetric shadow). The averaging map av_2 at degree 2 sends the ordered de Rham cohomology to Σ_2-coinvariants. The Σ_2-action is the composition of the geometric swap $w \mapsto -w$ with the R-matrix. On the symmetric channel, the R-matrix eigenvalue is $+\ /2 + O(\ ^2)$; on the antisymmetric channel, $-3\ /2 + O(\ ^2)$. The scalar surviving av_2 is $\kappa = 3(k+2)/4$ (equation (9.1)).

The kernel $\ker(\mathrm{av}_2)$ at genus 1 is richer than at genus 0: the B-cycle monodromy contributes an additional antisymmetric class (from the $e^{3\pi i/(k+2)}$ eigenvalue of M_B on $\bigwedge^2 V$) that is invisible to the symmetric theory.

10.5. **Degree** n**: elliptic braid group monodromy.** At degree n, the fundamental group of $\mathrm{Conf}_n^{\mathrm{ord}}(E_\tau)$ is the *elliptic braid group* EB_n, which is an extension

$$1 \to PB_n \to \mathrm{EB}_n \to \pi_1(E_\tau)^n \to 1, \tag{10.6}$$

where PB_n is the pure braid group on n strands and $\pi_1(E_\tau) \cong \mathbb{Z}^2$ contributes the A- and B-cycle monodromies for each point.

The KZB monodromy representation at degree n is

$$\rho_n^{\mathrm{KZB}} \colon \mathrm{EB}_n \to \mathrm{GL}(V^{\otimes n}).$$

The ordered chiral homology at degree n is the de Rham cohomology of the KZB local system on the Fulton–MacPherson compactification:

$$\int^{\mathrm{ord}}_{E_\tau, n} Y\ (\mathfrak{sl}_2) = H^*_{\mathrm{dR}}\big(\overline{\mathrm{FM}}_n^{\mathrm{ord}}(E_\tau),\ \nabla^{(n)}_{\mathrm{KZB}}\big) \otimes (s^{-1}\bar{Y}\ )^{\otimes n}.$$

Proposition 10.10 (Euler characteristic at low degrees)**.** *The twisted Euler characteristics of the KZB local system on* $\mathrm{Conf}_n^{\mathrm{ord}}(E_\tau)$*, with fibre* $V^{\otimes n}$ *(*$V = \mathbb{C}^2$*) for generic , are:*

$$\begin{aligned} n = 0 &: \quad \chi = \infty \quad (\mathbb{C}[\mathrm{qdet}] \textit{ is infinite-dimensional}), \\ n = 1 &: \quad \chi = 0 \quad (\textit{trivial local system on } E_\tau,\ \chi(E_\tau) = 0), \\ n = 2 &: \quad \chi = -4 \quad (\textit{rank } 4 \textit{ on } E_\tau \setminus \{0\}). \end{aligned}$$

Proof. At $n = 1$: $\mathrm{Conf}_1(E_\tau) = E_\tau$ and the KZB connection has no pairwise terms (there are no pairs $i < j$), so the local system is trivial of rank $d = 2$. The twisted Euler characteristic is $\chi = d \cdot \chi(E_\tau) = 2 \cdot 0 = 0$, consistent with $\dim H^0 = 2$, $\dim H^1 = 4$, $\dim H^2 = 2$ from the Künneth computation: $\chi = 2 - 4 + 2 = 0$.

At $n = 2$: the relative coordinate $w = z_1 - z_2$ identifies $\mathrm{Conf}_2^{\mathrm{ord}}(E_\tau)$ (modulo the E_τ-translation in the centre-of-mass) with $E_\tau \setminus \{0\}$, a genus-1 surface with one puncture. The KZB local system has rank $d^2 = 4$. The Euler characteristic of a rank-r local system on a surface of genus g with s punctures is $\chi = r(2 - 2g - s)$: here $\chi = 4(2 - 2 - 1) = -4$. For generic   , the monodromy has no invariant vectors and $H^0 = 0$; the puncture forces $H^2 = 0$; so $\dim H^1 = 4$. This is consistent with Proposition 10.8. □

Remark 10.11 (Higher degrees). For $n \geq 3$, $\mathrm{Conf}_n^{\mathrm{ord}}(E_\tau)$ is a smooth quasi-projective variety of complex dimension n. Its topological Euler characteristic vanishes: $\chi(\mathrm{Conf}_n^{\mathrm{ord}}(E_\tau)) = 0$ for all $n \geq 1$ (the product formula $\chi = \prod_{j=0}^{n-1}(\chi(E_\tau) - j) = 0$ since $\chi(E_\tau) = 0$). The twisted Euler characteristic of the KZB local system is also zero: $\chi(\mathrm{Conf}_n^{\mathrm{ord}}(E_\tau), \mathcal{L}_{\mathrm{KZB}}) = 0$ for $n \geq 3$. This does *not* mean the de Rham cohomology vanishes; it means the dimensions of the cohomology groups cancel in alternating sum. The individual Betti numbers with KZB coefficients require explicit computation of the elliptic braid group monodromy at each degree.

10.6. **The total ordered chiral homology and the averaging kernel.** Assembling all degrees, the total ordered chiral homology is

$$\int_{E_\tau}^{\mathrm{ord}} Y(\mathfrak{sl}_2) = \bigoplus_{n \geq 0} \int_{E_\tau, n}^{\mathrm{ord}} Y(\mathfrak{sl}_2).$$

At each degree, the averaging map $\mathrm{av}_n : \int_{E_\tau,n}^{\mathrm{ord}} \to (\int_{E_\tau,n}^{\mathrm{ord}})_{\Sigma_n}$ projects to the symmetric shadow. The kernel $\ker(\mathrm{av}) = \bigoplus_n \ker(\mathrm{av}_n)$ carries the genuinely E_1 data that is invisible to the symmetric theory.

Theorem 10.12 (Structure of $\int_{E_\tau}^{\mathrm{ord}} Y(\mathfrak{sl}_2)$)**.** *For generic* $\;\; = 1/(k+2)$ *with* $k \notin \{-2\} \cup \mathbb{Z}_{\leq 0}$*:*

(i) (Degree decomposition.) *The degree-by-degree structure is:*

n	$\int_{E_\tau,n}^{\mathrm{ord}} Y$	$\ker(\mathrm{av}_n)$	*Shadow*
0	$\mathbb{C}[\mathrm{qdet}\ T(u)]$	0	$\mathbb{C}[\mathrm{qdet}]$
1	$(s^{-1}\mathfrak{sl}_2)^{\oplus 4}$, *dim* 12	0	$(s^{-1}\mathfrak{sl}_2)^{\oplus 4}$
2	*KZB flat sections, dim* 4	$\bigwedge^2 V \oplus$ *B-cycle class*	$\kappa = 3(k+2)/4$
3	*elliptic braid monodromy*	$\supseteq \Lambda^3(\mathfrak{g})$	*traces*
n	EB_n*-monodromy of KZB*	*matrix-valued*	*scalar traces*

(ii) (Genus-1 enhancement of the kernel.) *The kernel* $\ker(\mathrm{av}_2)$ *at genus* 1 *is strictly larger than the kernel at genus* 0*: the B-cycle monodromy* q^Ω *introduces new antisymmetric classes in the* $\bigwedge^2 V$ *channel that have no genus-*0 *analogue. The Drinfeld associator at degree* 3 *is deformed by the KZB associator* Φ_{KZB}*, which is a genus-*1 *deformation of* Φ_{KZ}*.*

(iii) (Degeneration.) *In the limit* $\tau \to i\infty$ (*the B-cycle collapses and* E_τ *degenerates to* $\mathbb{C}^\times$)*, the KZB connection degenerates to the KZ connection, the B-cycle monodromy becomes trivial, and*

$$\lim_{\tau \to i\infty} \int_{E_\tau}^{\mathrm{ord}} Y(\mathfrak{sl}_2) = \int_{D^\times}^{\mathrm{ord}} Y(\mathfrak{sl}_2),$$

*recovering the genus-*0 *ordered chiral homology of §9.*

(iv) (Modular dependence.) *The ordered chiral homology varies with τ as a section of a vector bundle over $\mathbb{H}$ with flat connection $\nabla_\tau = \partial_\tau - \frac{1}{2\pi i}\sum_{i<j}\wp(z_{ij},\tau)\,\Omega_{ij}$. The* $\mathrm{SL}_2(\mathbb{Z})$*-action on τ acts on the ordered chiral homology through the modular transformation of the KZB connection. The kernel* $\ker(\mathrm{av})$ *transforms as a non-trivial representation of the modular group: the B-cycle monodromy is exchanged with the A-cycle monodromy under $\tau \mapsto -1/\tau$: the pair $(M_\gamma, M_B) = (q^{\Omega}, q^{-\Omega})$ is interchanged. The parameter $q = e^{2\pi i}$ itself is τ-independent; the modular group acts on the monodromy representation, not on q.*

Proof sketch. Parts (i)–(ii) follow from the degree-by-degree computation of §§10.2–10.5.

Part (iii): the Belavin r-matrix degenerates to the rational r-matrix as $\tau \to i\infty$ ($\wp_1(z,\tau) = \theta_1'(z|\tau)/\theta_1(z|\tau) \to 1/z$ and $\phi_\pm(z,\tau) \to 1/z$), the B-cycle shrinks, and the KZB flat sections degenerate to KZ flat sections. The degree-by-degree Euler characteristics are continuous under this degeneration.

Part (iv): the modular transformation of the KZB connection is the standard result of Bernard [5]: the $d\tau$ term ensures compatibility with the $\mathrm{SL}_2(\mathbb{Z})$-action, and the flat sections transform with the modular anomaly proportional to $\kappa(\mathcal{A})$ (equation (9.1)). $\square$

Remark 10.13 (Comparison with the quantum group). The object $\int_{E_\tau}^{\mathrm{ord}} Y\,(\mathfrak{sl}_2)$ computes the conformal blocks of the Yangian on E_τ. By the Yangian–quantum group deformation [34, Theorem 5.5.1], the genus-1 line-operator algebra is $U_q(\mathfrak{sl}_2)$ with $q = e^{2\pi i}$. The ordered chiral homology packages the full KZB monodromy data into a single invariant: it sees both the R-matrix (A-cycle, Yangian data) and the quantum group parameter (B-cycle, q-data) simultaneously. The symmetric shadow $(\int_{E_\tau}^{\mathrm{ord}})_{\Sigma_n}$ retains only the scalar traces (characters of $U_q(\mathfrak{sl}_2)$ representations), which are the WZW conformal blocks; the kernel $\ker(\mathrm{av})$ carries the matrix-valued monodromy from which knot invariants on the torus are extracted.

Remark 10.14 (Roots of unity). At integer level $k \in \mathbb{Z}_{>0}$, the quantum group parameter is $q = e^{2\pi i/(k+2)}$ and $q^{k+2} = 1$. The KZB local system acquires finite monodromy, and the ordered chiral homology at each degree becomes a finite-dimensional representation of the mapping class group of E_τ with n marked points. The $k+1$ integrable modules $V_0, \ldots, V_k$ of $U_q(\mathfrak{sl}_2)$ correspond to the truncation of the degree-by-degree decomposition: at degree $n > k$, the representation theory constrains certain components to vanish.

Remark 10.15 (The elliptic R-matrix as ordered datum). The Belavin elliptic r-matrix $r^{\mathrm{ell}}(z,\tau) = k \cdot r^{\mathrm{Belavin}}(z,\tau)$ is the degree-2 projection of the genus-1 Maurer–Cartan element $\Theta_{\mathcal{A}}^{(1)} \in \mathfrak{g}_{\mathcal{A}}^{\mathsf{E}_1}$. It satisfies the classical Yang–Baxter equation with spectral parameter (verified numerically in `elliptic_rmatrix_shadow.py`), and its averaging gives $\kappa = 3(k+2)/4$ (equation (9.1)), the same value as at genus 0. The agreement is forced: the Sugawara shift $\dim(\mathfrak{g})/2 = 3/2$ comes from the simple-pole self-contraction, a local computation independent of the global topology of the base curve.

10.7. **Recovery of the Verlinde formula.** At integer level $k \geq 1$, the quantum group parameter $q = e^{2\pi i/(k+2)}$ is a primitive $(k+2)$-th root of unity. The representation category of $U_q(\mathfrak{sl}_2)$ is semisimple with $k+1$ integrable modules $V_0, \ldots, V_k$, and the KZB local system has finite monodromy. The ordered chiral chain complex of $V_k(\mathfrak{sl}_2)$ on a genus-g curve Σ_g therefore computes a finite-dimensional invariant at each degree, and the symmetric coinvariants recover the space of conformal blocks (Tsuchiya–Ueno–Yamada [46]). The following proposition shows that the resulting dimension is given by the Verlinde formula [47].

Proposition 10.16 (Verlinde formula from ordered chiral homology). *Let $k \geq 1$ be a positive integer, and let $S_{jl} = \sqrt{2/(k+2)}\ \sin\big(\pi(j+1)(l+1)/(k+2)\big)$ be the modular S-matrix for $\widehat{\mathfrak{sl}}_2$ at level k, where $j, l \in \{0, 1, \ldots, k\}$.*

(i) (Genus 0.) *The ordered chiral homology at genus 0 with no marked points recovers the unique vacuum:*

$$\dim H^0\big(\mathcal{M}_{0,0},\ V_k(\mathfrak{sl}_2)\big) = \sum_{j=0}^{k} S_{0j}^2 = 1.$$

This is the unitarity of the S-matrix: $\sum_l S_{jl}^2 = \delta_{j0}$ for $j = 0$ follows from the orthogonality of the sine functions.

(ii) (Genus 1.) *The ordered chiral homology at genus 1 recovers the number of integrable representations:*

$$\dim H^0\big(\mathcal{M}_{1,0},\ V_k(\mathfrak{sl}_2)\big) = \sum_{j=0}^{k} S_{0j}^0 = k+1.$$

By the Zhu algebra identification $A(V_k(\mathfrak{sl}_2)) \cong U(\mathfrak{sl}_2)/(e^{k+1}, f^{k+1})$, the $k+1$ simple modules of $A(V_k)$ are $V_0, \ldots, V_k$, and the genus-1 conformal blocks are their characters. From the ordered chiral homology: Proposition 10.2 computes the degree-0 center at integrable level as $\mathbb{C}^{k+1}$ (the center of the integrable quotient of Y at the root of unity $q = e^{2\pi i/(k+2)}$).

(iii) (General genus.) *At genus $g \geq 2$, the Verlinde formula gives:*

$$Z_g(k) := \dim H^0\big(\mathcal{M}_{g,0},\ V_k(\mathfrak{sl}_2)\big) = \sum_{j=0}^{k} S_{0j}^{2-2g}.$$

This is an integer for all $g \geq 0$, $k \geq 1$. The first values are:

k	Z_0	Z_1	Z_2	Z_3	Z_4
1	1	2	4	8	16
2	1	3	10	36	136
3	1	4	20	120	800
4	1	5	35	329	3,611
5	1	6	56	784	13,328

At $k = 1$: $S_{00} = S_{01} = 1/\sqrt{2}$, so $Z_g(1) = 2 \cdot (1/\sqrt{2})^{2-2g} = 2^g$.

(iv) (Handle attachment.) *The ordered bar complex factorization under non-separating sewing recovers the TQFT handle-attachment formula: each isospin channel j contributes a handle operator $H_j = S_{0j}^{-2}$, so*

$$Z_{g+1} = \sum_{j=0}^{k} S_{0j}^{-2} \cdot S_{0j}^{2-2g} = \sum_{j=0}^{k} S_{0j}^{2-2(g+1)}.$$

(v) (Separating factorization.) *Under separating degeneration $\Sigma_g \rightsquigarrow \Sigma_{g_1} \cup \Sigma_{g_2}$ with $g = g_1 + g_2$, the bar complex factorization gives*

$$Z_g = \sum_{j=0}^{k} S_{0j}^{2-2g_1} \cdot S_{0j}^{2-2g_2} \cdot S_{0j}^{-2} = \sum_{j=0}^{k} S_{0j}^{2-2g}.$$

Proof. (i) The identity $\sum_j S_{0j}^2 = 1$ is the $(0,0)$-entry of $S \cdot S^T = I$, which follows from the orthogonality relation $\sum_{j=0}^{k} \sin(\pi(j{+}1)a/(k{+}2)) \sin(\pi(j{+}1)b/(k{+}2)) = (k{+}2)\delta_{ab}/2$. From the ordered chiral homology: at genus 0 with no marked points, the conformal block space is the space of vacua $\mathbb{C} \cdot |0\rangle$, which is one-dimensional.

(ii) At genus 1, $S_{0j}^0 = 1$ for all j, so $Z_1 = k + 1$. This has three independent derivations:

- (S-matrix.) Direct computation: $\sum_{j=0}^{k} 1 = k + 1$.
- (Zhu algebra.) The Zhu algebra $A(V_k(\mathfrak{sl}_2))$ has $k + 1$ simple modules; each contributes a one-dimensional space of genus-1 conformal blocks (the character).
- (Ordered center.) Proposition 10.2 identifies the degree-0 center of $Y(\mathfrak{sl}_2)$ at integrable level k as $\mathbb{C}^{k+1}$ (the center of the integrable quotient). The averaging map av_0 is an isomorphism (nothing to symmetrise at degree 0), so all $k + 1$ dimensions survive.

(iii) The Verlinde formula [47] gives $Z_g = \sum_j S_{0j}^{2-2g}$ as the dimension of the space of conformal blocks on a genus-g curve with no insertions. From the ordered chiral homology: the symmetric coinvariants of the ordered chain complex compute the conformal blocks via the factorization property (TUY [46]). The KZB local system at integrable level decomposes into $k + 1$ channels labeled by integrable representations; each channel j contributes S_{0j}^{2-2g} to the partition function. The integrality of Z_g follows from the identification with the rank of the Verlinde bundle over $\mathcal{M}_g$.

(iv) Handle attachment: a genus-g surface with a non-separating cycle cut open gives a genus-$(g{-}1)$ surface with two extra marked points. The bar complex sewing map reglues these marked points, summing over the $k + 1$ intermediate representations with the propagator S_{0j}^{-2} (the inverse of the j-th eigenvalue of the sewing operator). This gives $Z_{g+1} = \sum_j H_j \cdot S_{0j}^{2-2g}$.

(v) Separating factorization: the bar complex factorization under separating degeneration (Theorem A for the ordered bar complex on $\mathrm{Ran}^{\mathrm{ord}}$) decomposes the genus-$g$ partition function into a sum over channels of the product of the two lower-genus contributions, normalized by the propagator S_{0j}^{-2}.

All numerical values in the table have been verified by direct computation in `verlinde_ordered_engine.py` using three independent code paths (direct S-matrix summation, quantum-dimension formula, and handle recursion), cross-checked against the Verlinde table of `verlinde_shadow_algebra.py` (198 tests, all passing). □

Example 10.17 ($\widehat{\mathfrak{sl}}_2$ at level 1: the smallest nontrivial case)**.** For $k = 1$ the integrable representations are V_0 (vacuum) and V_1 (spin-1/2). The modular S-matrix is $S = (1/\sqrt{2})\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, so $S_{0j} = 1/\sqrt{2}$ for $j \in \{0, 1\}$. The Verlinde formula gives

$$Z_g(1) \;=\; \sum_{j=0,1} S_{0j}^{2-2g} \;=\; 2 \cdot (1/\sqrt{2})^{2-2g} \;=\; 2^g.$$

Every piece of the ordered theory specialises cleanly in this case: at genus 0, $Z_0 = 1$ is the unique vacuum; at genus 1, $Z_1 = 2$ counts the two integrable modules; at genus g each handle contributes a factor of 2 (from $H_j = S_{0j}^{-2} = 2$). The kernel of the averaging map at degree n is $\dim\ker(\mathrm{av}_n) = 2^n - (n+1)$ ($\mathfrak{sl}_2$ fundamental, $d = 2$) by Proposition 2.6. The κ-value is $\kappa(V_1(\mathfrak{sl}_2)) = 3(1+2)/4 = 9/4$, and $F_1 = 9/96 = 3/32$; the ratio F_g/Z_g is a nontrivial invariant already at $g = 1$.

Remark 10.18 (Verlinde bundle vs shadow tower). The Verlinde dimension Z_g and the shadow partition function $F_g = \kappa \cdot \lambda_g$ are different invariants of the same chiral algebra. Z_g is the rank of the Verlinde bundle over $\mathcal{M}_g$ (an integer, from the S-matrix); F_g is the first Chern class integrated over $\overline{\mathcal{M}}_g$ (a rational number, from κ). For $V_k(\mathfrak{sl}_2)$: $\kappa = 3(k{+}2)/4$ (equation (9.1)), $F_1 = \kappa/24$, and $Z_1 = k + 1$. The two invariants encode complementary data: Z_g counts the conformal blocks (combinatorial, from the modular tensor category), while F_g measures the curvature (analytic, from the Hodge class $\omega_g = c_1(\lambda)$ on $\overline{\mathcal{M}}_g$; : ω_g lives on $\overline{\mathcal{M}}_g$, not on the curve Σ_g).

Remark 10.19 (The ordered refinement). The Verlinde dimension $Z_g = \dim(\int_{\Sigma_g}^{\mathrm{ord}} V_k)_{\Sigma_n}$ is the dimension of the symmetric coinvariants. The full ordered chiral homology $\int_{\Sigma_g}^{\mathrm{ord}} V_k(\mathfrak{sl}_2)$ is strictly larger: its degree-n component carries the full KZB monodromy representation of the elliptic braid group EB_n (Proposition 10.8, Theorem 10.12), from which the Verlinde dimension is recovered by taking Σ_n-coinvariants and restricting to $n = 0$. The kernel $\ker(\mathrm{av}_n)$ carries the matrix-valued monodromy data: R-matrices, $6j$-symbols, and (at genus ≥ 1) dynamical quantum group data.

11. Ordered chiral homology at genus 2: $\int_{\Sigma_2}^{\mathrm{ord}} Y(\mathfrak{sl}_2)$

The genus-1 construction of §10 replaces the rational propagator $1/z$ by the elliptic function $\wp_1 = \theta_1'/\theta_1$ and the KZ connection by the KZB connection. At genus 2, three qualitatively new features enter: (i) the prime form $E(z, w)$ on a genus-2 Riemann surface Σ_2 replaces the Jacobi theta function; (ii) the period matrix $\Omega \in \mathbb{H}_2 = \{Z \in M_2(\mathbb{C}) : Z = Z^t,\ \operatorname{Im} Z > 0\}$ is a 2×2 symmetric matrix, and the KZB connection coefficients involve the matrix $(\operatorname{Im}\Omega)^{-1}$ rather than the scalar $1/\operatorname{Im}(\tau)$; (iii) the fundamental group $\pi_1(\mathrm{Conf}_n^{\mathrm{ord}}(\Sigma_2))$ is the genus-2 surface braid group, an extension of the pure braid group by $\pi_1(\Sigma_2)^n \cong (\mathbb{Z}^4)^n$ with four generators per point (two A-cycles, two B-cycles).

11.1. **Setup: the genus-2 ordered chiral chain complex.**

Let Σ_2 be a smooth projective curve of genus 2 with Siegel period matrix $\Omega = \left(\begin{smallmatrix} \Omega_{11} & \Omega_{12} \\ \Omega_{12} & \Omega_{22} \end{smallmatrix}\right) \in \mathbb{H}_2$. Fix a symplectic basis $\{A_1, A_2, B_1, B_2\}$ of $H_1(\Sigma_2, \mathbb{Z})$ with intersection pairing $A_i \cdot B_j = \delta_{ij}$, and let $\omega_1, \omega_2 \in H^0(\Sigma_2, K_{\Sigma_2})$ be the normalised holomorphic differentials satisfying $\oint_{A_i} \omega_j = \delta_{ij}$, $\oint_{B_i} \omega_j = \Omega_{ij}$.

The bar propagator on Σ_2 is

$$\eta_{ij}^{(2)} = d \log E(z_i, z_j),$$

where $E(z, w)$ is the *prime form* on Σ_2: a section of $K_{\Sigma_2}^{-1/2} \boxtimes K_{\Sigma_2}^{-1/2}$ with a simple zero along the diagonal $z = w$ and no other zeros or poles. This is weight 1 (the bar propagator is always weight 1). Explicitly, in terms of the Riemann theta function $\theta[\delta](z|\Omega)$ with odd characteristic δ:

$$E(z, w) = \frac{\theta[\delta]\Big(\int_w^z \vec{\omega} \,\big|\, \Omega\Big)}{h_\delta(z)\, h_\delta(w)},$$

where $\vec{\omega} = (\omega_1, \omega_2)^t$, $\int_w^z \vec{\omega} \in \mathbb{C}^2$ is the Abel–Jacobi map, and h_δ is a half-differential depending on the odd spin structure δ. In the genus-1 limit ($\Omega_{12} \to 0$, $\Omega_{22} \to i\infty$), one component decouples and $E(z, w) \to \theta_1(z - w|\Omega_{11})/\theta_1'(0|\Omega_{11})$, recovering the elliptic prime form.

Remark 11.1 (The Szegő kernel at genus 2). The logarithmic derivative of the prime form,

$$S_2(z, w) = d_z \log E(z, w) = \partial_z \log \theta[\delta]\Big(\int_w^z \vec{\omega} \,\Big|\, \Omega\Big) \cdot \omega(z) - d_z \log h_\delta(z), \tag{11.1}$$

is the genus-2 Szegő kernel. It has a simple pole at $z = w$ with residue 1 (matching $1/(z - w)$ locally) and is quasi-periodic under the B-cycles:

$$S_2(z + B_\alpha, w) = S_2(z, w) - 2\pi i\, \omega_\alpha(z), \qquad \alpha = 1, 2. \tag{11.2}$$

The two-component quasi-periodicity increment $-2\pi i\, \omega_\alpha(z)$ replaces the scalar $-2\pi i\, dz$ of the genus-1 case; it is the source of the 2×2 matrix structure in the genus-2 KZB connection.

The ordered chiral chain complex at degree n on Σ_2 is

$$C_n^{\mathrm{ord}}(\Sigma_2, \mathcal{A}) = \Big(\Omega^*_{\log}\big(\overline{\mathrm{FM}}_n^{\mathrm{ord}}(\Sigma_2)\big) \otimes (s^{-1}\bar{\mathcal{A}})^{\otimes n}, \quad d_{\mathrm{total}}\Big)$$

with

$$d_{\mathrm{total}} = d_{\mathrm{dR}} + d_{\mathrm{bar}} - \nabla_{\mathrm{KZB}}^{(g=2)}.$$

Definition 11.2 (Ordered chiral homology on Σ_2)**.** The *ordered chiral homology of $\mathcal{A}$ on Σ_2* is

$$\int_{\Sigma_2}^{\mathrm{ord}} \mathcal{A} \;:=\; \bigoplus_{n\geq 0} H^*\big(C_n^{\mathrm{ord}}(\Sigma_2, \mathcal{A}),\ d_{\mathrm{total}}\big).$$

Curvature. The fiberwise bar differential at genus 2 acquires curvature (UNIFORM-WEIGHT):

$$d_{\mathrm{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_2,$$

where $\omega_2 = c_1(\lambda) \in H^2(\overline{\mathcal{M}}_2)$ is the Hodge class on $\overline{\mathcal{M}}_2$ (a class on the *moduli space*, not a form on the curve), and $\kappa(\mathcal{A}) = 3(k+2)/4$ for $\mathcal{A} = Y(\mathfrak{sl}_2)$. The period-corrected differential $D^{(2)} = d_{\mathrm{total}} - \kappa(\mathcal{A}) \cdot \omega_2 \cdot [\text{Hodge correction}]$ restores $(D^{(2)})^2 = 0$ at the total level.

11.2. **The genus-2 KZB connection.** At degree $n \geq 2$, the KZB connection on Σ_2 is the genus-2 analogue of (10.1) (Bernard [5], Calaque–Enriquez–Etingof [6] for the universal formulation at arbitrary genus). The connection has two components: a spatial part on $\mathrm{Conf}_n^{\mathrm{ord}}(\Sigma_2)$ and a modular part along the Siegel upper half-space $\mathbb{H}_2$.

The spatial part at degree n is

$$\nabla^{(2)}_{\mathrm{spatial}} = d - \quad \sum_{i<j} \big(S_2(z_i, z_j)\big)\, \Omega_{ij},$$

where $\quad = 1/(k+h^\vee) = 1/(k+2)$ for $\mathfrak{sl}_2$, $S_2(z_i, z_j) = d_{z_i} \log E(z_i, z_j)$ is the Szegő kernel (11.1), and Ω_{ij} is the Casimir acting in positions i, j. This reduces to the genus-1 spatial connection $\quad \Omega\, \wp_1(w, \tau)\, dw$ in the degeneration $\Sigma_2 \to E_\tau$ (one handle collapses).

The modular part is a connection along the Siegel directions $\Omega_{\alpha\beta}$ ($\alpha, \beta = 1, 2$):

$$\nabla^{(2)}_{\mathrm{modular}} = \sum_{\alpha \leq \beta} \Big(\partial_{\Omega_{\alpha\beta}} - \frac{1}{2\pi i} \quad \sum_{i<j} \partial_{z_i} \partial_{z_j} \log E(z_i, z_j) \cdot \omega_\alpha(z_i)\, \omega_\beta(z_j)\, \Omega_{ij}\Big) d\Omega_{\alpha\beta}.$$

At genus 1, the Siegel matrix reduces to the scalar $\tau = \Omega_{11}$, $\omega_1(z) = dz$, and the modular part reduces to $\partial_\tau - \frac{}{2\pi i} \wp(w, \tau)\, \Omega\, d\tau$ of equation (10.3). At genus 2, the modular part has *three* independent components $d\Omega_{11}, d\Omega_{12}, d\Omega_{22}$ (the three directions in $\mathbb{H}_2$). The flatness of the full KZB system follows from the genus-2 heat equation for the theta function:

$$\frac{\partial}{\partial \Omega_{\alpha\beta}} \theta[\delta](z|\Omega) = \frac{1}{2\pi i} \frac{\partial^2}{\partial z_\alpha\, \partial z_\beta} \theta[\delta](z|\Omega).$$

Remark 11.3 (The (2×2)-matrix structure). The genus-2 KZB system makes essential use of the matrix $(\mathrm{Im}\,\Omega)^{-1}$. In the de Rham description, the harmonic projection requires the inner product on $H^1(\Sigma_2)$ given by the Hodge star, which on a genus-2 surface is encoded by the 2×2 positive-definite matrix $\mathrm{Im}\,\Omega$. Its inverse

$$(\mathrm{Im}\,\Omega)^{-1} = \frac{1}{\det(\mathrm{Im}\,\Omega)} \begin{pmatrix} \mathrm{Im}\,\Omega_{22} & -\mathrm{Im}\,\Omega_{12} \\ -\mathrm{Im}\,\Omega_{12} & \mathrm{Im}\,\Omega_{11} \end{pmatrix}$$

is a genuine 2×2 matrix (not a scalar as at genus 1 where $(\mathrm{Im}\,\Omega)^{-1} = 1/\mathrm{Im}(\tau)$). The B-cycle monodromy at genus 2 involves the matrix exponential $\exp(-2\pi i \quad \Omega_{ij})$ with the Casimir acting through a 2×2 block corresponding to the two B-cycles, producing a *doubly-dynamical* parameter.

11.3. **Degree 0: the center.** At degree 0, there are no marked points and no configuration space; the ordered chiral homology reduces to the chiral center of $\mathcal{A}$, independent of the base curve.

Proposition 11.4 (Degree-0 ordered chiral homology on Σ_2). *For $\mathcal{A} = Y(\mathfrak{sl}_2)$:*

$$H^*\big(C_0^{\mathrm{ord}}(\Sigma_2, Y(\mathfrak{sl}_2))\big) = Z(Y(\mathfrak{sl}_2)) = \mathbb{C}[\mathrm{qdet}\ T(u)].$$

This is identical to degree 0 *at genus* 0 *and genus* 1 (*Proposition* 10.2).

Proof. The degree-0 factorisation $\mathcal{D}$-module is a skyscraper at the empty configuration. It depends only on the algebra $\mathcal{A}$, not on the base curve. □

11.4. **Degree 1: de Rham cohomology of Σ_2 with Yangian coefficients.** At degree 1, $\mathrm{Conf}_1^{\mathrm{ord}}(\Sigma_2) = \Sigma_2$ and the KZB connection has no pairwise terms ($i < j$ is empty), so the ordered chiral chain complex is

$$C_1^{\mathrm{ord}}(\Sigma_2, Y) = \Big(\Omega^*(\Sigma_2) \otimes s^{-1}\bar{Y},\ \ d_{\mathrm{dR}} + d_{\mathrm{bar}}\Big),$$

where $\bar{Y} = \ker(\varepsilon)$ is the augmentation ideal. The de Rham complex of Σ_2 has cohomology

$$H^0(\Sigma_2) = \mathbb{C}, \qquad H^1(\Sigma_2) = \mathbb{C}^4, \qquad H^2(\Sigma_2) = \mathbb{C},$$

generated respectively by 1; the four cycle forms ω_1, ω_2 (A-cycles) and $\bar{\omega}_1, \bar{\omega}_2$ (B-cycles, with normalisation involving $(\mathrm{Im}\,\Omega)^{-1}$); and the volume form. By the Künneth theorem, with bar cohomology $H^*(s^{-1}\bar{Y}, d_{\mathrm{bar}}) = s^{-1}\mathfrak{sl}_2$ concentrated in degree 1 (equation (10.2)):

Proposition 11.5 (Degree-1 ordered chiral homology on Σ_2).

$$H^*\big(C_1^{\mathrm{ord}}(\Sigma_2, Y(\mathfrak{sl}_2))\big) \cong H^*(\Sigma_2) \otimes s^{-1}\mathfrak{sl}_2 \cong (s^{-1}\mathfrak{sl}_2)^{\oplus 6},$$

with the six copies sitting in cohomological degrees 1, 2, 2, 2, 2, 3 (*from the tensor product of* $H^{0,1,1,1,1,2}(\Sigma_2)$ *with the degree-*1 *bar cohomology* $s^{-1}\mathfrak{sl}_2$). *The total dimension is* $6 \cdot 3 = 18$.

Proof. Since the KZB connection is trivial at degree 1 (no pairs $i < j$), the complex splits as a tensor product. The bar cohomology at degree 1 is $s^{-1}\mathfrak{sl}_2$ by Koszulness of $V_k(\mathfrak{sl}_2)$ (equation (10.2)). The de Rham Betti numbers of Σ_2 are (1, 4, 1), and the Künneth formula gives $6 = 1 + 4 + 1$ copies of $s^{-1}\mathfrak{sl}_2$. □

Remark 11.6 (Comparison with genus 0 and genus 1). At genus 0, the degree-1 cohomology is 3-dimensional (1 copy of $s^{-1}\mathfrak{sl}_2$ from $H^0(D^\times)$, with additional degree-1 contribution from the circle). At genus 1, the Betti numbers (1, 2, 1) give $12 = 4 \cdot 3$. At genus 2, the Betti numbers (1, 4, 1) give $18 = 6 \cdot 3$. The pattern at genus g is $\dim H^*(C_1^{\mathrm{ord}}) = 3(2g + 2)$ from $\dim H^*(\Sigma_g) = 2 + 2g$, reflecting the $2g$ extra 1-cycles on Σ_g.

11.5. **Degree 2: the genus-2 KZB local system.** At degree 2, the configuration space $\mathrm{Conf}_2^{\mathrm{ord}}(\Sigma_2) = \Sigma_2^2 \setminus \Delta$ carries the KZB connection with non-trivial pairwise interaction. After separating the centre-of-mass coordinate (translation on Σ_2 via the Jacobian $\mathrm{Jac}(\Sigma_2) = \mathbb{C}^2/(\mathbb{Z}^2 + \Omega\,\mathbb{Z}^2)$, which has dimension 2), the relative configuration is a punctured genus-2 surface: the KZB local system lives on $\Sigma_2 \setminus \{0\}$, a genus-2 surface with one puncture.

The ordered chiral chain complex at degree 2 is

$$C_2^{\mathrm{ord}}(\Sigma_2, Y) = \Big(\Omega^*_{\log}\big(\overline{\mathrm{FM}}_2^{\mathrm{ord}}(\Sigma_2)\big) \otimes (s^{-1}\bar{Y})^{\otimes 2},\ \ d_{\mathrm{dR}} + d_{\mathrm{bar}} - \nabla_{\mathrm{KZB}}^{(g=2)}\Big).$$

The collision residue is governed by the genus-2 r-matrix

$$r^{(g=2)}(z_i, z_j) = \quad S_2(z_i, z_j)\,\Omega = \frac{1}{k + h^\vee}\, d_{z_i} \log E(z_i, z_j)\,\Omega,$$

where $\;\; = 1/(k + h^\vee) = 1/(k + 2)$ and Ω is the $\mathfrak{sl}_2$ Casimir (in the KZ normalisation consistent with equation (10.1)).

Decomposition by $\mathfrak{sl}_2$-isospin. On $V \otimes V = \mathrm{Sym}^2 V \oplus \wedge^2 V$ with $V = \mathbb{C}^2$:

$$\Omega\big|_{\mathrm{Sym}^2} = +\tfrac{1}{2}, \qquad \Omega\big|_{\wedge^2} = -\tfrac{3}{2},$$

exactly as at genus 0 and genus 1. The KZB system decomposes into scalar equations on each isospin channel, now with genus-2 coefficients.

Monodromy. The fundamental group $\pi_1(\Sigma_2 \setminus \{0\})$ has presentation

$$\pi_1(\Sigma_2 \setminus \{0\}) = \big\langle a_1, b_1, a_2, b_2, \gamma \;\big|\; [a_1, b_1]\,[a_2, b_2]\,\gamma = 1 \big\rangle$$

with a_α, b_α the A- and B-cycle generators ($\alpha = 1, 2$) and γ the puncture loop. The KZB connection produces a monodromy representation

$$\rho^{(2)} : \pi_1(\Sigma_2 \setminus \{0\}) \to \mathrm{GL}(V \otimes V) \tag{11.3}$$

with monodromy matrices:

- Puncture loop: $M_\gamma = \exp(2\pi i \;\; \Omega)$ (identical to genus 0 and genus 1, capturing the local collision residue).
- A-cycles: M_{a_α} ($\alpha = 1, 2$); the A-cycle monodromy is trivial at leading order since $S_2(z, w)$ is periodic under A-cycles: $S_2(z + A_\alpha, w) = S_2(z, w)$.
- B-cycles: M_{b_α} ($\alpha = 1, 2$); the quasi-periodicity (11.2) gives a shift by $-2\pi i\, \omega_\alpha$, producing
$$M_{b_\alpha} = \exp\!\big(-2\pi i \;\; \Omega \cdot \omega_\alpha\big), \qquad \alpha = 1, 2.$$
At genus 1, $\omega_1 = dz$ is a scalar and $M_B = M_\gamma^{-1}$; at genus 2, the two B-cycle monodromies M_{b_1}, M_{b_2} are independent and involve the two holomorphic differentials ω_1, ω_2 separately.

The constraint from the fundamental group relation is

$$[M_{a_1}, M_{b_1}]\,[M_{a_2}, M_{b_2}]\,M_\gamma = I.$$

Remark 11.7 (Dynamical parameters at genus 2). The B-cycle monodromies at genus 2 depend on a *two-component* dynamical parameter $\lambda = (\lambda_1, \lambda_2) \in \mathfrak{h}^* \oplus \mathfrak{h}^*$, replacing the single dynamical parameter $\lambda \in \mathfrak{h}^*$ of the genus-1 theory (Remark 10.7). The dynamical Yang–Baxter equation is replaced by a system of two coupled equations, one for each B-cycle. This is the genus-2 instantiation of the Felder dynamical R-matrix theory [32, 33].

Proposition 11.8 (Degree-2 ordered chiral homology on Σ_2). *The degree-2 ordered chiral homology $H^*(C_2^{\mathrm{ord}}(\Sigma_2, Y))$ is the de Rham cohomology of the KZB local system on $\mathrm{Conf}_2^{\mathrm{ord}}(\Sigma_2)$ with coefficients in $(s^{-1}\bar{Y})^{\otimes 2}$. On evaluation modules $V_a \otimes V_b$ ($V = \mathbb{C}^2$, so the local system has rank $d^2 = 4$):*

(i) *The KZB local system on $\Sigma_2 \setminus \{0\}$ (a genus-2 surface with one puncture) has monodromy representation (11.3) with generators $M_{a_1}, M_{b_1}, M_{a_2}, M_{b_2}, M_\gamma$.*

(ii) *For generic $\;\;$, the monodromy has no invariant vectors: $H^0_{\mathrm{dR}} = 0$.*

(iii) *The Euler characteristic of a rank-r local system on a surface of genus g with s punctures is $\chi = r(2 - 2g - s)$. For $g = 2$, $s = 1$, $r = d^2 = 4$:*
$$\chi = 4 \cdot (2 - 4 - 1) = -12.$$

(iv) *Since $H^0 = 0$ for generic $\;\;$ and $H^2 = 0$ (the puncture forces vanishing), we obtain:*
$$\dim H^1_{\mathrm{dR}}(\Sigma_2 \setminus \{0\},\, \nabla_{\mathrm{KZB}}) = 12.$$

Proof. The surface $\Sigma_2 \setminus \{0\}$ has genus $g = 2$ and $s = 1$ punctures. The Euler characteristic of a local system of rank r is $\chi = r(2 - 2g - s)$: $\chi = 4(2 - 4 - 1) = 4 \cdot (-3) = -12$. For generic (equivalently, generic level k), the monodromy around the puncture is $M_\gamma = \exp(2\pi i \quad \Omega)$ with eigenvalues $e^{\pi i/(k+2)}$ (on $\mathrm{Sym}^2 V$, dimension 3) and $e^{-3\pi i/(k+2)}$ (on $\bigwedge^2 V$, dimension 1); generically neither is 1, so $\ker(M_\gamma - I) = 0$ and $H^0 = 0$. The four B-cycle monodromies M_{b_1}, M_{b_2} impose further nontrivial constraints, reinforcing $H^0 = 0$. Poincaré–Verdier duality on $\Sigma_2 \setminus \{0\}$ (a non-compact surface) gives $H^2 = (H^0_c)^\vee = 0$ for generic monodromy; alternatively, the puncture forces $H^2 = 0$ by a residue argument. The alternating sum gives $0 - \dim H^1 + 0 = -12$, so $\dim H^1 = 12$. The 12-dimensional space decomposes along the isospin channels: $\mathrm{Sym}^2 V$ (rank 3) contributes $3 \cdot 3 = 9$, and $\bigwedge^2 V$ (rank 1) contributes $1 \cdot 3 = 3$; total $9 + 3 = 12$. □

Remark 11.9 (Comparison of H^1 dimensions). The degree-2 H^1 dimension grows linearly with genus:

g	$\chi(\Sigma_g \setminus \{0\})$	$\chi = r \cdot (2 - 2g - 1)$	$\dim H^1$
0	1	$4 \cdot 1 = 4$	4
1	-1	$4 \cdot (-1) = -4$	4
2	-3	$4 \cdot (-3) = -12$	12
g	$1 - 2g$	$4(1 - 2g)$	$4(2g - 1)$

The jump from $\dim H^1 = 4$ at $g = 1$ to $\dim H^1 = 12$ at $g = 2$ reflects the richer monodromy: at genus 1, the monodromy group is generated by three elements (M_γ, M_a, M_b) with one relation; at genus 2, by five elements $(M_\gamma, M_{a_1}, M_{b_1}, M_{a_2}, M_{b_2})$ with one relation, allowing a 12-dimensional space of flat sections with prescribed singular behaviour at the puncture. The genus-0 entry $\dim H^1 = 4$ uses $\chi(\mathbb{P}^1 \setminus \{0, \infty\}) = 0$ for the once-punctured sphere (the relative coordinate lives on $\mathbb{C}^\times$), so $\chi = 0$ and the H^1-dimension does not follow from χ alone; it is computed directly from KZ flat sections (cf. §10.4).

Remark 11.10 (The symmetric shadow at genus 2). The averaging map av_2 at degree 2 sends the 12-dimensional ordered de Rham cohomology to Σ_2-coinvariants. The scalar surviving av_2 is $\kappa = 3(k+2)/4$ (equation (9.1)), the same value as at genus 0 and genus 1: the averaging map is a local computation (residue extraction), independent of the global topology. The kernel $\ker(\mathrm{av}_2)$ at genus 2 is richer than at genus 1: the four B-cycle monodromies contribute independent antisymmetric classes that are invisible to the symmetric theory.

11.6. **Degree n: genus-2 surface braid group monodromy.** At degree n, the fundamental group of $\mathrm{Conf}_n^{\mathrm{ord}}(\Sigma_2)$ is the genus-2 *surface braid group* $\mathrm{SB}_n(\Sigma_2)$, which fits into the extension

$$1 \to PB_n \to \mathrm{SB}_n(\Sigma_2) \to \pi_1(\Sigma_2)^n \to 1,$$

generalising the elliptic braid group extension (10.6) (at genus 1, $\pi_1(E_\tau) \cong \mathbb{Z}^2$ contributes two generators per point; at genus 2, $\pi_1(\Sigma_2) \cong \langle a_1, b_1, a_2, b_2 \mid [a_1, b_1][a_2, b_2] = 1 \rangle$ contributes four generators per point).

The ordered chiral homology at degree n is

$$\int_{\Sigma_2, n}^{\mathrm{ord}} Y(\mathfrak{sl}_2) = H^*_{\mathrm{dR}}\big(\overline{\mathrm{FM}}_n^{\mathrm{ord}}(\Sigma_2),\ \nabla^{(n)}_{\mathrm{KZB}}\big) \otimes (s^{-1}\bar{Y})^{\otimes n}.$$

Proposition 11.11 (Euler characteristic at low degrees, genus 2)**.** *The twisted Euler characteristics of the KZB local system on* $\mathrm{Conf}_n^{\mathrm{ord}}(\Sigma_2)$, *with fibre* $V^{\otimes n}$ $(V = \mathbb{C}^2)$ *for generic* , *are:*

$$\begin{aligned} n = 0 :&\quad \chi = \infty \quad (\mathbb{C}[\mathrm{qdet}] \textit{ is infinite-dimensional}),\\ n = 1 :&\quad \chi = -4 \quad (\textit{trivial local system of rank } 2 \textit{ on } \Sigma_2,\ \chi(\Sigma_2) = -2),\\ n = 2 :&\quad \chi = -12 \quad (\textit{rank } 4 \textit{ on } \Sigma_2 \setminus \{0\}). \end{aligned}$$

For $n \geq 3$, *the Euler characteristic of* $\mathrm{Conf}_n^{\mathrm{ord}}(\Sigma_2)$ *with coefficients in a rank-*d^n *local system is:*

$$\chi\big(\mathrm{Conf}_n^{\mathrm{ord}}(\Sigma_2),\ \mathcal{L}_{\mathrm{KZB}}\big) = d^n \cdot \prod_{j=0}^{n-1}\big(\chi(\Sigma_2) - j\big) = 2^n \cdot \prod_{j=0}^{n-1}(-2 - j). \tag{11.4}$$

In particular: $n = 3$ *gives* $8 \cdot (-2)(-3)(-4) = -192$; $n = 4$ *gives* $16 \cdot (-2)(-3)(-4)(-5) = -1920$.

Proof. At $n = 1$: $\mathrm{Conf}_1(\Sigma_2) = \Sigma_2$ and the KZB connection is trivial (no pairwise terms). The local system has rank $d = 2$ and the Euler characteristic is $\chi = d \cdot \chi(\Sigma_2) = 2 \cdot (-2) = -4$. (In contrast to genus 1 where $\chi(E_\tau) = 0$ gives $\chi = 0$, the negative Euler characteristic of Σ_2 makes the twisted Euler characteristic non-zero even at degree 1.)

At $n = 2$: Proposition 11.8.

At $n \geq 3$: the product formula for the Euler characteristic of ordered configuration spaces gives $\chi(\mathrm{Conf}_n^{\mathrm{ord}}(\Sigma_2)) = \prod_{j=0}^{n-1}(\chi(\Sigma_2) - j) = \prod_{j=0}^{n-1}(-2 - j)$, and the multiplicativity of the Euler characteristic with respect to local systems gives (11.4). (Unlike genus 1, where $\chi(E_\tau) = 0$ kills the product at the first factor and gives $\chi(\mathrm{Conf}_n^{\mathrm{ord}}(E_\tau)) = 0$ for all $n \geq 1$, here $\chi(\Sigma_2) = -2 \neq 0$ and the product is non-vanishing for all n.) □

11.7. **Summary: genus 0 vs. genus 1 vs. genus 2.**

	Genus 0 ($\mathbb{P}^1$ or $D^\times$)	Genus 1 (E_τ)	Genus 2 (Σ_2)
Base curve	$\mathbb{P}^1$	$E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$	Σ_2
Period matrix	--	$\tau \in \mathbb{H}$, scalar	$\Omega \in \mathbb{H}_2$, 2×2
Bar propagator	$d\log(z_1 - z_2)$	$d\log E(z,w) = \wp_1\, dz$	$d\log E(z,w)$ (prime form)
weight	1	1	1
Connection	KZ	KZB	KZB (genus 2)
π_1 per point	0	$\mathbb{Z}^2$ (A, B-cycle)	$\mathbb{Z}^4$ (A_1, B_1, A_2, B_2)
B-cycle monodromy	--	$M_B = e^{-2\pi i \quad \Omega}$	$M_{b_\alpha} = e^{-2\pi i \quad \Omega\cdot\omega_\alpha}$
Dynamical parameter	--	$\lambda \in \mathfrak{h}^*$, one component	$(\lambda_1, \lambda_2) \in (\mathfrak{h}^*)^2$
Degree 0	$\mathbb{C}[\mathrm{qdet}]$	$\mathbb{C}[\mathrm{qdet}]$	$\mathbb{C}[\mathrm{qdet}]$
Degree 1 dim	3 (on $D^\times$)	$12 = 4 \times 3$	$18 = 6 \times 3$
Degree 2: dim H^1	4	4	12
Degree 2: χ	4 ($g = 0$, $s = 2$)	-4	-12
$\chi(\mathrm{Conf}_n)$, $n \geq 1$	varies	0 (all $n \geq 1$)	$\prod_{j=0}^{n-1}(-2 - j) \neq 0$
Curvature d_{fib}^2	0	$\kappa(\mathcal{A}) \cdot \omega_1$	$\kappa(\mathcal{A}) \cdot \omega_2$
lives on	--	$\overline{\mathcal{M}}_1$	$\overline{\mathcal{M}}_2$
Degeneration	--	$\tau \to i\infty$: $E_\tau \to \mathbb{C}^\times$	$\Omega_{12} \to 0$, $\Omega_{22} \to i\infty$: $\Sigma_2 \to E_\tau$

Remark 11.12 (Euler characteristics and degeneration). At genus 1, $\chi(E_\tau) = 0$ forces $\chi(\mathrm{Conf}_n^{\mathrm{ord}}(E_\tau)) = 0$ for $n \geq 1$; at genus 2, $\chi(\Sigma_2) = -2$ gives rapidly growing Euler characteristics $(-4, -12, -192, \ldots)$ providing genuine lower bounds on de Rham cohomology dimensions. In the separating degeneration $\Sigma_2 \to E_\tau \cup E_{\tau'}$, the KZB splits and ordered chiral homology factorises: $\int_{\Sigma_2}^{\mathrm{ord}} Y \xrightarrow{\mathrm{res}} \int_{E_\tau}^{\mathrm{ord}} Y \otimes \int_{E_{\tau'}}^{\mathrm{ord}} Y$. The non-separating degeneration carries genuinely genus-2 information (the off-diagonal Ω_{12}).

12. Comparison with related work

Two existing programmes address the integration of braided algebraic structures over surfaces: Ben-Zvi–Brochier–Jordan integrate braided monoidal *categories*, while Latyntsev constructs chiral homology for factorisation *coalgebras*. Ordered chiral homology occupies a different position: it integrates chiral *algebras* at the chain level, applies uniformly to all four shadow classes, and does not require a finite ribbon category.

Conjecture 12.1 (BBJ decategorification). *Let $\mathfrak{g}$ be a finite-dimensional simple Lie algebra, k an admissible rational level (so that $\mathrm{Rep}_q(\mathfrak{g})$ is a finite ribbon category with $q = e^{2\pi i/(k+h^\vee)}$), Σ a closed oriented surface, and X a smooth projective curve of the same genus.*

(i) (Decategorification quasi-isomorphism.) *There is a quasi-isomorphism*

$$\mathrm{HH}_*\Big(\int_\Sigma \mathrm{Rep}_q(\mathfrak{g})\Big) \simeq \int_X^{\mathrm{ord}} V_k(\mathfrak{g}), \tag{12.1}$$

where the left side is the categorical Hochschild homology of the BBJ categorical factorisation homology and the right side is the ordered chiral homology of the affine vertex algebra.

(ii) (E_3-intertwining.) *The quasi-isomorphism* (12.1) *intertwines the E_3-structure on the derived chiral centre $Z_{\mathrm{ch}}^{\mathrm{der}}(V_k(\mathfrak{g}))$ with the E_3-structure on the categorical Hochschild cochains of the categorical invariant.*

(iii) (Scope: admissible rational levels.) *The comparison requires k admissible rational, so that $\mathrm{Rep}_q(\mathfrak{g})$ is a finite ribbon category (the BBJ construction uses the ribbon structure and the finiteness condition). At irrational or generic k, the categorical side is not defined; ordered chiral homology applies without restriction.*

(iv) (Kernel.) *The kernel of the decategorification map* (12.1) *detects higher categorical data: objects of $\int_\Sigma \mathrm{Rep}_q(\mathfrak{g})$ that are invisible at the chain level (i.e. whose categorical Hochschild homology vanishes).*

Remark 12.2 (BBJ and Latyntsev). BBJ [2, 3] integrate braided monoidal categories over surfaces; ordered chiral homology integrates chiral algebras over curves. Conjecture 12.1 asserts that categorical Hochschild homology of the former recovers the latter at admissible rational levels; at generic k (where no finite ribbon category exists), ordered chiral homology applies without restriction. Latyntsev [31] constructs chiral homology for factorisation coalgebras; the chain-level comparison with the present construction (particularly for irregular singularities, class M) is open.

Remark 12.3 (Wilson lines by ambient dimension). The type of Wilson line, the form of the R-matrix, and the resulting Yang–Baxter equation are controlled by the ambient space dimension. The following table summarizes the principal cases.

Ambient	R-matrix	YBE	Algebra	Integrable system
$\mathbb{R}^3$	constant R	quantum $R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}$	$U_q(\mathfrak{g})$	Jones polynomial
$\mathbb{R}\times\mathbb{C}$	$r(z)$, 1 spectral	CYBE $[r_{12}, r_{13}]$ + cyc = 0	$Y(\mathfrak{g})$	Gaudin
$\mathbb{R}^2\times\mathbb{C}$	$R(u)$, spectral	quantum spectral YBE	$U_q(L\mathfrak{g})$	XXZ
$\mathbb{C}^2$	$r(z,w)$, 2 params	2d CYBE	$\mathfrak{g}[u,v]$	Hitchin on surface
$\mathbb{R}\times\mathbb{C}^2$	$R(u,v)$, 2 spectral	quantum + dynamical	$U_{q,t}(\widehat{\widehat{\mathfrak{g}}})$	eCM + Hitchin
$\mathbb{C}^3$ (CY)	$r + \mathbb{Z}\times\mathbb{Z}$ grading	6d YBE	E_1-chiral QG	DT/BPS

This paper works in the $\mathbb{R}\times\mathbb{C}$ setting (Costello's 4d Chern–Simons [12–15]), producing the Yangian $Y(\mathfrak{g})$ as the line-operator algebra. The extensions to $\mathbb{R}^2\times\mathbb{C}$ (quantum loop algebras), $\mathbb{C}^2$ (double current algebras), $\mathbb{R}\times\mathbb{C}^2$ (quantum toroidal algebras), and $\mathbb{C}^3$ (6d holomorphic Chern–Simons defects) are frontier directions; ordered chiral homology provides the natural integration theory for the $\mathbb{R}\times\mathbb{C}$ column.

13. Further results

Three results from the monograph [34] that complement the constructions of this paper.

Theorem 13.1 (Verlinde polynomial family; [34, Thm. ??]). *The Verlinde dimension* $Z_g(k) = \sum_{j=0}^{k} S_{0j}^{2-2g}$ *for* $\widehat{\mathfrak{sl}}_2$ *at level* k *is a polynomial* $P_g(n)$ *of degree* $3(g-1)$ *in* $n = k+2$ *with* $P_g(n) = n^{g-1}(n^2-1)\cdot R_{g-2}(n^2)$. *Through* $g = 6$*:* $P_2 = n(n^2{-}1)/6$, $P_3 = n^2(n^2{-}1)(n^2{+}11)/180$, $P_4 = n^3(n^2{-}1)(2n^4{+}23n^2{+}191)/7560$. *Leading coefficient:* $\zeta(2g{-}2)/(2^{g-2}\pi^{2g-2})$. *Rational generating function:* $G_n(x) = \sum_{j=1}^{n-1} 1/(1-a_j x)$, $a_j = n/(2\sin^2(\pi j/n))$.

Proposition 13.2 (Critical level: monodromy trivialises; [34, Prop. ??]). *At* $k = -h^\vee$ *for* $V_{-2}(\mathfrak{sl}_2)$*:* $\kappa = 0$, *ALL monodromy is trivial (eigenvalues* $-1, +3$ *are integers), the degree-2 ordered chiral homology on* $V\otimes V$ *has* $\dim H^1_{\mathrm{ord},2}$ *doubling from* 4 *to* 8 *(a finite computation conserving* $\chi_{\mathrm{ord},2} = -4$*), Koszulness fails, and the FULL symmetric bar cohomology spreads to* $\Omega^*(\mathrm{Op}_{\mathfrak{sl}_2}(D))$*, which is infinite-dimensional in every cohomological degree (Feigin–Frenkel 1992, Frenkel–Teleman 2006; distinct from the finite degree-2 ordered slice). The spectral sequence mechanism:* $d_k = d_{\mathrm{crit}} + (k+2)\,\delta$*; at* $\lambda = k+2 = 0$*, the* d_1 *page vanishes.*

Proposition 13.3 (Genus-2 conformal blocks by degree; [34, Prop. ??]). *At integer level* k*, the degree-2 conformal block count on* Σ_2 *is* $\mathrm{CB}_{2,2}(k) = 2k(k+1)(k+2)/3$ *(cubic in* k*). At* $k = 1$*:* 4 *(triplet truncated). At* $k = 2$*:* 16*. At* $k = 3$*:* 40.

14. Conclusion and open problems

This paper defines ordered chiral homology $\int_X^{\mathrm{ord}} \mathcal{A}$ for E_1-chiral algebras on algebraic curves and establishes the chiral quantum group equivalence (Theorem 6.5) relating vertex R-matrices, A_∞-structures, and chiral coproducts as three faces of the ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$. The representation-theoretic kernel formula $\dim\ker(\mathrm{av}_n) = d^n - \binom{n+d-1}{d-1}$ (Proposition 2.6) quantifies the exponential growth of ordered data versus the polynomial growth of the symmetric shadow. Explicit computations across all four shadow classes verify the formality bridge and the regular/irregular dichotomy. The genus-1 and genus-2 extensions (§10–11) produce the KZB connection and the quantum group parameter $q = e^{2\pi i}$.

The following problems remain open.

(a) (E_3 identification: resolved for simple $\mathfrak{g}$.) Theorem 8.6 matches the derived chiral centre and the CFG algebra over $(k+h^\vee)H^3(\mathfrak{g})[[k+h^\vee]]$. Extension to non-simple $\mathfrak{g}$ (where $H^3(\mathfrak{g})$ is higher-dimensional) remains open.

(b) (Ordered chiral centre of the chiral Yangian.) Conjecture 2.4: the passage from the kernel bound $d^n - \binom{n+d-1}{d-1}$ to the actual bar cohomology; whether the degree-1 Gerstenhaber bracket recovers $\mathfrak{g}$ for all simple Lie algebras.

(c) (Holomorphic CS invariants of curves.) Does $\int_X^{\mathrm{ord}} \mathrm{CE}_k^{\mathrm{ch}}(\mathfrak{g})$ recover $Z_{\mathrm{CS}}(X, \mathfrak{g}, k)$ as the Σ_n-coinvariant shadow?

(d) ($\mathcal{W}_{1+\infty}$: Stokes multipliers.) The remaining open question is the Stokes multipliers of the irregular KZ connection for $\mathcal{W}_N$ (Poincaré rank $N-1$), and whether they encode the higher-spin BPS spectrum. The antipode obstruction is resolved (Proposition 6.22).

(e) (Higher genus.) At $g \geq 3$, the curvature obstruction $d_{\mathrm{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g$ (UNIFORM-WEIGHT) requires a period-corrected differential with $(g \times g)$-matrix $(\mathrm{Im}\,\Omega)^{-1}$ coefficients. Factorisation at boundary divisors of $\overline{\mathcal{M}}_g$ is open.

(f) (Beyond $\mathfrak{sl}_2$.) Trigonometric and elliptic R-matrices for $\mathfrak{sl}_N$ ($N \geq 3$): regular/irregular/elliptic trichotomy correlated with quantum group types.

(g) (Categorification.) Conjecture 12.1: the BBJ decategorification $\mathrm{HH}_*(\int_\Sigma \mathrm{Rep}_q(\mathfrak{g})) \simeq \int_X^{\mathrm{ord}} V_k(\mathfrak{g})$ at admissible rational levels.

(h) (Topological enhancement at higher genus.) Does the $\mathsf{E}_3^{\mathrm{top}}$ structure on $\mathrm{CE}_k^{\mathrm{ch}}(\mathfrak{g})$ (Proposition 8.11) extend to genus $g \geq 2$?

References

[1] V. I. Arnold, *The cohomology ring of the group of dyed braids*, Mat. Zametki **5** (1969), 227–231.

[2] D. Ben-Zvi, A. Brochier, and D. Jordan, *Integrating quantum groups over surfaces*, J. Topol. **11** (2018), 874–917; arXiv:1501.04652.

[3] D. Ben-Zvi, A. Brochier, and D. Jordan, *Quantum character varieties and braided module categories*, Selecta Math. (N.S.) **24** (2018), 4711–4748; arXiv:1606.04769.

[4] A. Beilinson and V. Drinfeld, *Chiral algebras*, American Mathematical Society Colloquium Publications, vol. 51, AMS, Providence, RI, 2004.

[5] D. Bernard, *On the Wess–Zumino–Witten models on the torus*, Nucl. Phys. B **303** (1988), 77–93.

[6] D. Calaque, B. Enriquez, and P. Etingof, *Universal KZB equations: the elliptic case*, in Algebra, Arithmetic, and Geometry (Manin Festschrift), Progress in Math., vol. 269, Birkhäuser, 2009, pp. 165–266.

[7] T. Kadeishvili, *The algebraic structure in the homology of an $A(\infty)$-algebra*, Soobshch. Akad. Nauk Gruzin. SSR **108** (1982), 249–252.

[8] M. Kontsevich, *Operads and motives in deformation quantization*, Lett. Math. Phys. **48** (1999), 35–72; arXiv:math/9904055.

[9] D. Tamarkin, *Formality of chain operad of little discs*, Lett. Math. Phys. **66** (2003), 65–72; arXiv:math/9809164.

[10] V. G. Drinfeld, *Hopf algebras and the quantum Yang–Baxter equation*, Soviet Math. Doklady **32** (1985), 254–258.

[11] V. G. Drinfeld, *Quasi-Hopf algebras*, Leningrad Math. J. **1** (1990), 1419–1457.

[12] K. Costello, J. Francis, and O. Gwilliam, *Chern–Simons factorization algebras and knot polynomials*, arXiv:2602.12412, 2026.

[13] K. Costello and O. Gwilliam, *Factorization algebras in quantum field theory*, vols. 1–2, Cambridge University Press, 2017 and 2021.

[14] K. Costello, E. Witten, and M. Yamazaki, *Gauge theory and integrability, I*, Notices ICCM **6** (2018), no. 1, 46–119; arXiv:1709.09993.

[15] K. Costello, E. Witten, and M. Yamazaki, *Gauge theory and integrability, II*, Notices ICCM **6** (2018), no. 1, 120–146; arXiv:1802.01579.

[16] A. D'Agnolo and M. Kashiwara, *Riemann–Hilbert correspondence for holonomic D-modules*, Publ. Math. Inst. Hautes Études Sci. **123** (2016), 69–197.

[17] P. Etingof and D. Kazhdan, *Quantization of Lie bialgebras, I*, Selecta Math. (N.S.) **2** (1996), 1–41.
[18] P. Etingof and D. Kazhdan, *Quantization of Lie bialgebras, V: Quantum vertex operator algebras*, Selecta Math. (N.S.) **6** (2000), 105–130.
[19] P. Etingof and A. Varchenko, *Solutions of the quantum dynamical Yang–Baxter equation and dynamical quantum groups*, Comm. Math. Phys. **196** (1998), 591–640.
[20] E. Frenkel and D. Ben-Zvi, *Vertex algebras and algebraic curves*, second ed., Mathematical Surveys and Monographs, vol. 88, AMS, Providence, RI, 2004.
[21] J. Francis, *The tangent complex and Hochschild cohomology of* E_n*-rings*, Compos. Math. **149** (2013), 430–480.
[22] D. Gaiotto and M. Rapčák, *Vertex algebras at the corner*, J. High Energy Phys. **2019**(1), 160; arXiv:1703.00982.
[23] R. Hotta, K. Takeuchi, and T. Tanisaki, *D-modules, perverse sheaves, and representation theory*, Birkhäuser, Boston, 2008.
[24] J. Lurie, *Higher algebra*, available at `math.ias.edu/~lurie/`, 2017.
[25] S. Jindal, S. Kaubrys, and A. Latyntsev, *Critical CoHAs, vertex coalgebras, and deformed Drinfeld coproducts*, arXiv:2603.21707 (2026).
[26] D. Joyce, *Enumerative invariants and wall-crossing formulae in abelian categories*, Lecture notes, 2018; see also *Configurations in abelian categories. IV*, Adv. Math. **217** (2008), 125–204.
[27] M. Kashiwara, *D-modules and microlocal calculus*, Translations of Mathematical Monographs, vol. 217, AMS, 2003.
[28] A. Z. Khan and K. Zeng, *Poisson vertex algebras and three-dimensional gauge theory*, arXiv:2502.13227, 2025.
[29] D. Maulik and A. Okounkov, *Quantum groups and quantum cohomology*, Astérisque, vol. 408, Société Mathématique de France, 2019.
[30] M. Kontsevich and Y. Soibelman, *Cohomological Hall algebra, exponential Hodge structures and motivic Donaldson–Thomas invariants*, Commun. Number Theory Phys. **5** (2011), 231–352; arXiv:1006.2706.
[31] A. Latyntsev, *Factorisation quantum groups*, arXiv:2312.07274, 2023.
[32] G. Felder, *Conformal field theory and integrable systems associated to elliptic curves*, in Proc. ICM Zürich 1994, Birkhäuser, 1995, pp. 1247–1255.
[33] G. Felder and A. Varchenko, *On representations of the elliptic quantum group* $E_{\tau,\eta}(\mathfrak{sl}_2)$, Commun. Math. Phys. **181** (1996), 741–761; q-alg/9601003.
[34] R. Lorgat, *Modular Koszul duality*, in preparation, 2026.
[35] R. Lorgat, A_∞ *chiral algebras and 3D holomorphic-topological QFT*, in preparation, 2026.
[36] R. Lorgat, *Calabi–Yau categories, quantum groups, and BPS algebras*, in preparation, 2026.
[37] A. Molev, *Yangians and classical Lie algebras*, Mathematical Surveys and Monographs, vol. 143, AMS, Providence, RI, 2007.
[38] M. Rapčák, Y. Soibelman, Y. Yang, and G. Zhao, *Cohomological Hall algebras, vertex algebras and instantons*, Commun. Math. Phys. **376** (2020), 1803–1873; arXiv:1810.10402.
[39] T. Procházka and M. Rapčák, $\mathcal{W}$*-algebra modules, free fields, and Gukov–Witten defects*, J. High Energy Phys. **2019**, no. 5, 159; arXiv:1808.08837.
[40] Z. Mebkhout, *Une équivalence de catégories et une autre équivalence de catégories*, Compos. Math. **51** (1984), 51–62; 63–68.
[41] C. Sabbah, *Introduction to Stokes structures*, Lecture Notes in Math., vol. 2060, Springer, Berlin, 2013.
[42] D. Sinha, *Manifold-theoretic compactifications of configuration spaces*, Selecta Math. (N.S.) **10** (2004), 391–428.
[43] O. Schiffmann, A. Mellit, A. Minets, and E. Vasserot, *Coherent sheaves on surfaces, COHAs and deformed* $W_{1+\infty}$*-algebras*, arXiv:2311.13415, 2023.
[44] O. Schiffmann and E. Vasserot, *Cherednik algebras, W-algebras and the equivariant cohomology of the moduli space of instantons on* $\mathbb{A}^2$, Publ. Math. Inst. Hautes Études Sci. **118** (2013), 213–342; arXiv:1202.2756.
[45] W. Wasow, *Asymptotic expansions for ordinary differential equations*, Interscience Publishers, New York, 1965; reprinted Dover, 1987.
[46] A. Tsuchiya, K. Ueno, and Y. Yamada, *Conformal field theory on universal family of stable curves with gauge symmetries*, Adv. Studies Pure Math. **19** (1989), 459–566.
[47] E. Verlinde, *Fusion rules and modular transformations in 2d conformal field theory*, Nuclear Phys. B **300** (1988), 115–138.
[48] Y. Yang and G. Zhao, *Cohomological Hall algebras and affine quantum groups*, Selecta Math. (N.S.) **24** (2018), 1093–1119; arXiv:1604.01865.

Perimeter Institute for Theoretical Physics, Waterloo, ON N2L 2Y5, Canada, Department of Physics, University of Waterloo, Waterloo, ON N2L 3G1, Canada

Email address: `rlorgat@perimeterinstitute.ca`